MATH ROCK ESSENTIALS



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A **GUITARIST'S** GUIDE TO **MATH ROCK**

A Guitarist's Guide to Math Rock

Hello, Steve here, hope you are doing well. In this guide, you will find a wealth of materials to help you learn, write, and play math rock (and for styles often associated with math rock) on your guitar.

The guide is set up so you can start from any section without needing to finish any other part first. But, for the best results, I recommend following it step by step in order.

Example videos are included for each exercise giving you a visual and auditory reference to help you learn each one. These can be found by clicking the link beneath each exercise. The examples are played slowly, making them easier to digest.

Best of luck and thanks for the support,

Steve

As a bonus, I have created a folder containing all the exercises in PDF and Guitar Pro file format: Click this link to download them

Focus

Before we dive into the lessons, I want to emphasize the critical importance of focus in your guitar practice. It may sound simple and obvious, but it is incredibly effective and often underestimated. Focusing means being fully present in the moment when you are playing or practicing. This practice will not only make you play better but also help you notice details that you might otherwise miss when your mind is wandering.

Consider this: an hour of concentrated practice, where you pay close attention to your hands, the feel and sound of what you are playing, the piece you are practicing, or the music you are composing, is much more valuable (in my view) than three or four hours of distracted, interrupted practice. So, I encourage you to make an effort to focus your attention on your guitar during your practice sessions.

If you find your mind drifting, or if you are feeling tired and unfocused, it might be a sign that your thoughts are elsewhere. In such cases, it is often more productive to set the guitar aside and practice when you can give it your full attention. So, put aside thoughts of other tasks, games, socializing, work, and so on, during your practice time. By doing this, I hope you will start to notice significant improvements in your guitar playing abilities. With that said, let's dive right in!

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Section 1: Math Rock Guitar Techniques

Segment 1A – Finger Tapping

How to Practice?

Before we dive into the core math rock guitar techniques, I would like to share an effective practice strategy. This approach is aimed at learning new techniques or refining existing ones. It may sound counterintuitive, but it involves intentionally making mistakes. This process of generating errors is essential for skill improvement. Here is how it works:

1. Generate Errors: The brain processes errors after each practice session. When you return to practice, you will begin to see improvements in that skill. Therefore, aim to perform as many repetitions as possible of the technique you are practicing within your dedicated practice time.

2. Use a Metronome: A metronome should be your ally in this process. Start with a tempo where you can comfortably play the exercise, then gradually increase it (typically by 3 bpm increments) until you reach a point where you are no longer comfortable. This is the sweet spot for improvement because you will be generating more errors in your repetitions, pushing your brain's neural plasticity to adapt to this new challenge. Hopefully, by the next practice session, that challenging tempo will become easier, and your technique will be cleaner.

3. Focused Practice: When practicing a technique (skill), it is crucial to have a specific focus, even if it is just one aspect of the technique. For instance, you might dedicate a practice session to observing the movement of one of your hands or focusing on the positioning of one of your hands. Notice where there is minimal resistance and what feels most comfortable. By rotating your focus in different practice sessions, you can systematically improve various aspects of your technique, leading to a well-rounded skill set.

In summary, I encourage you to try this practice strategy for the upcoming techniques and exercises. After each practice session, take a few minutes to allow your brain to simply reflect on what you have practiced. Good luck and may your practice sessions be productive and rewarding!

Guitar Techniques Needed to Learn, Write, and Play Math Rock

The following guitar techniques are what I believe will best assist you in learning & playing math rock (In no particular order):

- Alternate Picking
- Strumming
- Legato Techniques: hammer-ons, pull-offs, legato slides, etc
- Bends
- Harmonics: natural, tapped, artificial
- Vibrato
- Muting: palm muting & string muting (this one is important!)
- String Skipping
- Staccato

- Fingerpicking
- Finger Tapping
- Hybrid Picking

The last three techniques are distinctively applied by math rock guitarists and have become closely associated with this style. As a result, this section will focus solely on these three techniques. Please note that I will not be delving into the other techniques mentioned here, as there are plenty of other learning materials that cover them more comprehensively.

It is also important to recognize that this is just one part of the equation. To effectively play and write math rock music, we need to consider two more essential elements: firstly, a solid understanding of the relevant music theory specific to math rock (which we will explore in Section 2), and secondly, the necessity of learning and mastering the songs, riffs, and chords within this style. In essence, successful math rock musicians need to combine technique, pertinent music theory, and familiarity with established riffs and songs to excel in this genre.

Finger Tapping

Finger tapping is considered a staple of math rock guitar techniques by many. It is widely employed in the style, and in similar styles that are often attributed to math rock (Midwest emo, math core, neo soul, Nu Prog and so on). The technique is employed somewhat differently when compared to more conventional forms of finger tapping, and this difference is what makes it unique to mathy styles of guitar playing.

I can only speculate on its rise in prominence, perhaps it is due to wanting to play rhythm and lead guitar at the same time, or it could simply just be to look flashy. Either way, it can sound excellent when done right and look hella (pun intended) impressive too! It has developed into being as synonymous with the style as much as the use of odd time signatures has come to be.

Here is a list of a few guitarists whom of which regularly employ the technique in their writing and playing:

Yvette Young (Covet)

Tim Collis (TTNG)

Brock Benzel (Invalids)

Dylan Mattheisen (Tiny Moving Parts)

Marcos Mena (Standards)

Tim Henson (Polyphia)

Nick Reinhart (Tera Melos)

In the following section, you will find exercises, technique pointers, and handy tips to get you on your way to tapping like a mathematical pro!

Finger Tapping Technique Types Explored

For this segment we will be exploring two handed tapping & tapping with chords.

Tablature key so you know which hand to be using for tapping in each exercise:

Ⓣ	tap with your fretting hand (hammer-on from nowhere)
T	tap with your strumming hand

Finger Tapping Technique Pointers

Evenness of tapping: try to keep volumes even as possible at first. Later, you can experiment with dynamics in your tapping, but first I recommend trying to be as dynamically even as possible because this will improve your control.

Developing finger strength balance

To achieve better finger strength balance, I recommend dedicating more time to your ring and little fingers on both hands. The exercise below is an excellent starting point for improving finger evenness and building strength.

1. Position your fingers on the fifth to eighth frets of the fretboard.

2. Use your fretting hand to tap the strings in this order: index, middle, ring, and little finger.

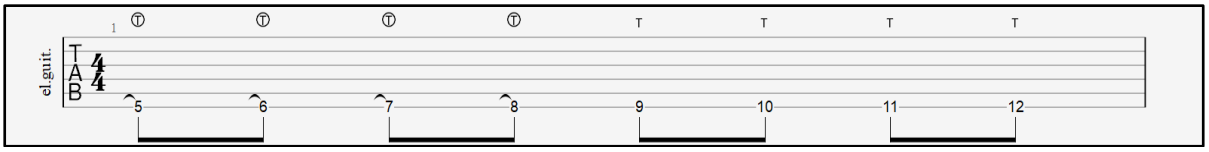
3. Move to the ninth to twelfth frets and use your strumming hand to tap in the same finger order.

4. A useful tip for your little finger on your strumming hand is to place your ring finger on top of it (apply pressure to your little finger's nail) to add more strength to the tap.

5. Once you feel comfortable on one string, try moving up and down all the strings.

6. Ensure you tap each note for the specified duration (8th notes: count 1 & 2 & 3 & 4 &), and make sure each note sounds correctly.

7. Do not forget to use a metronome to help with timing and consistency.



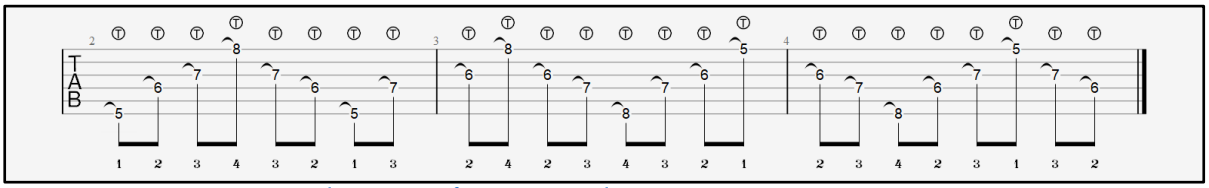
1.A - Evenness in Tapping Exercise

Finger Strength & Dexterity Development

As you practice finger tapping, your tapping dexterity will naturally improve. To get you started though, I suggest the following exercises. These also double up as excellent warm-up routines!

The following exercise is adapted from a John Petrucci exercises a guitar teacher shown me years ago. It has some good stretches and notice how the exercises becomes inverse of itself as it progresses and arrives back at the original fingering at the beginning of the repeat. I have added which fingers you should be using next to each note head in the exercise. Here is a key for which numbers refer to which finger on your fretting hand:





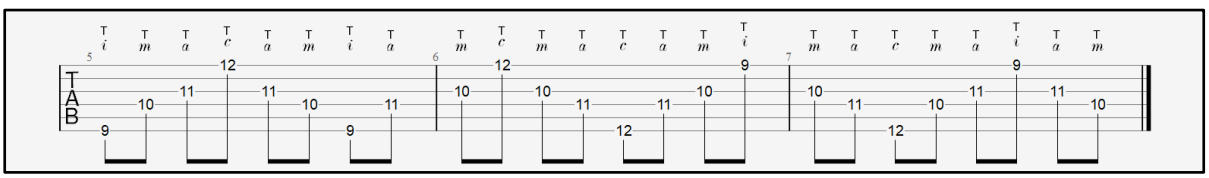
1.B - Finger Tapping Dexterity Stretching Exercise for Fretting Hand

Tip: when tapping, use the underside of your index finger to mute unwanted string ring (when you are not using it to tap of course!) by lightly touching strings that could potentially ring. To clarify, you are not to push down on the strings, just apply a light pressure for it to be enough to stop unwanted strings from ringing out. This is something you should develop if you have not already because it will make your tapping sound much cleaner. Also, when you are tapping on higher strings, G, B, or E for example, you should lightly rest the palm of your strumming hand on the lower strings (E, A, D) to mute them.

For our strumming hand, we have the same exercises but higher up the fretboard as this is more common territory for that hand. It will feel awkward at first, and you will probably hit some bum notes, nevertheless, persevere, it will get better, trust me!

Once again, I have added which fingers you should be using next to each note head on the tab (classical style):





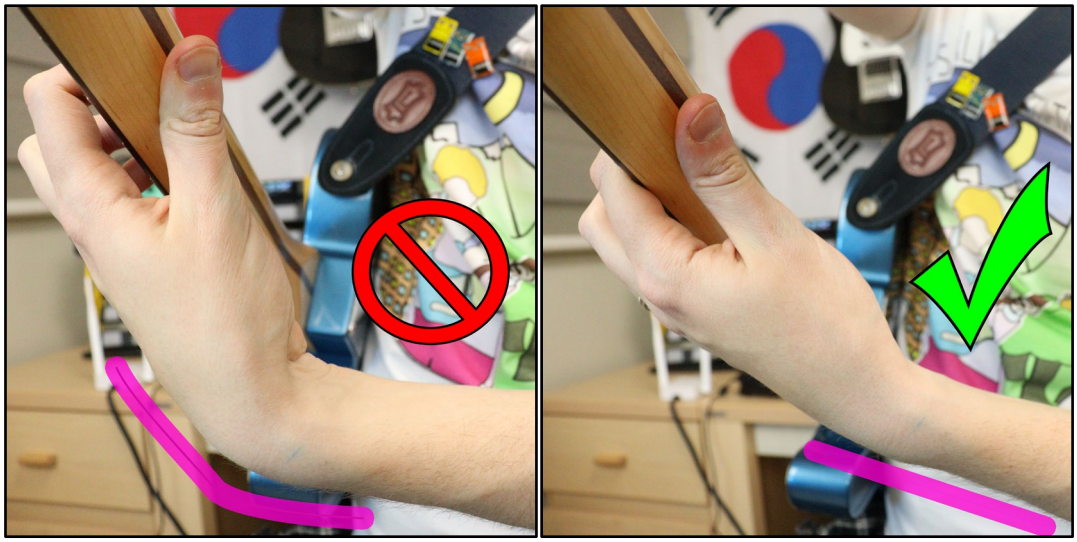
1.C - Finger Tapping Dexterity Stretching Exercise for Strumming Hand

Tip: use your other hand to mute unwanted string ring by placing it above your strumming hand (that is doing the tapping) and touch the strings lightly to mute them.

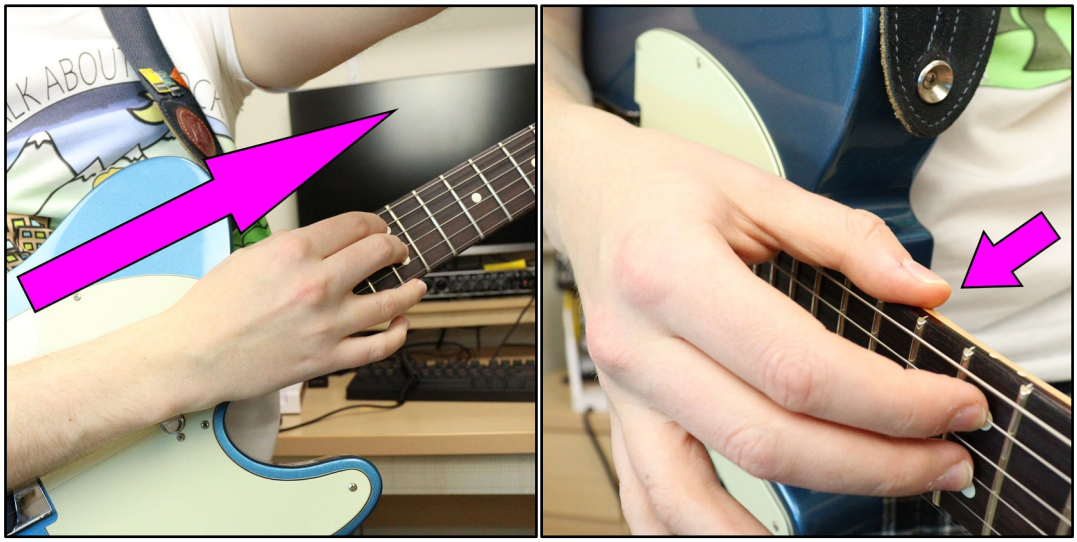
Hand/Wrist Positioning for Finger Tapping

To mitigate any unnecessary strain on your fingers, hands, and wrists I have a few tips to help!

First, try to keep your fretting hand's wrist straight as possible, it is not always possible but aim to keep it so:



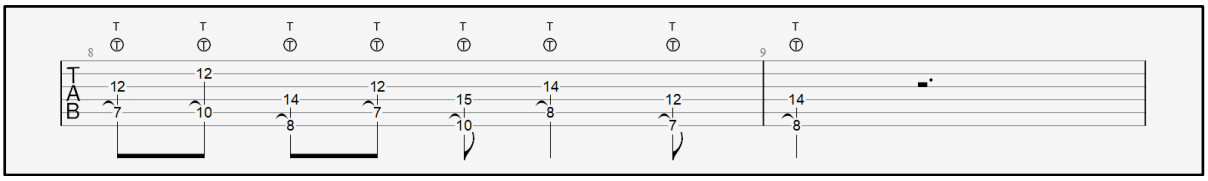
Lastly, for your strumming hand, try to get your hand in-line with your fretboard (see left picture below for clarification) so that your hand and fingers are facing towards the nut of your guitar. You will find this more comfortable and easier to play than having you hand and fingers perpendicular to the fretboard. I like to anchor my thumb on the top side of the fretboard when finger tapping for stability, but this is not always an option (see the picture on the right below for clarification).



Two Handed Tapping & Alternating Finger Tapping

Two handed tapping mainly consist of alternating tapping (alternating between each hand when tapping). This is the most prominent version of tapping employed by math rock guitarists.

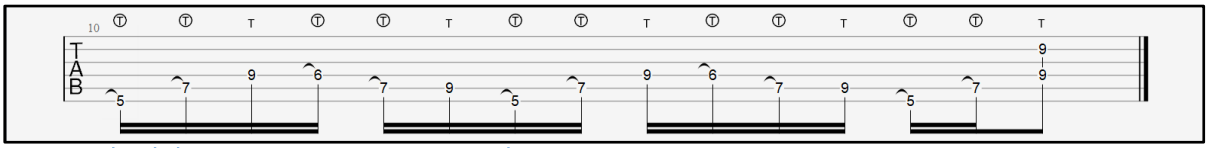
Occasionally, you might encounter a situation where you need to tap notes simultaneously with both hands, which, in my opinion, can be quite challenging! For instance, when attempting this section in my finger-tapping cover of "Santa Claus is Coming to Town," I personally find it more difficult to coordinate tapping with both hands simultaneously. A helpful tip to make this exercise easier is to first learn each part separately and then gradually combine the two.



1.D - Two Handed Tapping at the Same Time Exercise

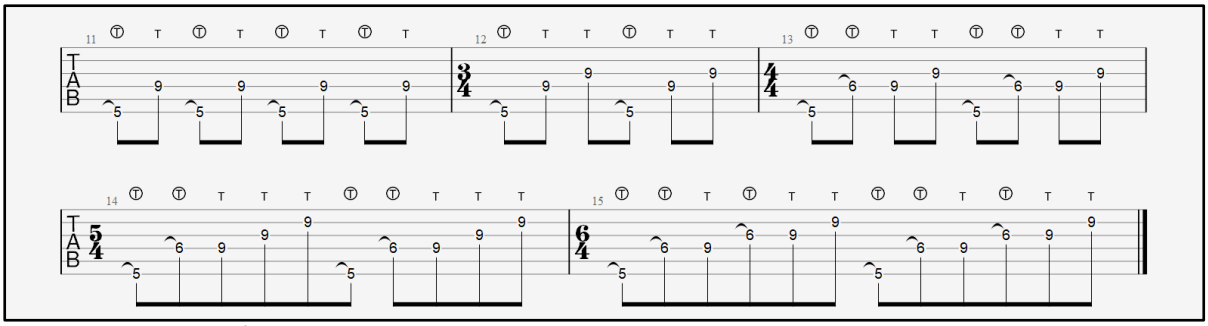
Below, you will find a bunch of exercises aimed at improving your two-handed tapping technique. In later sections, we will explore chords, arpeggios, and scales to create finger tapping ideas, but for now let's concentrate on getting better at the technique.

Exercise 1.E goes easy on the strumming hand and has a small ascending and descending pattern in it.



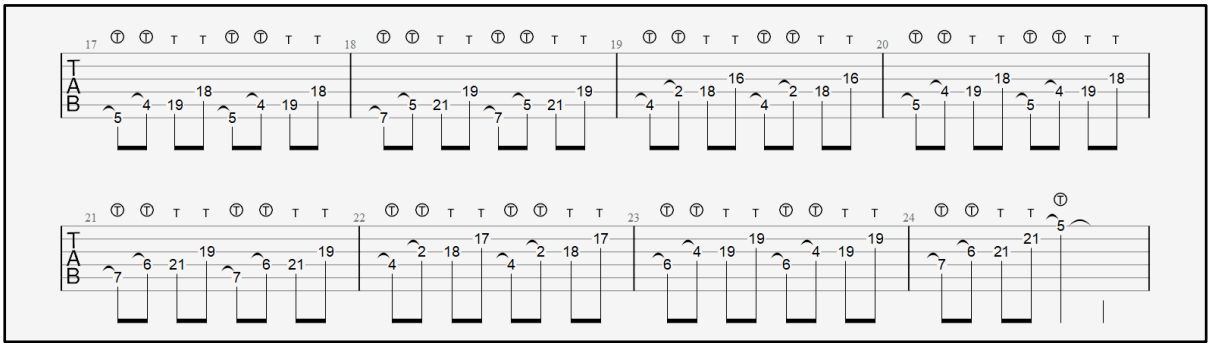
1.E - A Two-handed Tapping Exercise to Get You Started

Exercise 1.F is built around a major seven arpeggio. It starts with two notes and adds an extra note in each bar that follows. This is a good exercise to demonstrate simply how you can break an arpeggio into parts for tapping ideas.



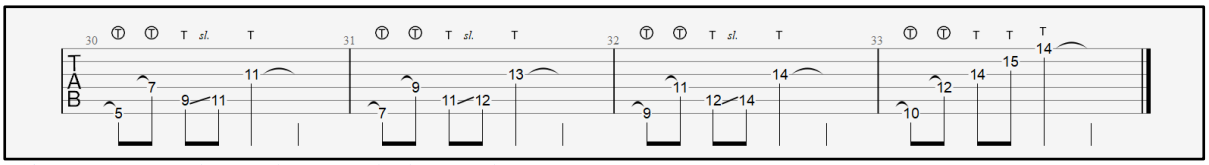
1.F - A Major 7 Arpeggio Shape Tapping Exercise

Exercise 1.G is based around extended seven chord arpeggios. I have split each arpeggio between both hands. As each bar progresses a new arpeggio of the major scale is introduced. Take your time with each bar, then glue it all together when you are comfortable with each individual arpeggio.

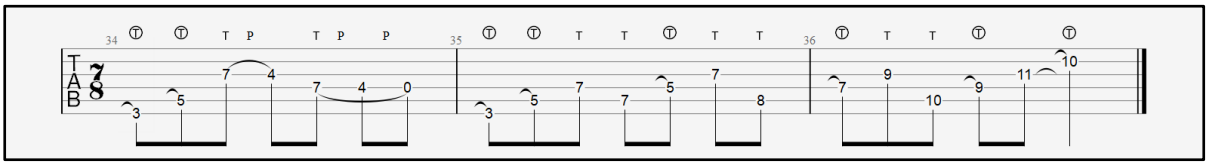


1.G - Tapping Extended Seventh Chord Arpeggio Shapes of the Entire Major Scale

I have included the following two exercises (NS1 & NS2) as next step exercises. They are more like the type of tapping you are to come across, but they still have an exercise feel to them. Never-the-less, I recommend them as the next step to take, and they will set you up nicely to attempt the test exercises at the end of this section.



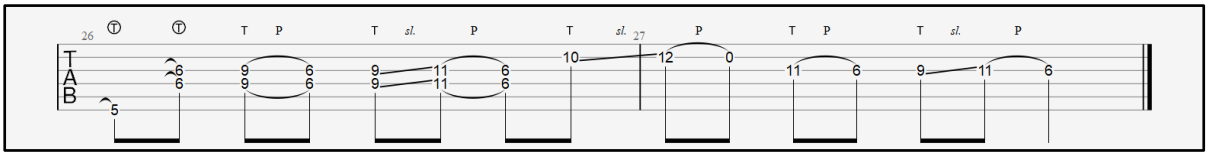
NS1 - Next Step Exercise: tapping arpeggios in a creative manner



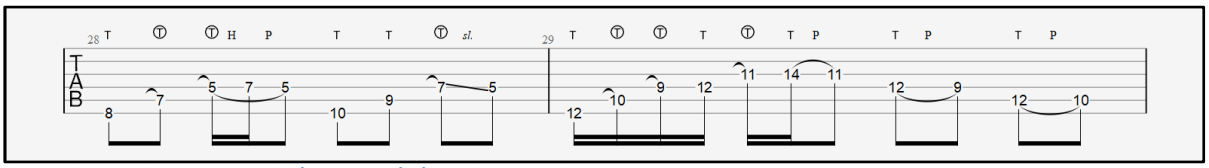
NS2 - Next Step Exercise: A repeating phrase with a tapped arpeggio ending

Test Yourself: Two Handed Finger Tapping Exercises in Context

As you have reached the end of the two-hand finger tapping exercises, take a moment to reflect on your progress so far. Challenge yourself by learning the following two exercises. By mastering these, you are already ahead of many other guitarists who either have not learned or have put minimal effort into finger tapping. Pause for a moment, and give yourself a well-deserved pat on the back, you have earned it.



1.I - Mountains – Future – Outro Tapping Part – Two Handed Tapping in Context Example



1.J - A Tiny Moving Parts Inspired Two Handed Tapping Exercise

To round off the two-hand tapping segment, I have listed a bunch of songs that use two-handed tapping at some point within the song. They are a great way to expand your tapping repertoire, and through osmosis, you will start to include things that you learn into your own playing and song writing through already established math rock content.

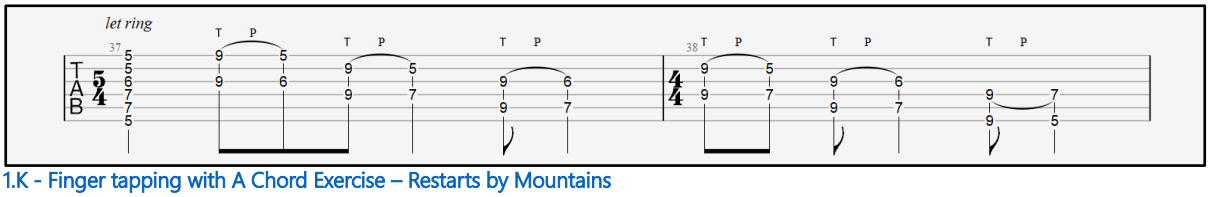
Band	Song	Difficulty
Maps & Atlases	Everyplace is a house Intro/Verse	Beginner (Dave Davison's part)
Tiny Moving Parts	Always Focused Intro	Intermediate
Don Cabellero	Fire Back About Your New Baby's Sex	Intermediate
Tera Melos	Last Smile for Jaron Intro	Intermediate
Tera Melos	40 Rods to The Hogs Head Intro	Advanced
Covet	Pelagic Intro	Advanced
Standards	Kale & Strawberries Intro	Advanced

Finger Tapping with Chords

This technique is basically what the name implies; strum or arpeggiate a chord and tap notes along with it. You could also fret a chord or a particular set of notes, but not pick or strum the shape, and instead tap notes and pull off letting the fretted note or notes of your chord ring out underneath.

This is my favourite tapping technique. It is particularly useful for bands with one guitarist because it fills out the sound by potentially playing harmony and a melody at the same time. The technique mainly consists of tapping single notes, or combinations of two, and sometimes, though rarely, three notes at the same time then pulling off, which lets the notes of the chord you are fretting ring out as if they were open notes. There are no hard rules on which fingers to use for tapping, but generally you will feel most comfortable using your index and middle fingers when not using a pick, and if you are using a pick, try your middle and ring fingers.

Here is an exercise to get you started. Hold an A major barre chord, strum the chord and then use two fingers to tap and pull off letting the notes of the A major chord ring out underneath.

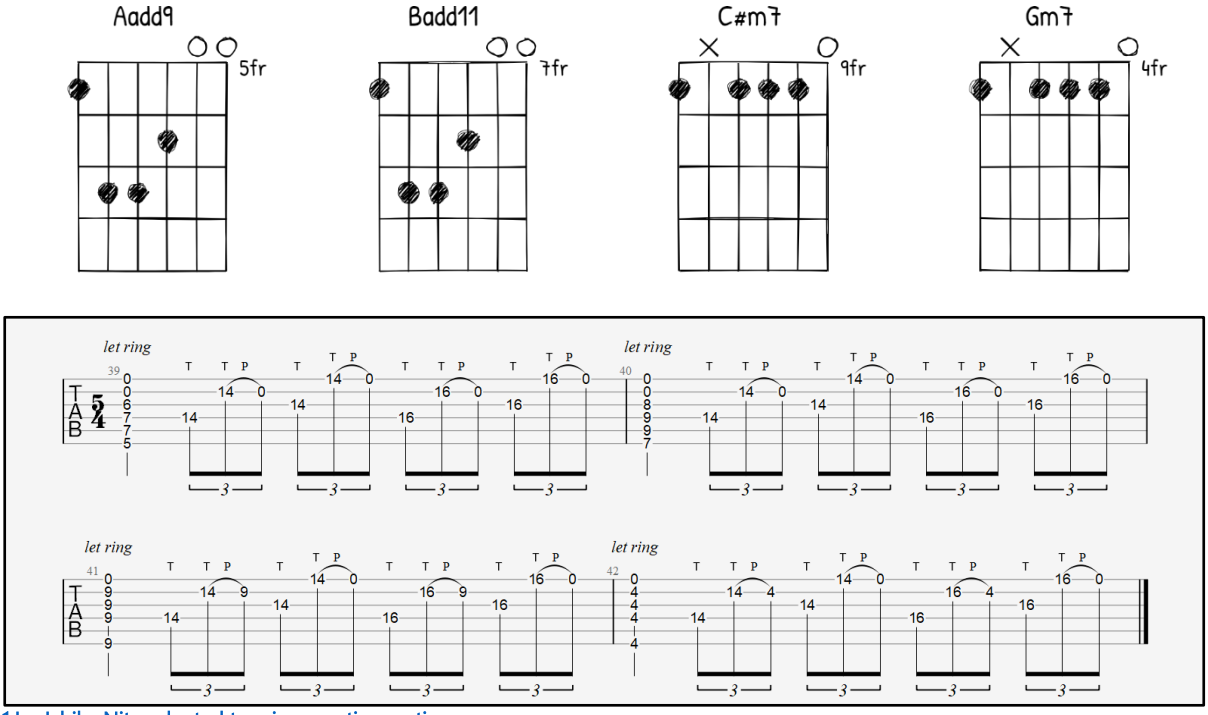


1.K - Finger tapping with A Chord Exercise – Restarts by Mountains

For exercise 1.L, I adapted an exercise from the amazingly talented guitarist Ichika Nito. For this exercise, I recommend practicing the chord changes first (I have listed them above the exercise for clarity), and then attempt the tapping parts. This way you are not trying to master two

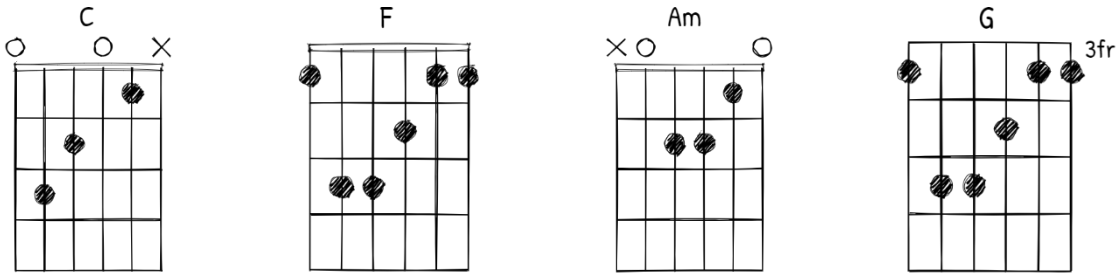
things at once, which will make the exercise more manageable. Remember to anchor your thumb on the side of the fret board. Then, when you are comfortable, try doing the same exercise with a pick and tap the tapped parts with your middle and ring finger.

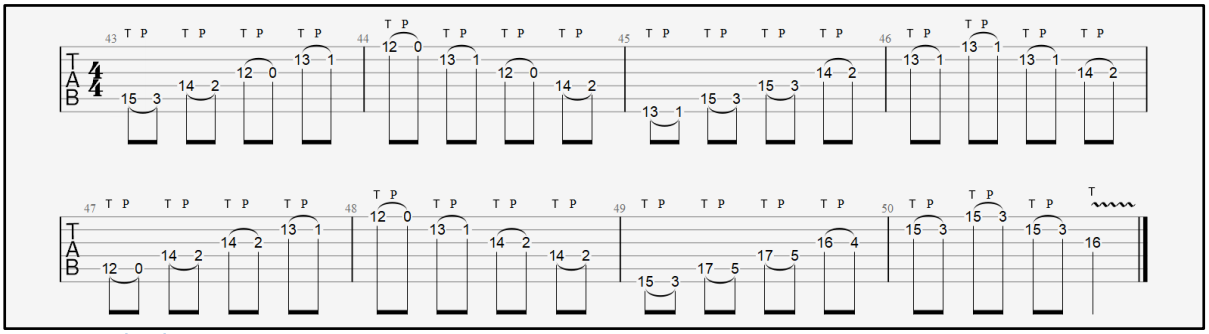
By practicing both tapping approaches, you will develop your tapping technique for both hybrid picking and fingerpicking techniques. Lastly, notice that it is in 5/4 time, so there are five beats instead of the usual four, and notice you are tapping out a triplet rhythm on each beat after strumming each chord.



1.L - Ichika Nito adapted tapping practice routine

For exercise 1.M, fret and hold the chord shapes then tap and pull-off. The tapped notes are octaves of each chord. This octave higher tapping method is one option you can employ for finger tapping with chords, but it in context it will be better to mix it up with other notes to make it sound less like an exercise. Again, I have included the chord shapes for clarity:

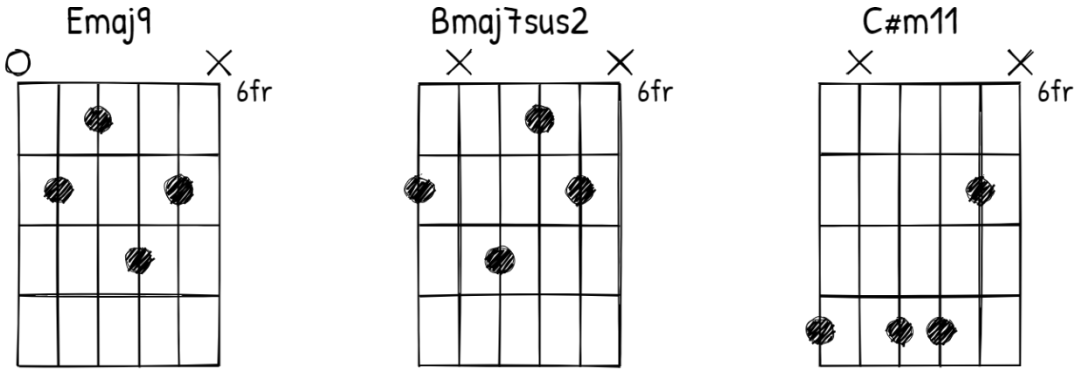


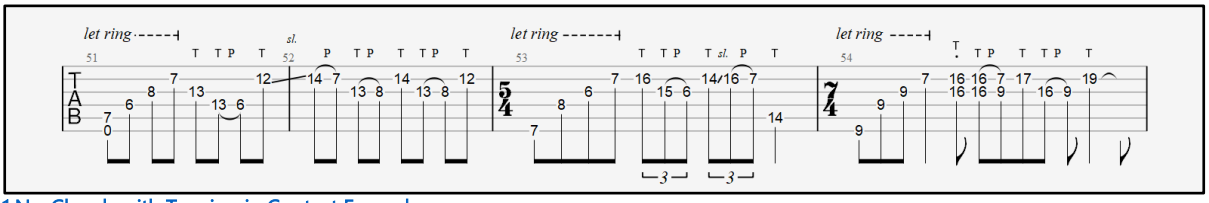


1.M - Open Chords Tapping Exercise

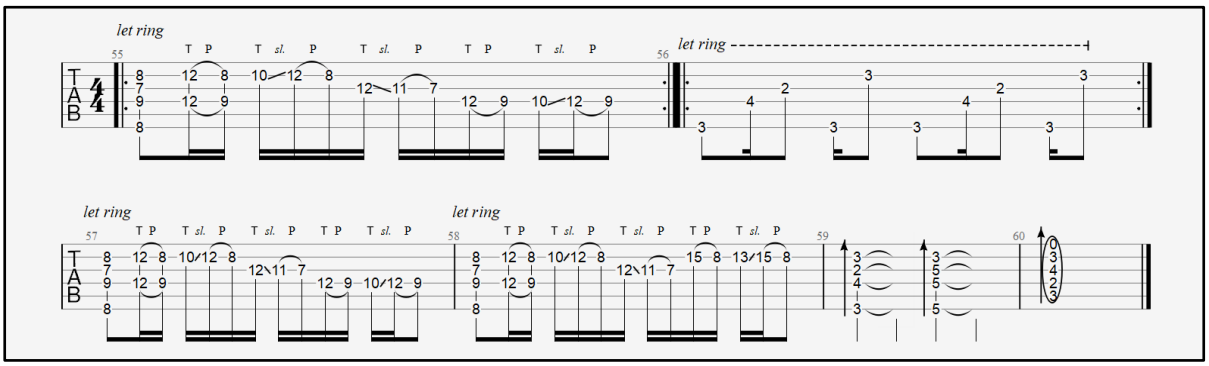
Test Yourself: Finger Tapping with Chords Exercises in Context

Lastly, let us conclude this section with two examples of how finger tapping with chords can be used in a more musical fashion. For a nice little task, try working them out without watching the videos. This is better practice than just copying the videos. However, if you struggle to do this, I encourage you keep applying yourself the best to your ability, make any progress you can before checking out the video example!





1.N - Chords with Tapping in Context Example



1.O - Chords with Tapping in Context Example

Segment 1B – Fingerpicking

Fingerpicking

Fingerpicking is an incredibly valuable technique that helps simplify intricate picking parts and makes it easier to play complex pieces involving frequent string skipping. Furthermore, this approach lends a distinct character to our guitar playing, making it suitable for clean tones or acoustic guitars, introducing unique timbres to our playing. Notably, in the realm of math rock, guitarists like Tim Collis of TTNG and Yvette Young of Covet are masters of this technique. There is a wealth of creative possibilities awaiting exploration with this method, so let's dive right in.

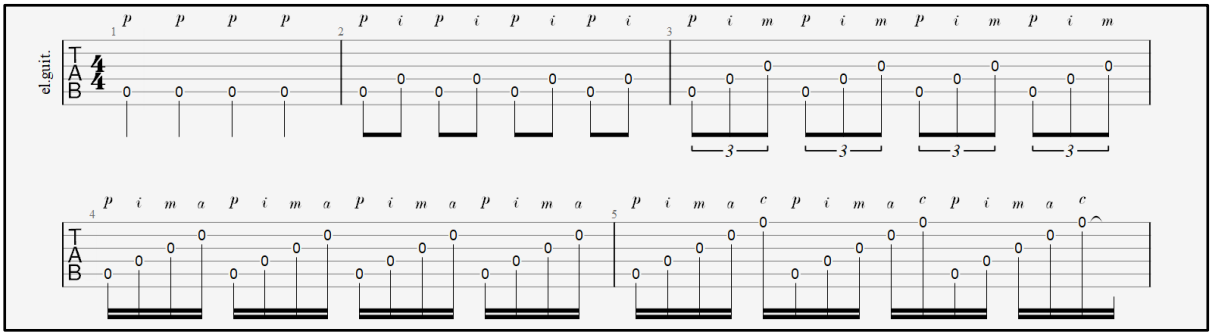
Once again, I have added suggested fingering next to each note head, and here is a key for which letter refers to which finger on you picking hand (think piMac to help with memorization).



Developing Dexterity & Finger Strength

To begin with, I recommend resting the palm of your picking hand lightly on the bridge of your guitar so that your fingers are poised above the strings and ready to pluck them. Your hand should not have to move much at all because your fingers should be spaced out enough to cover all the strings adequately. If you feel strain, perhaps you are rushing, not warming up enough, or positioning your wrist incorrectly. Your hand and wrist should feel relaxed, so if they are not, you may be doing something incorrectly, so try to address this as you work your way through the following exercises.

To begin with, exercise 2.A is an addition exercise where an extra finger is introduced as the exercise progresses. Once you are comfortable with playing it, try it backwards too.

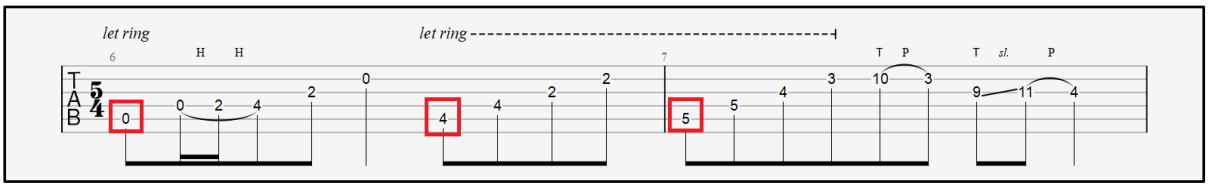


2.A - Fingerpicking Addition Exercise

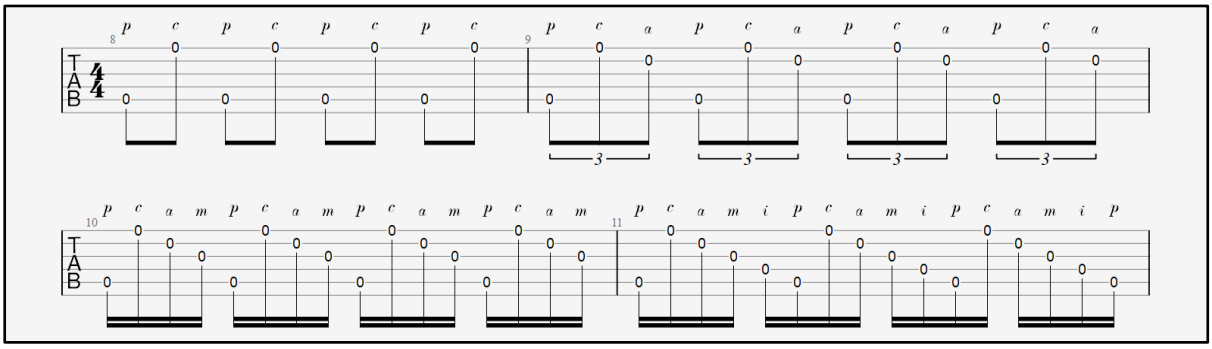
Fingerpicking Patterns

When you observe math rock guitarists who frequently use fingerpicking, you will begin to notice a common pattern: they often begin their riffs and licks from a lower note on the guitar, typically this will be the root note of a chord. Sometimes, they let this note ring out, especially if

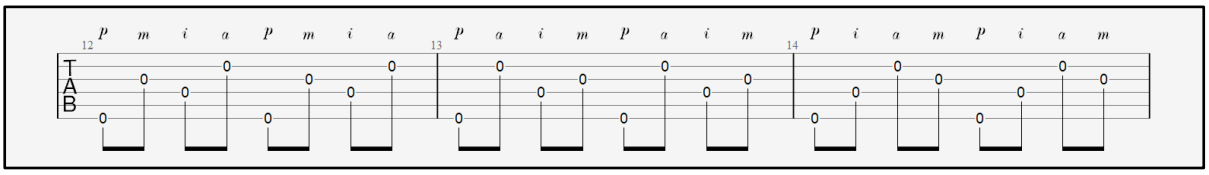
it is an open string. Other times, they do not. This approach is used to create a sense of harmony within the overall musical idea, suggesting harmonic movement in the progression, if there is one. Typically, this is followed by an arpeggio that captures the essence or "feel" of the underlying chord. For example, if we look at the following TTNG inspired idea. The circled lower notes in red show where each musical phrase is built from, and the notes that follow each circled note make up an arpeggiated chord and in turn, an arpeggiated melody of sorts.



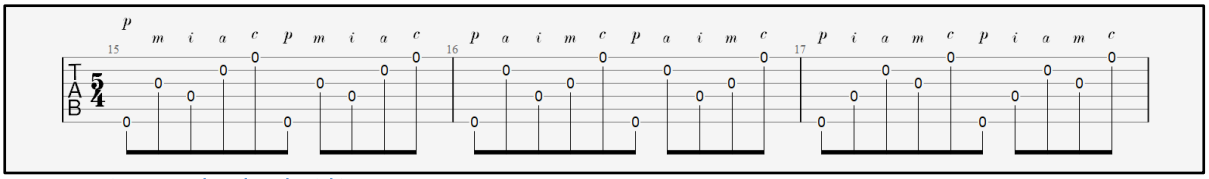
Exercises 2.B, 2.C, and 2.D are beneficial for getting started with picking patterns you are likely to come across, albeit in a somewhat simplified manner. Exercise 2.B adds a finger for each bar, exercise 2.C is designed to practice some basic picking patterns with three fingers and a thumb (which I find is more commonly used than all fingers and a thumb), and lastly exercise 2.D makes use of the same patterns but includes the little finger. Notice the change to 5/4 time to fit in the extra finger. Remember, precision over speed is vital, and use a metronome! Once you are comfortable with playing these picking patterns, try applying them to some chord shapes you like to use.



2.B - Finger Addition Picking Pattern

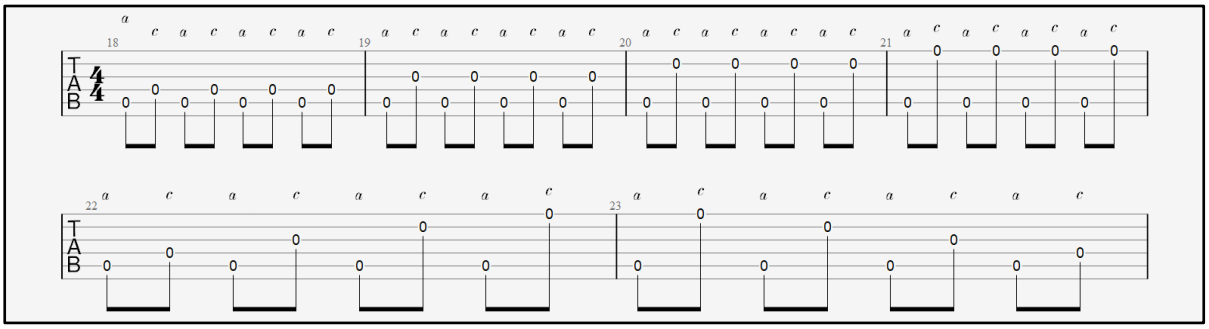


2.C - Three Fingers and A Thumb Picking Pattern



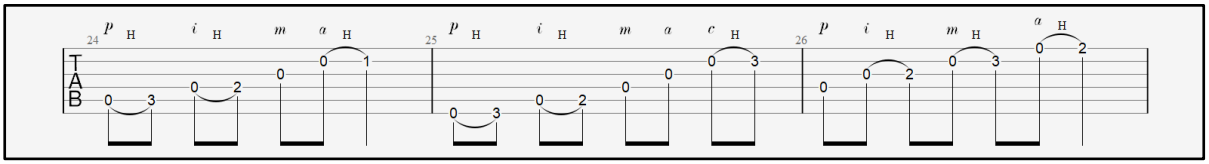
2.D - Four Fingers and A Thumb Picking Pattern

As you work through these exercises, you will quickly notice that your ring and little fingers are not as skilful as your thumb, index, and middle fingers. This is mainly because they share a ligament, which limits their dexterity. You can use the following exercise to enhance their strength and agility. While these two fingers may always feel a bit awkward, remember that precision is more important than speed. With consistent practice, you will see improvements over time. Although they may never match the strength and dexterity of your other fingers, we should still aim to make them as strong and agile as possible.



2.E - Ring and Little Finger Strength and Dexterity Exercise

Let's move on to start applying some notes on the fretboard. A common technique utilized with fingerpicking are hammer-ons and pulls-offs. Exercise 2.F will get you started with this concept. The exercise is based around three basic open chord shapes: C major, G major, and D major. Once you can play the exercise comfortably, try playing it by using pull-offs instead. And I encourage you to try a combination of both.

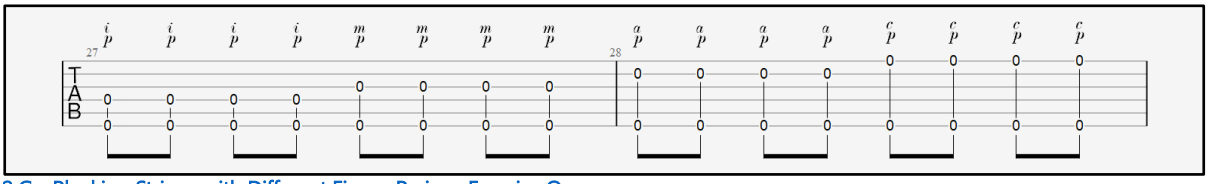


2.F - Hammering on Exercise

Finger Control & Plucking Chords

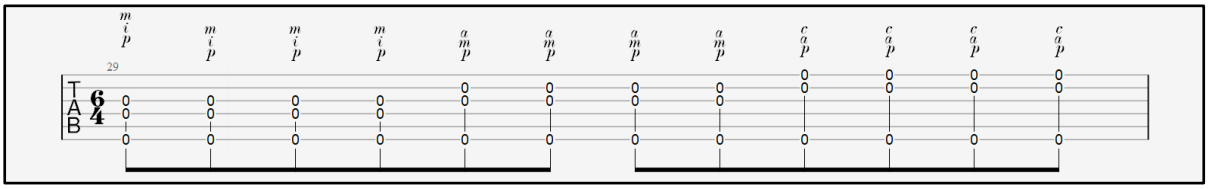
Next, let's start plucking strings at the same time. This is an excellent approach for providing a different feel to your chords because strings are all plucked simultaneously vs a pick dragging (strummed) over the strings. It can give you better control over dynamics, better accuracy, as well as a different timbre.

To get started, exercise 2.G combines different finger parings (look at the note heads for which finger to use) paired with your thumb.



2.G - Plucking Strings with Different Finger Parings Exercise One

Exercise 2.H expands on the previous exercise with the inclusion of another finger. Once you are comfortable with this, try adding more fingers until you are using all of them.

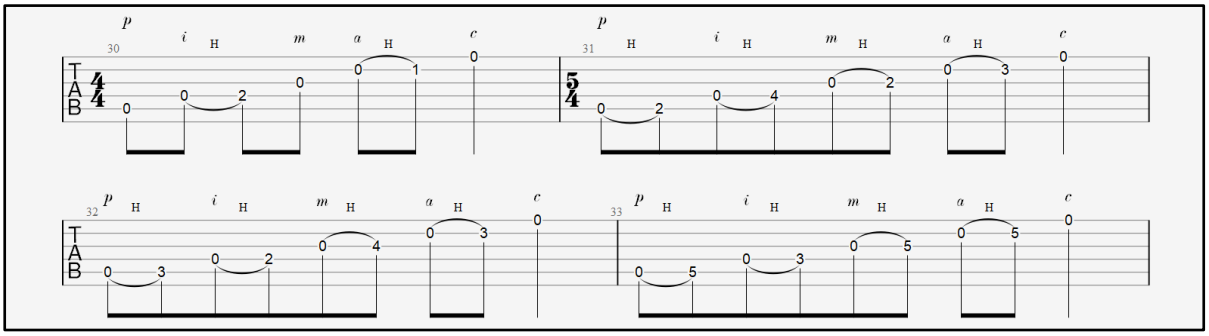


2.H - Plucking Strings with Different Finger Parings Exercise Two

Test Yourself: Fingerpicking Exercises in Context

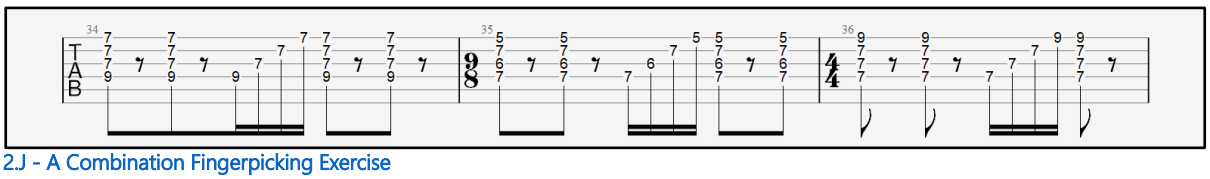
Let's end this section with some exercises to put fingerpicking in context and give ourselves a little test.

Exercise 2.I demonstrates arpeggiating chords through fingerpicking, incorporating hammer-ons. While it is a simple example, it is an excellent starting point because this hammer-on technique is frequently used by math rock guitarists. Do not forget to incorporate pull-offs and a mix of both techniques! Keep your fingers positioned in chord shapes, ready for hammer-ons or pull-offs, as it will assist in executing the arpeggios smoothly. For cases where this approach does not fit, consider the optimal finger placement order to make it easier to play.



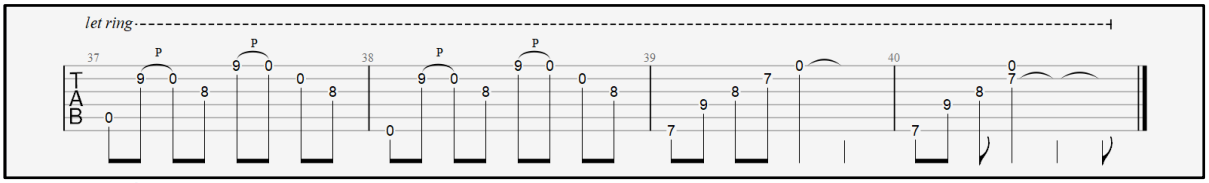
2.I - Arpeggios with Fingerpicking Exercise

Exercise 2.J is an example of what you can do when plucking chords and arpeggiating them in the same arrangement. You will likely come across this way of playing in a variety of fingerpicked songs by math rock bands.



2.J - A Combination Fingerpicking Exercise

Lastly, exercise 2.K is a nice post-rocky sounding example. The string skipping jump from the initial note in the first two bars would be a bit tricky if you were to use a pick but is made somewhat less challenging by using fingerpicking. To make this exercise easier, I recommend working out your finger placement first.



2.K - Fingerpicking in Context Exercise

Some examples of bands and parts of songs you can tackle:

Band	Song	Difficulty
TTNG	I'll forgive you for throwing that rock cos that dance was kinda funny	Beginner/intermediate
Delta Sleep	Camp Adventure	Beginner/intermediate
American Football	Never Meant (Originally played with a pick but it is a good fingerpicking exercise nonetheless)	Beginner/Intermediate
COVET	Parachute	Intermediate
TTNG	Chinchilla	Advanced

Segment 1C - Hybrid Picking

Hybrid Picking

Hybrid picking is where we make use of a pick and fingers at the same time. It is not as prominent in math rock as other techniques we have explored but still very useful. I for one love this technique and employ it regularly. I like to use it because I can get the best of both worlds; the strengths of fingerpicking and having the option of using a pick at the same time. I implore you to learn this technique if you have not been doing so already.

This technique, when developed, can help you play fast precise licks, especially ones involving string skipping with a lot less effort. In addition, you can combine the pick with fingers for plucking chords (just like fingerpicking). Overall, it is a great compromise between using only fingers or only a pick.

Some examples of guitarists who use hybrid picking:

Mario & Erick of CHON – Book

Tim & Scott of Polyphia – Look but Don't Touch Playthrough

Notice how Tim Henson reaches the higher strings with more precision by using his fingers to take care of string skipping parts of the song. You can see the same thing happening in the open riff to G.O.A.T.

Alan of Tangled Hair – Yeah... It does look like a spider

Pay close attention to the singing parts and how all the strings resonate simultaneously when Alan plucks them, in contrast to when he strums during the louder sections. Additionally, starting from the two-minute mark, observe how Alan's use of hybrid picking enables him to execute that section, achieving the distinct sound it produces due to the accessibility afforded by this technique.

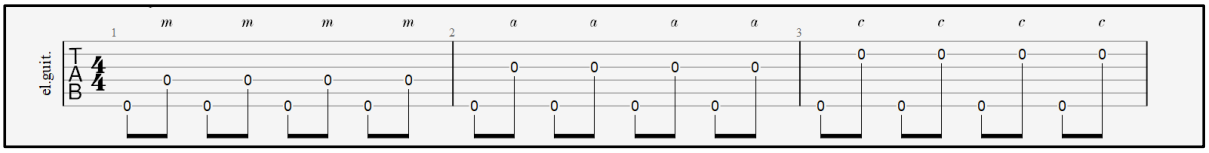
Hybrid Picking Dexterity Exercises

Below you will find a series of exercises I have designed to help you become accustomed with the technique.

To clarify, you will be holding a pick between your thumb and index finger (as you normally would) and using your remaining free fingers (middle, ring, and little finger) to pick notes. Compared to fingerpicking, I have noticed I rarely use my little finger when hybrid picking. That is not to say you will not need to use it, just a reminder that it is fine not to.

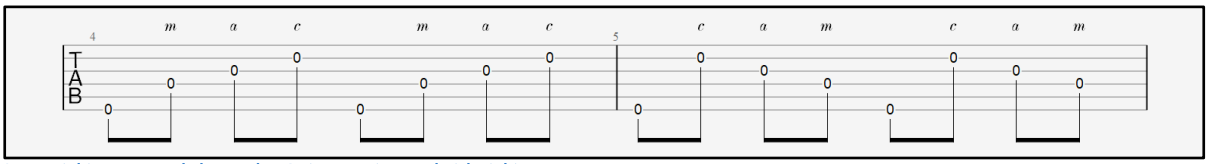
To being with, exercise 3.A is designed to get you started with the string skipping potentials of hybrid picking. I have labelled which fingers to use (remember Mac for this one). The lowest string will be played with your pick.

Keep your picking hand steady by resting it on the bridge, and notice how you hand and wrist are not jumping about like you would if you were to pick all these notes using only a pick.



3.A - An Introduction to String Skipping Using Hybrid Picking

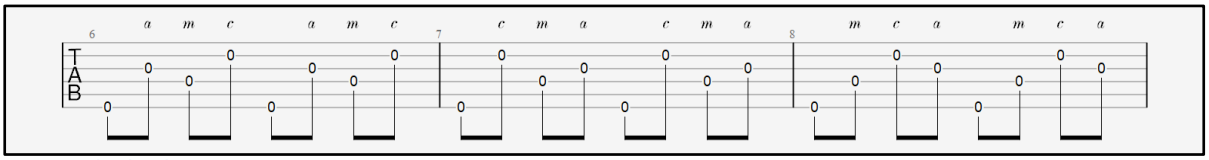
I designed Exercise 3.B to get your fingers familiar with picking up and down in an arch like motion. This kind of arching melody motion is common in math rock.



3.B - Picking up and down the Strings Using Hybrid Picking

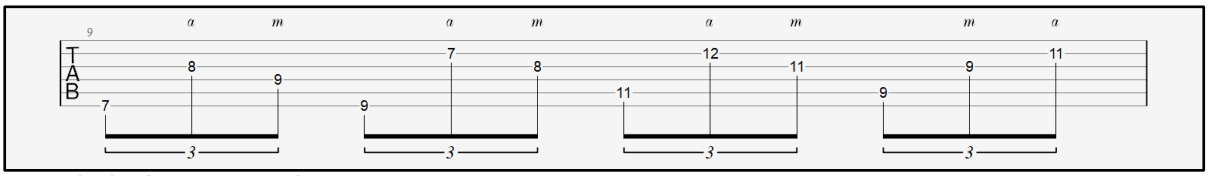
Bonus tip for exercise 3.B: once you are comfortable with playing it precisely, try picking up and down as fast as you can (without hurting yourself of course!). You can develop this into a quick raking like action across the strings (like sweep picking but with fingers included). It is very satisfying once you can do it!

Exercise 3.C is a slight variation on the previous exercise. It is designed to get your fingers moving in different patterns.



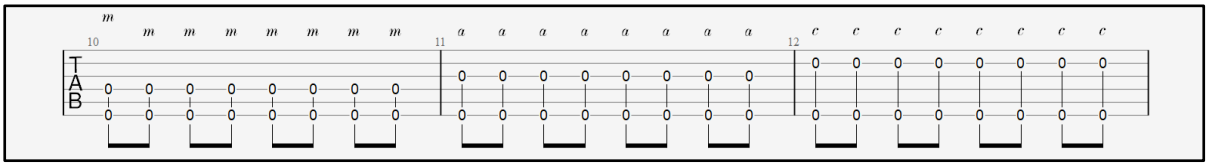
3.C - A Hybrid Picking Pattern Exercise

Exercise 3.D applies a picking patten along with some arpeggios.



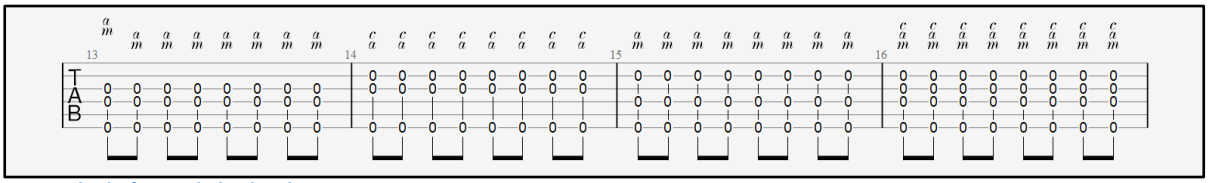
3.D - Hybrid Picking Exercise with Arpeggios

Let's move on to plucking strings at the same time. Exercise 3.E gets you started with plucking two strings at a time using different finger combinations along with a pick. This may feel a little strange at first but that is completely normal.



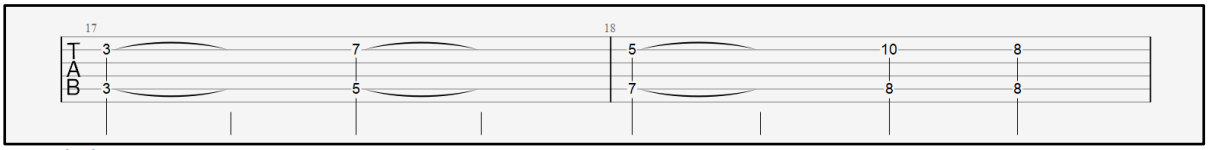
3.E - plucking two strings at a time hybrid picking exercise

Exercise 3.F expands on the previous exercise by including the use of two fingers with a pick and in the last bar the inclusion of all three spare fingers!



3.F - multiple fingers hybrid picking exercise

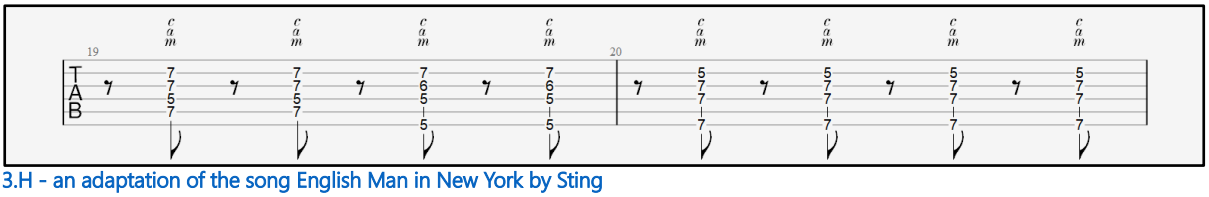
Exercise 3.G uses just two notes from chord shapes (highest and lowest in this case) to create some interesting ideas.



3.G - plucking two strings exercise

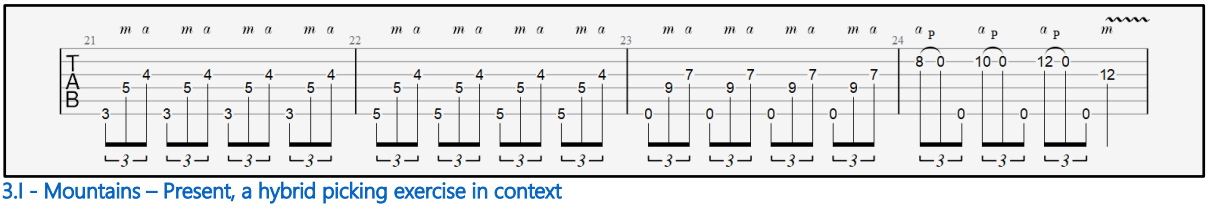
Test Yourself: Hybrid Picking Exercises in Context

Exercise 3.H is an adaptation of the main chord progression to the song English Man in New York by Sting. Use your pick to play the lowest note of each chord and use your fingers to pluck the rest of the chord shape.



3.H - an adaptation of the song English Man in New York by Sting

Lastly, exercise 3.I uses a repeating ascending pattern in triplets. Again, like exercises before, use the pick for the lower notes. Notice in the last bar, that by using hybrid picking, the string skipping becomes somewhat easier than trying to play it with just a pick.



3.I - Mountains – Present, a hybrid picking exercise in context

General Technique

To conclude the technique section, it is also crucial to assess your overall general technique. Are you bending your wrist(s) excessively, leading to unnecessary strain? Is your posture causing you to bend your back too much, with your fretting arm sticking out too far? Is your playing as clean as it could be? If not, consider why (possibly due to rushing or insufficient practice). Are you tensing up during specific techniques? If so, determine the cause. Is it because you are trying to play too fast or not allowing yourself to relax, essentially attempting to "run before you can walk"? Making small adjustments to your technique can yield significant benefits in your playing!

Section 2 - Beginner Music Theory, Chords and Chord Progressions, and Alternative Tunings for Math Rock

Segment 2A - Understanding Basic Music Theory & Chord Theory

In this section, I will provide an overview of music theory relevant to math rock and related music styles. We will explore fundamental principles of music theory, offering you a solid starting point to expand your knowledge. Before we dive in, you might have two essential questions:

How Much Music Theory Knowledge is Needed to Learn, Write, and Play Math Rock?

- Key signatures (what sharps and flats are in each key: accidentals)
- Semitones and tones
- Chords that are built from each note in a key (diatonic)
- Intervals (space between two notes)
- Chord progressions
- Melody & harmony
- Rhythm (patterns of strong and weak pulses)
- Time signatures
- Major & minor scales
- Chord types - Triads, extended chords, sus chords, etc
- Repetition and variation
- Strumming patterns

What Music Notation Concepts Should I Know to Best Learn, Write, and Play Math Rock?
(These are all related to reading music on a stave)**:**

- Time signatures (simple, compound, odd)
- Key signatures
- Note divisions (whole, half, quarter, eighth, sixteenth, triplets, etc)
- Reading rhythm
- Rests
- Ties and dots
- Repetition signs
- Notation symbols for guitar: hammer-ons, pull-offs, slides, bends, staccato, tapping (left and right hand), - guitar pro is a great way to learn these notation symbols!
- Dynamical markers

Learning these concepts is not as challenging as you might imagine, and the effort you invest in understanding them is highly rewarding. I recommend using this guide as a starting point and also obtaining a grade 1 music theory book, then gradually progressing through to grade 5. I myself have achieved a grade 5 in music theory, which is often suggested as the goal for those learning popular music styles. Beyond grade 5, music theory exams tend to focus more on classical music.

Here's a description of grade 5 music theory for those who are interested: "A pass at this level shows that you have a good understanding of music notation, scales, keys, time, intervals, transposition, basic harmony, chords, general knowledge about orchestral instruments, and composition."

Major and Minor

Let's start with major and minor key signatures. Most modern math rock is quite melodic and usually uses chords and notes from one key signature (be this a conscious decision or not) and typically is comprised of harmony and melody you typically hear in popular music styles. This is somewhat different to earlier variations of math rock music, which is often referred to as being more dissonant in nature. I have linked an article below that was written by Nickk Hunter of Fecking Bahamas which clarifies these differences better than I can here.

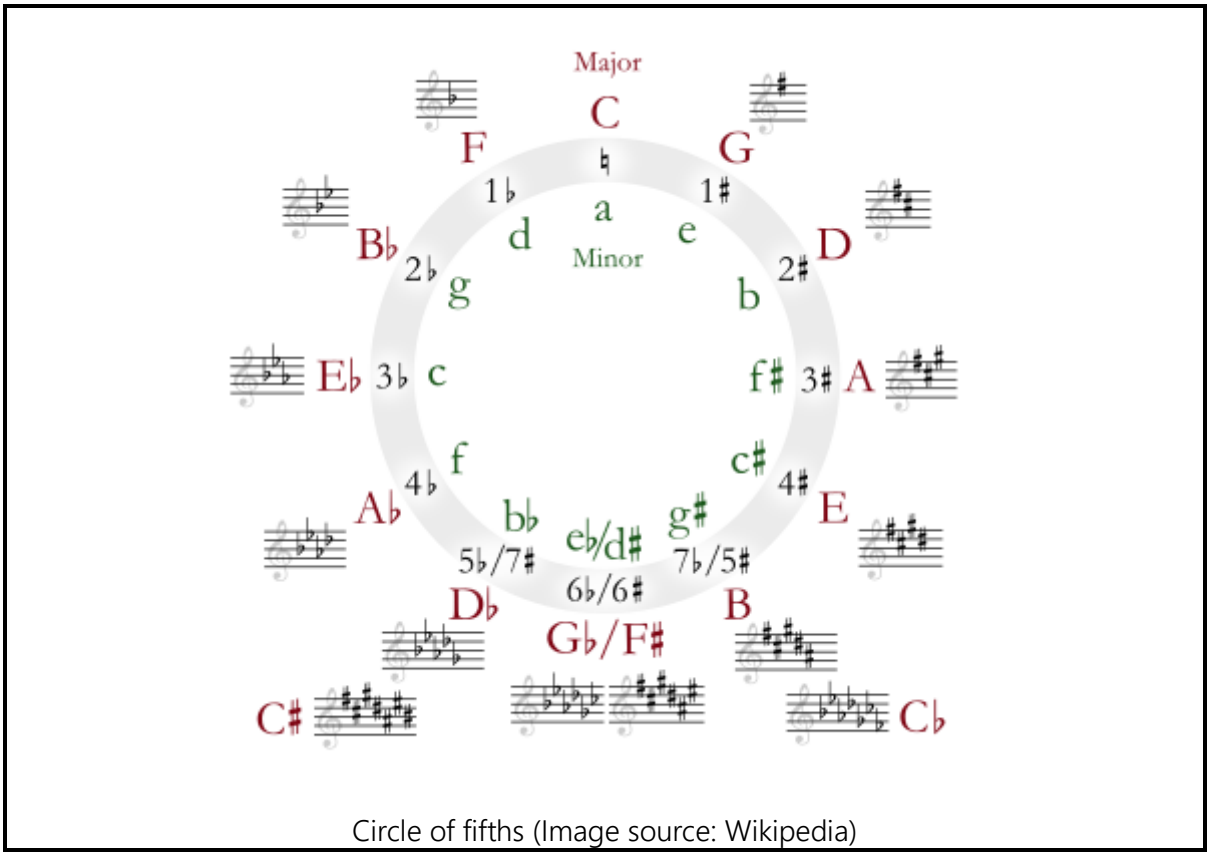
Of course, this is not to say that every current math rock band and every song they write is more melodic than their predecessors. Even though it is a niche style of music there are many outliers to any claim that I could make.

I will begin by explaining major and minor keys, what notes they consist of, and how you can create chords from these notes. Most importantly, I will demonstrate why understanding this basic concept can greatly impact your ability to generate song ideas.

Key Signatures

A key signature consists of seven notes. Whether it is a major or minor key, a specific set of notes is used, which determines the sharps (#) or flats (b) for that key. The circle of fifths below

shows the sharps and flats for each major and minor key.



Circle of fifths (Image source: Wikipedia)

The outside of the circle (red) shows major keys, and the inside (green) shows minor keys. Notice how each major key has a relative minor key vice versa. For example, C major and A minor are relative of each other, meaning they share the same accidentals (which is no sharps or flats), same notes, and chords, but they start from different notes.

What Notes Are Available and How Many?

When counting in sharps, we have these twelve notes: A, A#, B, C, C#, D, D#, E, F, F#, G, and G#. And when counting in flats, we have these twelve notes: A, Bb, B, C, Db, D, Eb, E, F, Gb, G, and Ab. It is important to note that A# and Bb (and others like C# and Db, D# and Eb) are the same notes. These are called enharmonic equivalents, and which one you use (sharps or flats) depends on your choice of key signature. A key signature will use seven of the twelve available notes. For example, C major uses C D E F G A B and back to C (the octave above).

Do not worry about learning all the keys right now, you will pick them up the more you use them, and you can always refer to the circle of fifths to know what sharps or flats are in the key you are using. To add clarity, I recommend watching this video about keys and the circle of fifths.

Lastly, you may be asking 'how do I know which key I am in when writing or playing a piece of music?' I would like you to have a little think about how you could work this out, but I will show you a method soon!

Why Is It Useful to Know Key Signatures?

There are many benefits to knowing what key we are using when writing or playing songs. To start with, it is an excellent tool for knowing what chords, notes, and scales/arpeggios are available to us.

To elaborate, in the key of C major, we have these notes:

1	2	3	4	5	6	7	8 (1)
C	D	E	F	G	A	B	C

First off, if you consult the circle of fifths chart above and find C major, you can see there are no sharps or flats (accidentals) in the key of C major. Now you know the appropriate notes to use for a song or an idea in the key of C major (you could play any of the above notes and not hit a bum one). However, careful selection of notes based on music theory can tell us what notes will 'fit' nicely along with chords we choose, and which notes will not (this is not to say there are only correct and incorrect notes to use for a melody, but rather we will get different results from choosing different notes).

How Basic Chords Work

By knowing the notes in a key signature, we can start to build chords. (For the sake of simplicity, we will stick to standard tuning).

Continuing with the key of C major, let's build some basic triads (basic chords – those open chord shapes you first learn when learning guitar). We have the chords:

C	D	E	F	G	A	B
Major	Minor	Minor	Major	Major	Minor	Dim
I	ii	iii	IV	V	vi	vii°

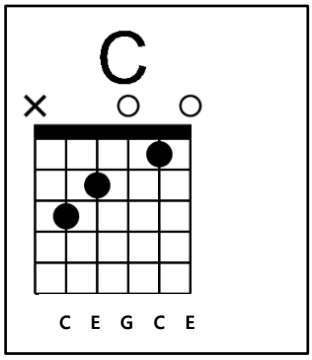
Chords built from a major key follow this order: Major, Minor, Minor, Major, Major, Minor, Diminished. These are often referred to by roman numerals (see above). Upper case for major, and lower case for minor or diminished; I ii iii IV V vi vii°

Chords built in a minor key follow this order: Minor, Diminished, Major, Minor, Minor, Major, and Major or i ii° III iv v VI VII

Here is the key of A minor, which is the relative minor key of the key of C major (same chords but in a different order):

A	B	C	D	E	F	G
Minor	Dim	Major	Minor	Minor	Major	Major
i	ii°	III	iv	v	VI	VII

A chord needs to use at least three notes to be considered a chord, and the distances between the notes in a chord (intervals = spaces) makes it they type of chord it is. (Two notes played at the same time are referred to as intervals, but we could outline a chord with just two notes). Basic chords are made up of three notes, and you are probably already familiar with open chords such as C major:



As you can see, even though there are five strings used to play this chord, it only contains three notes: C, E, and G.

The spelling for basic major chords is 1, 3, 5 - (Root, Major 3rd and Perfect 5th are the intervals counting from the C, the root note).

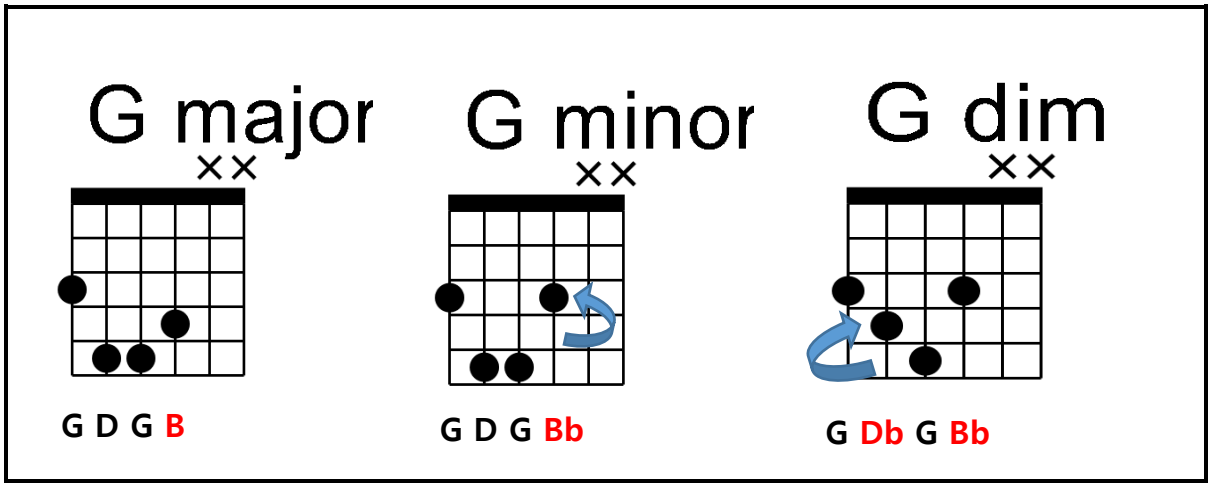
If we refer to the C major scale, we can see where the notes that make up a C major chord come from. I have highlighted them in red.

1 C	2 D	3 E	4 F	5 G	6 A	7 B	8 (1) C

The same applies for basic minor and diminished chords but their spellings are slightly different:

Major	1 – 3 – 5
Minor	1 – b3 – 5 **(flat 3rd)**
Diminished	1 – b3 – b5 **(flat 3rd, flat 5th)**

You can see this difference in action (the moving third degree and/or fifth degree) below:



In a G major chord, there is a major 3rd (B), and when we change to the G minor chord: a flattened 3rd degree, a Bb. Lastly, in the diminished chord we have a flattened 3rd and a flattened 5th Db, and Bb. These intervallic differences are what makes these chords sound the way they do.

How Do I Know Which Key Signature Is Used?

Here is a method that I learned during my time studying music at university:

If a key signature is not given, you can often figure out the key by analysing the harmonic elements such as chords, melody, and how different melodic instruments interact.

For example, if you had an intro section of a song using chords: C major, F major, and G major, we can work out the key from this information. I would deduce that this song is in the key of C major, this is because these three chords are all contained in the key of C major, and because the F major and G major chords are right next to each other this tells me they are most likely to be the fourth and fifth chord of a major key (or the VI VII if we were in A minor):

C Major I	D Minor ii	E Minor iii	F Major IV	G Major V	A Minor vi	B Dim vii

You can apply the same method if you see two minor chords together or two minor chords that occur together but are spaced out in the chord progression. For example, a C major – D minor – F major – E minor progression (I – ii – IV – iii), the D minor and E minor are spaced out in the progression, but they are next to each other alphabetically in the key of C major.

Of course, this method is not bulletproof, especially when you have ambiguous information or less information to work with. For example, let's say you have just two chords: C major and G major, this could be either the key of C major or G major:

C Major I	D Minor ii	E Minor iii	F Major IV	G Major V	A Minor vi	B Dim vii

G Major I	A Minor ii	B Minor iii	C Major IV	D Major V	E Minor vi	F# Dim vii

In cases such as these we generally need more information or take extra steps to work it out. First, I would check if there was an F# in the melody or a riff, this would tell us it is likely to be G major because that key contains an F#. Another way you can tell is by extending the chords to sevenths, you would either get a C major 7 and G dominant 7 (extended chords elaborated on in the following section) in the key of C major, or a G major 7 and C major 7 in the key of G major. I would try out both sets of chords and hear which set of chords fits the song better. From this, you can better deduce whether it was the key of G or C major. Of course, it could be

either key, and it could be your decision to choose one or even make use of both!

It is also worth keeping in mind that even though a song may be situated within a particular key, there may be chords and notes borrowed from other key signatures. There are methods we can use to help identify borrowed chords. And you will see how this is usually done in the borrowed chord segment towards the end of this section.

Task

Below, I have given few chord progressions, and I would like you to work out the key of each one. The answers are provided at the end of this section. And keep in mind that some answers may be ambiguous or use more than one key. Fun times! Good luck!

1. E major – A major – G# minor – F# minor – E major

2. D minor – G7 – C major

3. D major – E minor – F# diminished – G major

4. C major – G major – A minor – E minor

5. G major – B minor – C major – C minor

Segment 2B – Chords and Chord Progressions for Math Rock

Extended Chords

By now, you should have (or already had) an understanding of how basic chords are built and how they function in major and minor keys. The next step is to take these basic chords and make them into extended chords. Extended chords are used a ton in mathy and emo styles of music. They add a bit of extra flair to basic chords, so they have the potential to play with emotions and feelings in different ways.

To begin with, we will look at how extended chords are built, the theory behind them, learn extended chord shapes, and learn some chord progressions using extended chords.

Extended chords are made by adding extra notes to basic chord spellings. Common extensions begin with the seventh chords. Again, like basic major, minor and diminished chords, extended chords have their own spellings. Below are the chord spellings for extended seven chords, the first possible extension:

Major 7th	1 – 3 – 5 – 7
Dominant 7th	1 – 3 – 5 – b7 (flat 7th)
Minor 7th	1 – b3 – 5 – b7 (flat 3rd and flat 7th)
Minor7b5	1 – b3 – b5 – b7 (flat 3rd, flat 5th, and flat 7th)

You may have noticed that we have added two new chord types: dominant seven, and a minor seven flat five chords. Dominant chords have a strong sound to them and feel they need to resolve to another chord. M7b5 chords add more suspense to the basic diminished chord. They

are useful as tension builders, and for slotting in-between chords in chord progressions for some suspenseful colouring.

Beyond extended sevens, there are extended nines, elevens, and even thirteenth chords! Realistically it is up to you to choose which extended chords you want to learn and use. I recommend learning all the shapes provided in this guide and overtime find which work for you. For example, I found I largely prefer to use extended major and minor nines, and minor seven and minor eleven chords.

Before we continue, you may be saying to yourself "hang-on, there are only seven notes in a scale, so where are the 9ths, 11ths, 13ths coming from?" Let me elaborate.

Counting through the scale of C major we eventually arrive at the octave (the same note but higher in pitch). Think of those octave shapes commonly used in pop-punk songs.

1	2	3	4	5	6	7	8 (1)
C	D	E	F	G	A	B	C

If we continue counting past the octave, notes 2, 4, or 6 are referred to as 9th, 11th, and 13th degrees when used in chords. This is because they are extended above the octave (hence the term). Notes: 1st, 3rd, 5th are referred to as normal.

1	9 (2)	3	11 (4)	5	13 (6)	7
C	D	E	F	G	A	B

It is important to note that extended chords like 9, 11, and 13 are named that way only if they include the seventh degree. For instance, if you have a chord with the notes 1, 3, 5, and 9, it is simply called an Add9 chord, like Cmajadd9 for C major. The seventh degree is crucial for the sound of these extended chords.

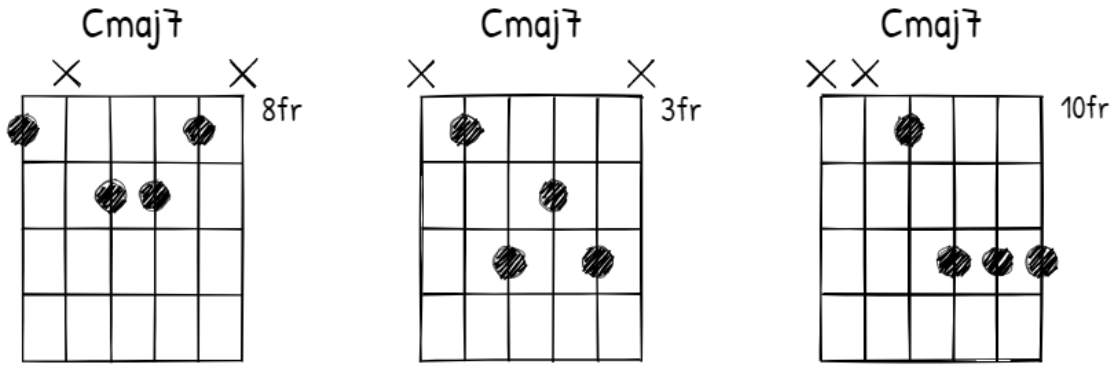
The reason I mention this is that with larger extended chords, you have more notes, but limited fingers and fretboard space to play them all. So, some 'less important' notes are often left out to make it playable. However, it is noteworthy that the seventh degree is usually retained in these chords:

Chord	Chord Spelling	Note(s) Usually Dropped	Resulting notes
Major 7	1 – 3 – 5 – 7	None	n/a
Major 9	1 – 3 – 5 – 7 – 9	5th	1 – 3 – 7 – 9
Minor 11	1 – b3 – 5 – b7 – 9 – 11	5th & 9th	1 – b3 – b7 – 11
Dom 13	1 – 3 – 5 – b7 – 9 – 11 – 13	5th & 11th sometimes 9th.	1 – 3 – b7 – (9) – 13

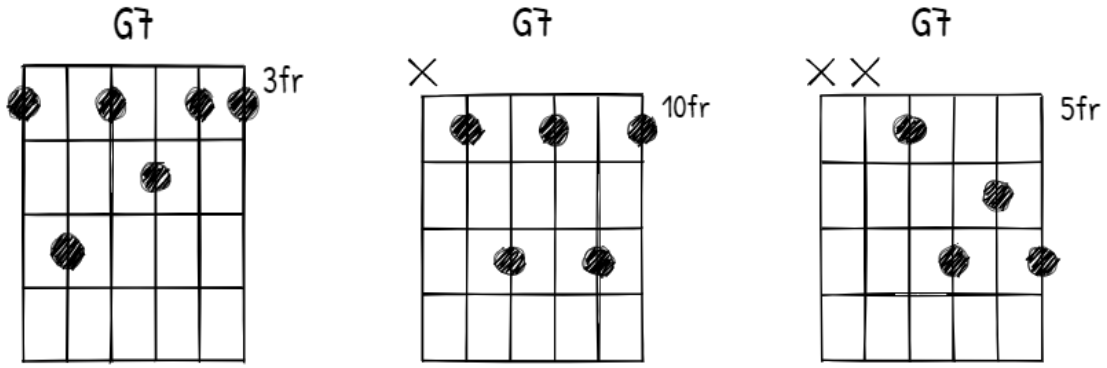
Seventh Chord Voicings & Practice Progressions

Below are extended seven chord voicings starting from the sixth, fifth, and fourth strings. Learning in all three positions will help with changing chords chord progressions more smoothly (minimizing jumping around the fretboard) and provide you with more options. So, take your time to learn them, and keep in mind there are other ways of voicing these chords. These are shapes I like, and I think you will too.

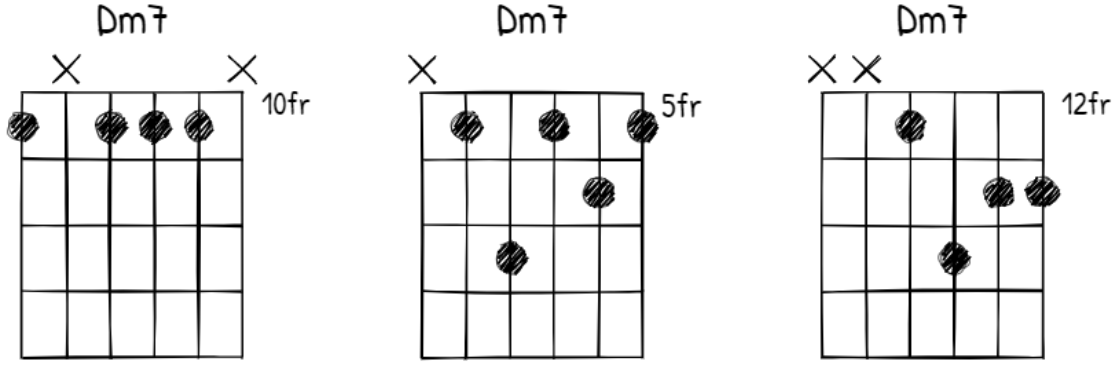
Major 7 Voicings



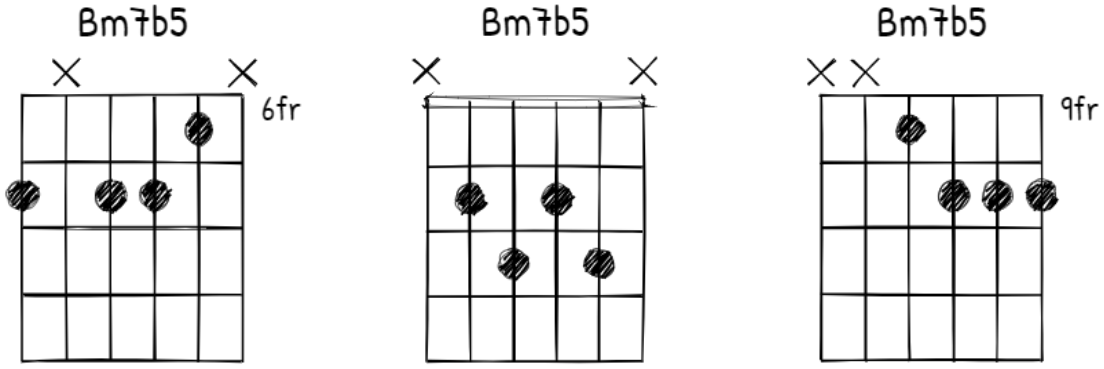
Dominant 7 Voicings



Minor 7 Voicings



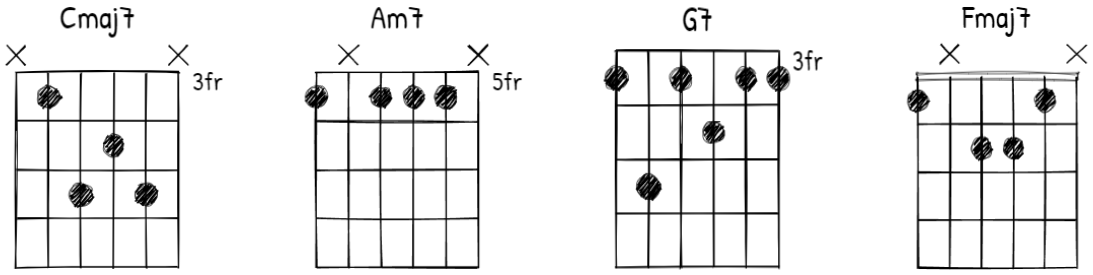
Minor 7 Flat 5 Voicings



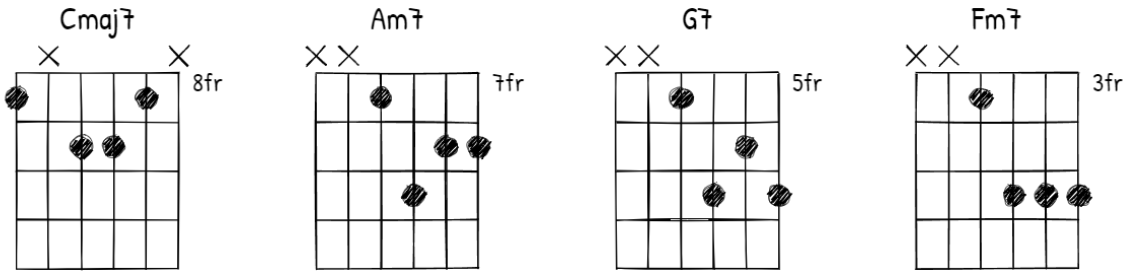
In common music practice, changes between chords in a progression are usually accomplished by moving to the closest available voicing of a particular chord. There are a couple of reasons for this: smoothness of overall feel (does not sound erratic), and less effort for the guitar player. However, interesting things can be achieved with bigger chords jumps, and this happens quite often in math rock. So, do not take this as a rule that must always be followed.

Here is a chord progression in the key of C major (CM7, Am7, G7, FM7) represented as I, vi, V, IV. I have shown three ways to play this progression on the guitar neck, starting from different initial chords. Try playing them using your preferred strumming patterns (left to right). Additionally, experiment with other ways to play the progression, possibly with larger chord jumps or spacing between the chords.

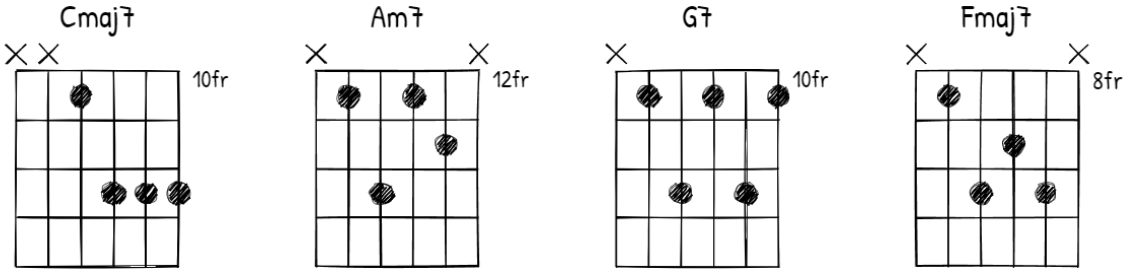
1



2

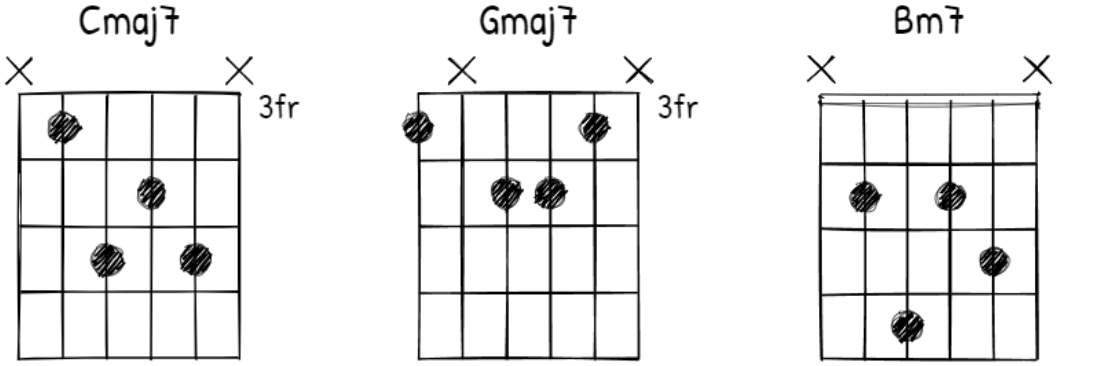


3

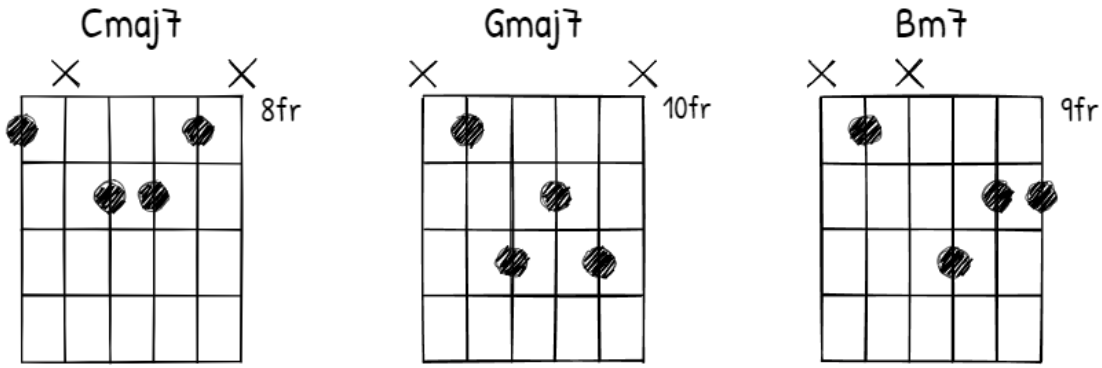


Here is one more. This time a IV, I, iii progression in G major. I have given you two of the progressions. One starting from the sixth string, and the other from the fifth string. Can you complete the last chord progression by finding the closest chord shapes to move to?

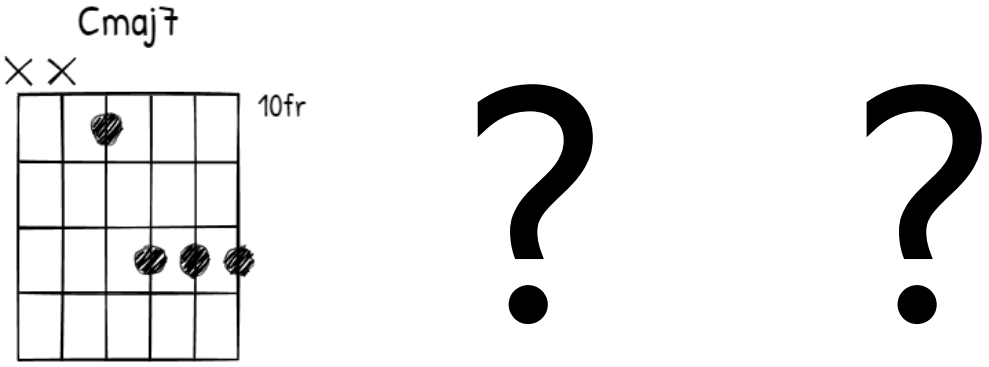
1



2



3



In the next segment, we look at extended nine chord variations. While there are other variations beyond this, elevens and thirteenth chords, for now (and probably for most of the time) you will find extended seven and nine chords the most useful and applicable to math rock and emo influenced styles of math rock.

Ninth Chord Voicings & Practice Progressions

Extended nine chords are hands down my favourite chord type to use. They have something about them that makes them that just a bit more special than extended seventh chords. That been said, I almost always use a mixture of chord types in my chord progressions, because some chord shapes or chord types work better in a particular place in a progression than others.

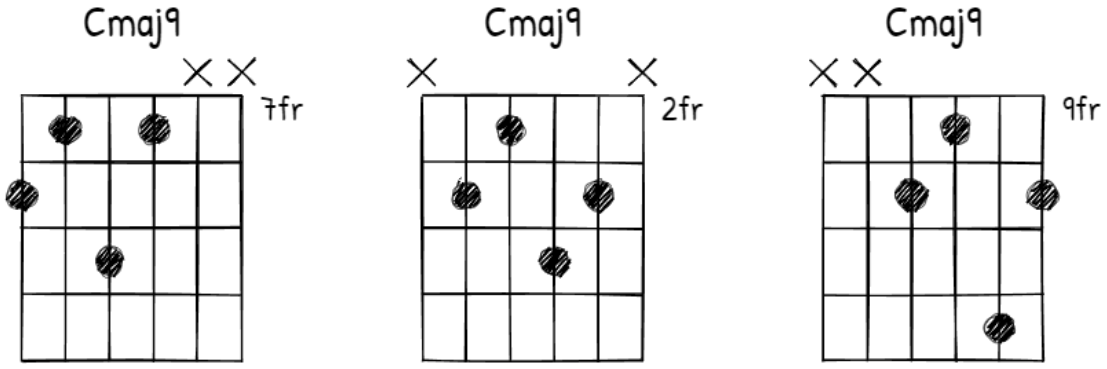
Ninth chords have the spelling 1, 3, 5, 7, 9 and like extended seventh chords, I have included the spellings for each variation for clarity:

Major 9th	1 – 3 – 5 – 7 – 9
Dominant 9th	1 – 3 – 5 – b7 – 9 (flat 7th)
Minor 9th	1 – b3 – 5 – b7 – 9 (flat 3rd and flat 7th)
Minor 9b5	1 – b3 – b5 – b7 – 9 (flat 3rd, flat 5th, and flat 7th)

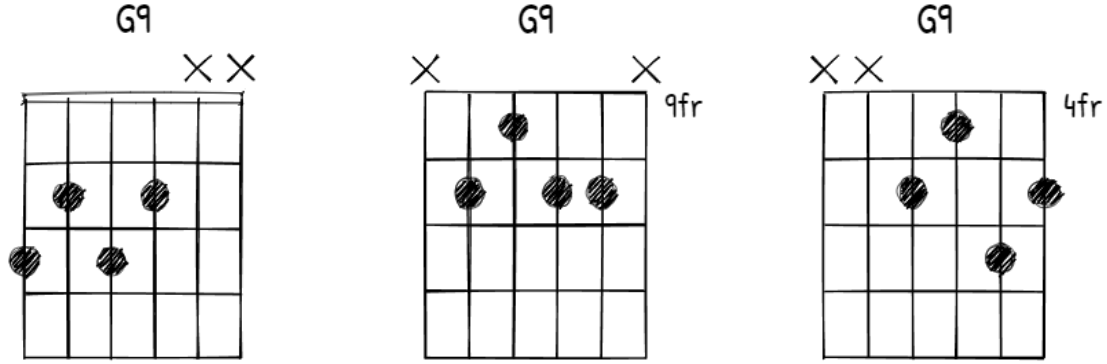
As you can see, the ninth degree of the chord is not altered (sharp or flat) in any of the extended ninth chords. The same is true for extended elevens and thirteenth chords. This is because the guide tones of a chord are its third and seventh. Even without a root you can almost completely define a chord by just knowing its third and seventh. If you look at the spellings chart above, you can see changes to the third or/and seventh as we change from major nine through to minor nine flat five where a flattened fifth is applied.

Note: The minor nine flat five is a little different because of its flattened fifth, it still contains the same guide notes as a minor nine chord: a flattened third and seventh, but until we add the fifth (either a natural fifth or a flattened fifth) into the chord we will not know for sure if the chord is a minor or a half-diminished chord (m7b5).

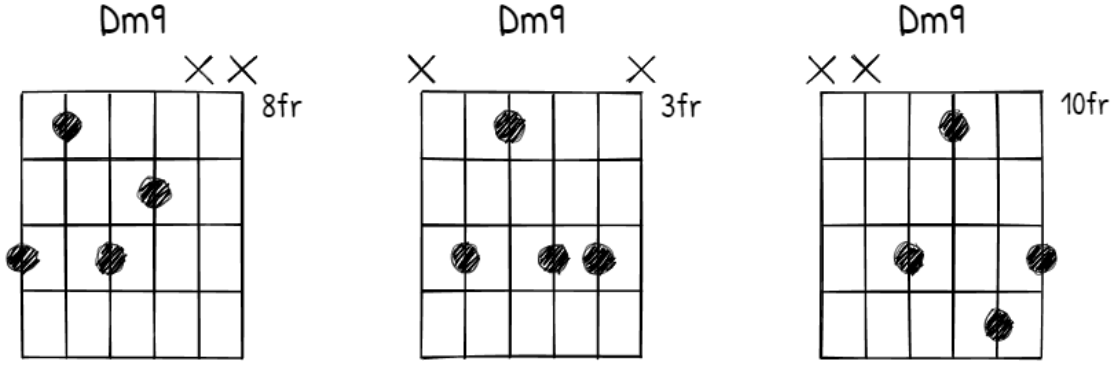
Major 9 Voicings



Dominant 9 Voicings



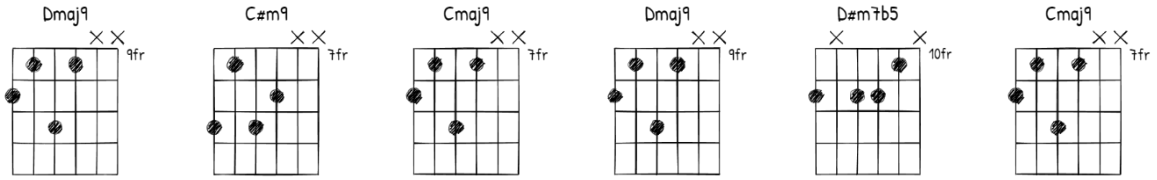
Minor 9 Voicings



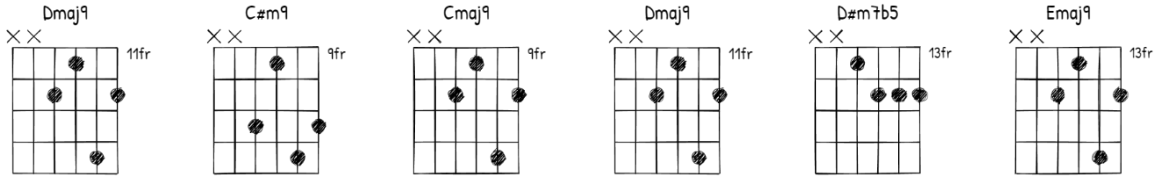
I have left out the minor nine flat five variations because they require some serious finger acrobatics. Instead, I recommend using the minor seven flat five voicings, because they are much easier to play.

The progressions below will get you on your way to changing between extended nine chords shapes more fluidly.

1



2



Other Chord Variations Useful for Math Rock

There are many more chords and chord types to explore beyond what we have discussed. Even without delving into altered chords (like E7#9) and chord inversions, there is still a vast world of chords to discover. In this section, I will highlight some chords that I find particularly useful in mathy styles of music, and I will provide guidance on when and where to incorporate them.

Suspended Chords

A suspended chord is a chord where the third degree has been omitted and usually replaced with either a **major second** or a **perfect fourth** (as indicated in the chart below). They have neither a major nor minor sound to them, rather, more of an 'open sound' that often feel like it needs to be resolved (hence the name suspended).

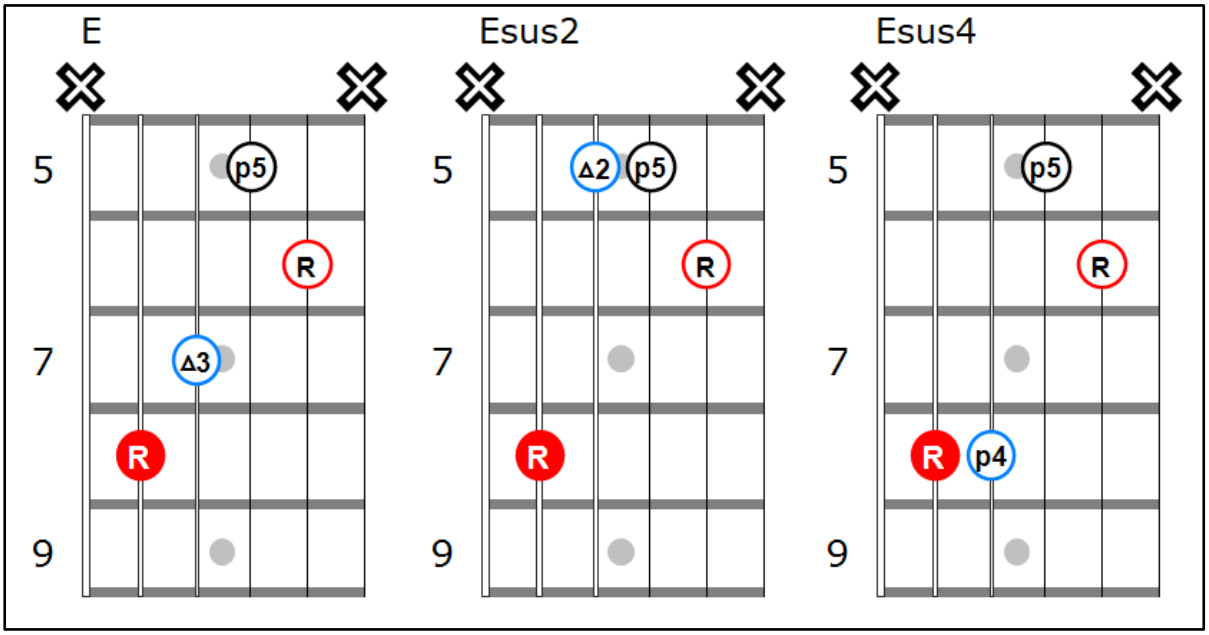
Suspended chord spellings:

Sus2	1 – 2 – 5
Sus4	1 – 4 – 5

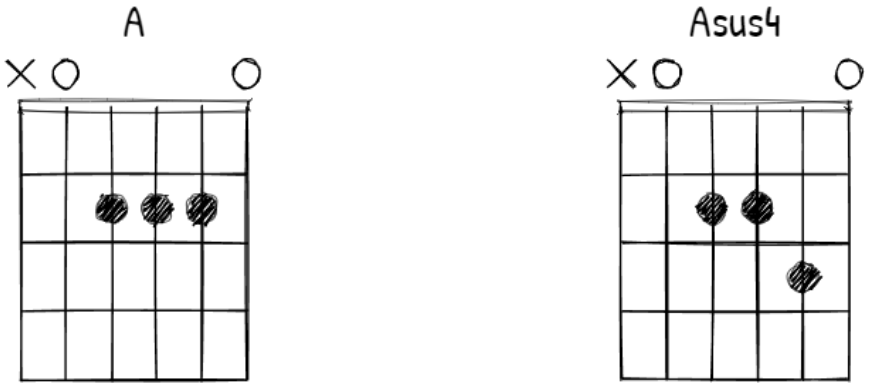
Example in C major:

1	2	3	4	5	6	7
C	D	E	F	G	A	B

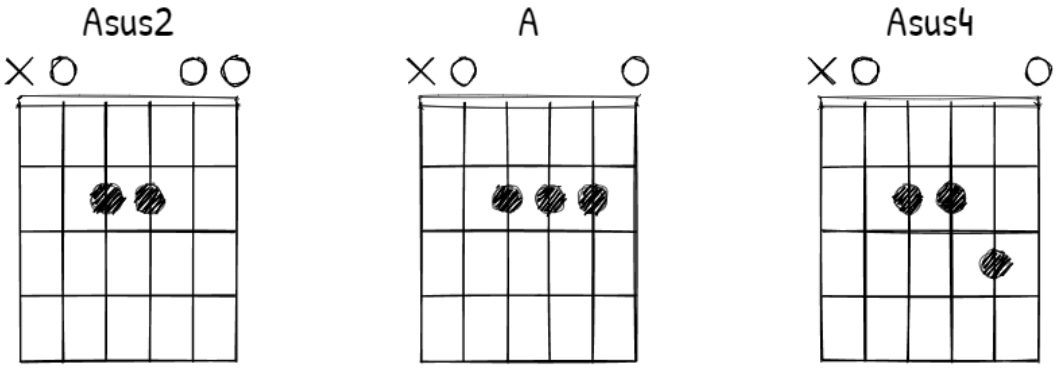
This example illustrates the replacement of the third degree (indicated by the blue coloured note):



The most popular usage of suspended chords in math rock and Midwest emo is to embellish an existing chord in a progression, creating movement within the chord itself, or use them to situate the listener in a feeling of uncertainty. For the former, if you were to play an A major chord followed by an Asus4 you will hear how this embellishes the original A major chord, giving it a different feel and flavour. In addition, you can play this backwards (Asus4 to A) for a more resolved sound:



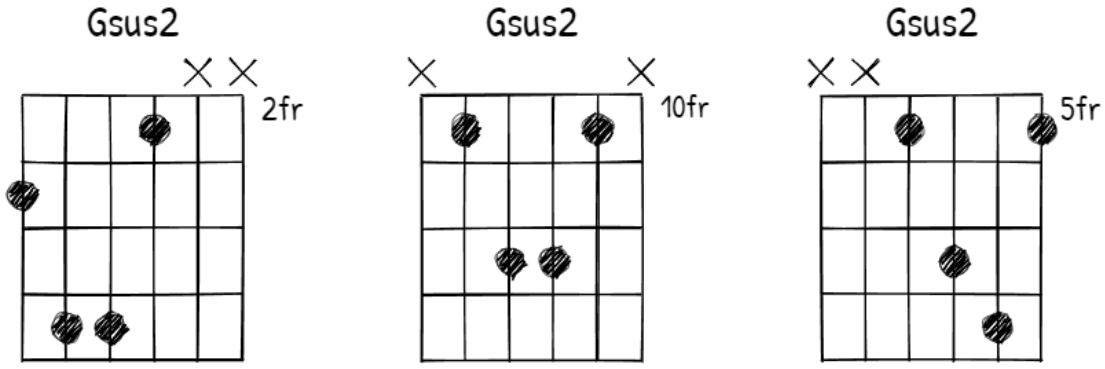
You could even add an Asus 2 to the progression:



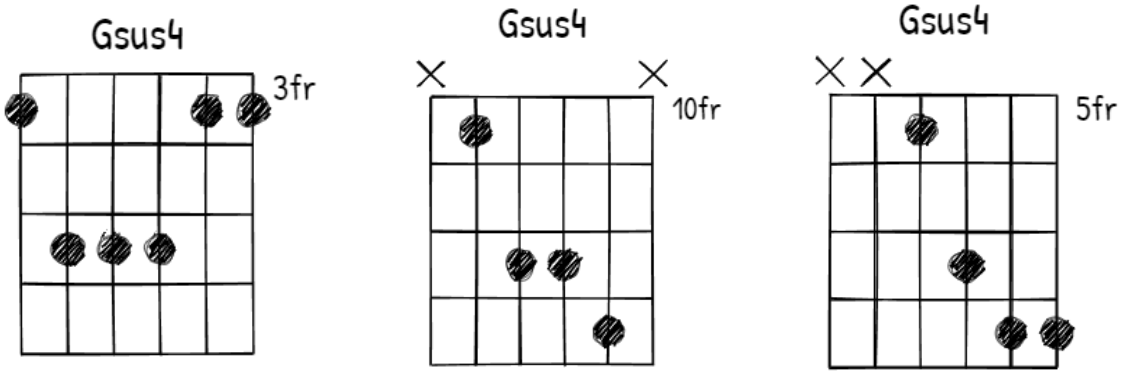
Listen how the slight change of the ascending higher note alters the overall feel of the chord.

Below, you will find chord shapes for suspended second, and suspended fourth chords, with root notes starting from the sixth, fifth, and fourth strings for maximum coverage.

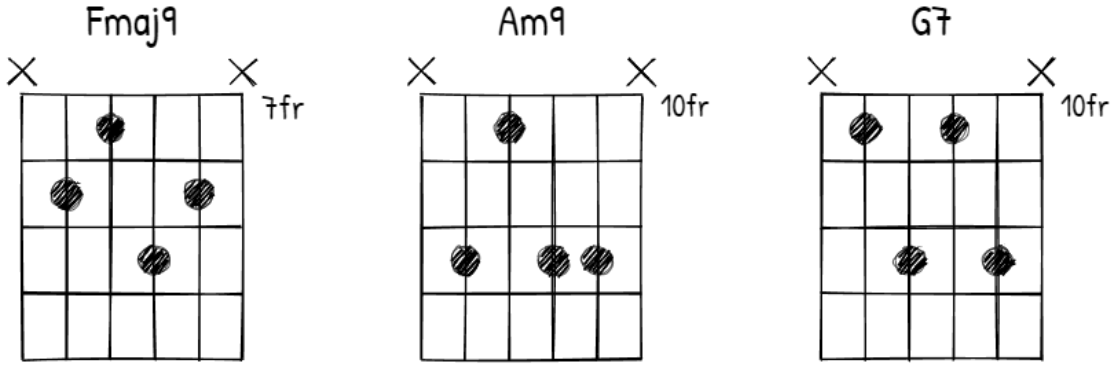
Sus2 Chord Shapes



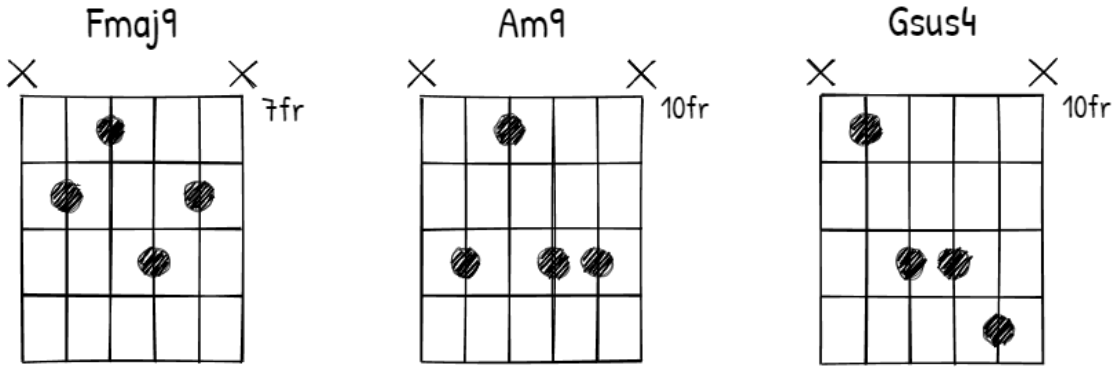
Sus4 Chord Shapes



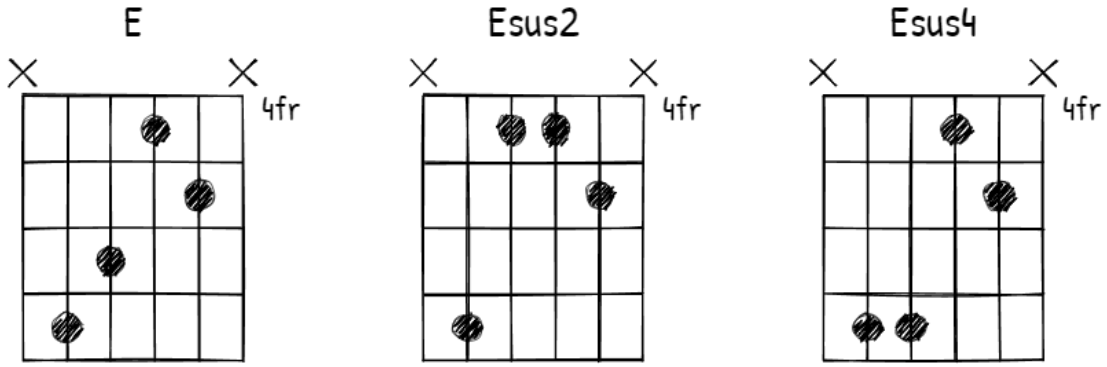
Another useful application of these chords is to use them to replace dominant chords that sound out of place in a chord progression or to replace an unwanted dominant sound in a progression. Remember, dominant seven chords are built from the fifth degree (V chord) of a key (scale). Try both chord progressions below to hear this in action. Personally, I find the dominant chord sounds a little out of place for mathy sounds (it is used a lot though), but the suspended four chord replacement makes the progressions much more suitable for the desired sound I was looking for.



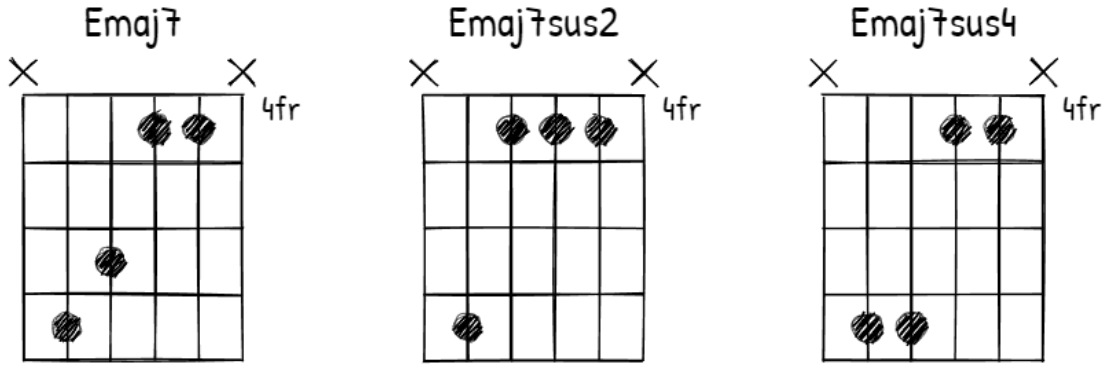
Vs



I would like to teach you one last sus chord variation starting from the A string root.



I find this voicing much more appropriate sounding for Midwest emo, and especially if we bring in the major seventh degree it makes it even more so:



One last point on Sus chords. It is worth pointing out that not every triad chord in a key can be replaced with either a sus 2 or sus 4 chord. I have made a handy chart below so you can see which chords can be substituted with a sus chord.

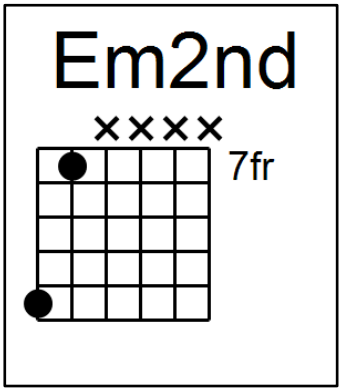
Key	Chord Type	Sus 2	Sus 4
I	Major	O	O
ii	Minor	O	O
iii	Minor	X	O
IV	Major	O	X
V	Major	O	O
vi	Minor	O	O
vii	Diminished	X	X

Major and Minor 6 and 6add9 Chords

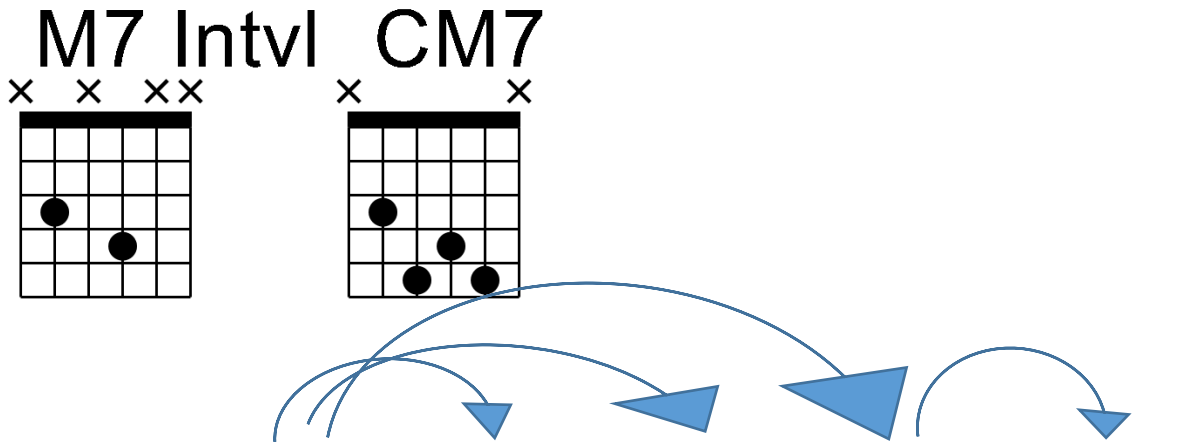
- Major and Minor 6
- Major and Minor add9 Chords
- Major and Minor 6add9 Chords.

The chords listed above offer creative alternatives to extended chords while avoiding the seventh degree. They add harmonic interest to your music without the complexity of including the seventh degree.

Major 6, add 9, or major 6add9 chords are often used because they are a great substitute for major seven chords when it comes to providing a more resolved stable sound. This is because a major seven chord contains a minor second interval between the root and major seventh interval. I.e. the major seventh degree in the chord is only a minor second (a semitone – one fret on a guitar) away from the root. See the chart below for clarification. Minor second intervals do not sound pleasant. If you play the example below you will hear a minor second interval in action.



This sound is contained in a major seven chord, but because it is more spaced out than the interval you just played, it does not sound as nasty. And when the other notes are added, it becomes less obvious that the interval is there.



Interval C major 7	Root 1	Major 3rd 3	Perfect 5th 5	Major 7th 7	Minor 2nd 1

To get around this, players will sometimes use added chords that do not include the major seventh (minor second) interval. This can ease tension by creating a more resolved sounding chord, that are, harmonically speaking, more interesting than basic chords. This can be useful for finishing a song on a resolved chord or writing parts with less desired tension.

It is worth keeping in mind that minor 6 and minor 6add9 chords can have a more dissonant sound compared to their minor 7 counterparts. Therefore, it is a good idea to experiment with them to achieve the specific musical results you desire.

Another useful application of these chords is to substitute basic major, minor, or diminished chords or to use in conjunction (like we looked at earlier) for a more colourful sound.

Major, minor 6 or 6/9 chords have similar applications to suspended chords. However, they differ in sound, and tend to have less tension that suspended chords. It is pretty much personal preference as to how, when, and where to use them.

Taking a quick dive into their chord spellings, you will notice how they differ from suspended chords:

Major 6	1 – 3 – 5 – 6
Minor 6	1 – b3 – 5 – 6
Major 6add9	1 – 3 – 5 – 6 – 9
Minor 6add9	1 – b3 – 5 – 6 – 9

The third degree has returned, and a sixth or a sixth and a ninth degree are added, depending on the chord.

For example, the notes of a C major six chord would be as follows:

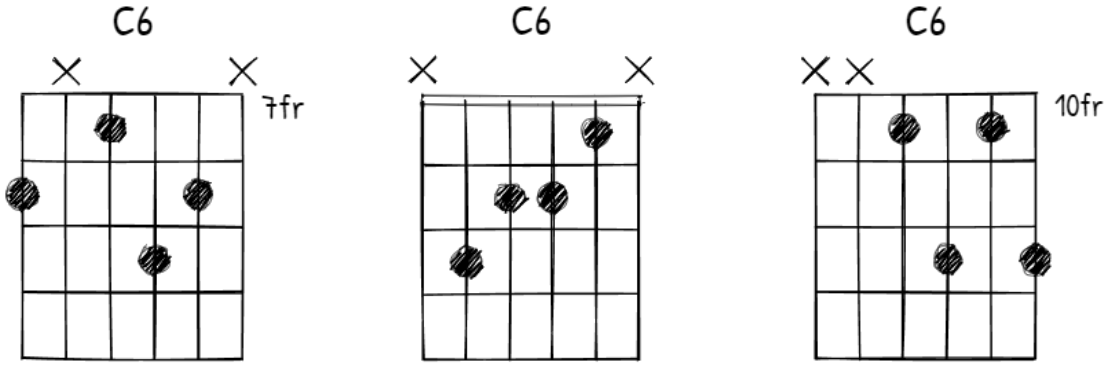
1 C	2 D	3 E	4 F	5 G	6 A	7 B

And a C6add9 chord:

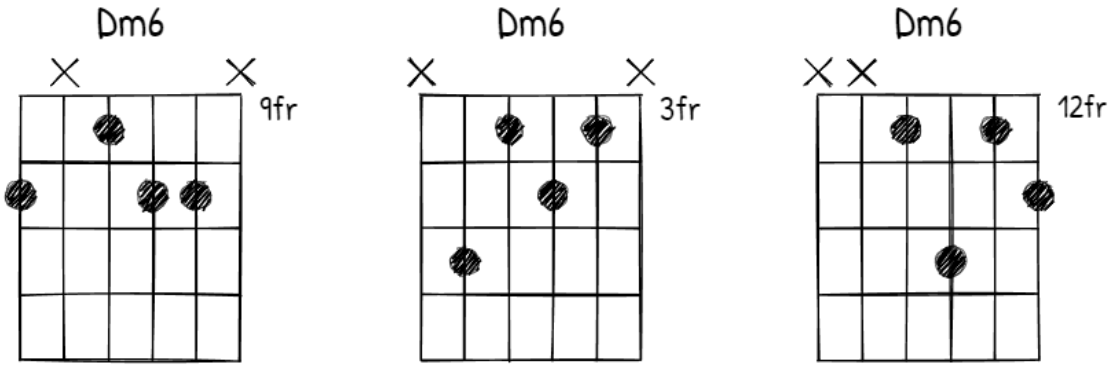
1 C	9 (2) D	3 E	4 F	5 G	6 A	7 B

Below you will find the chord shapes for major, minor 6 and 6/9 chords.

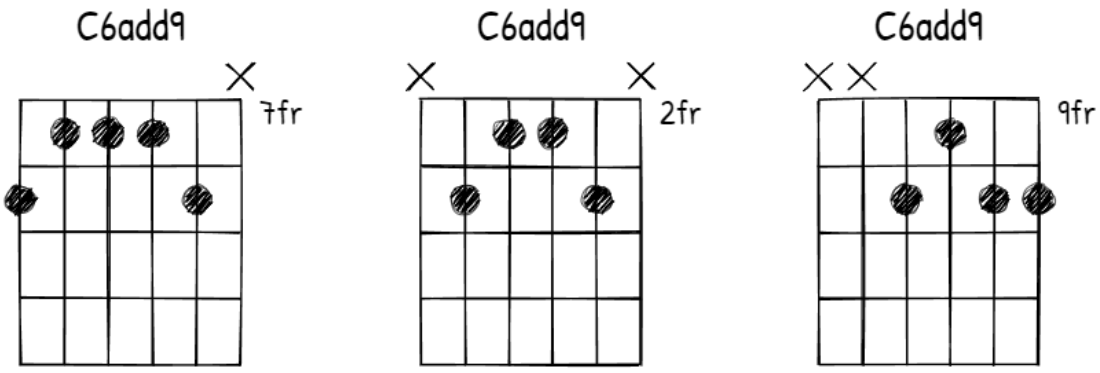
Major six shapes



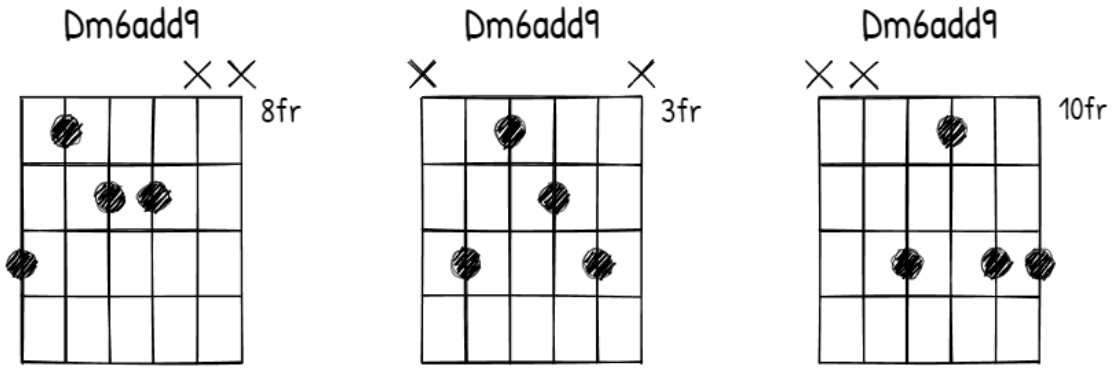
Minor six shapes



Major 6add9 shapes



Minor 6add9 shapes



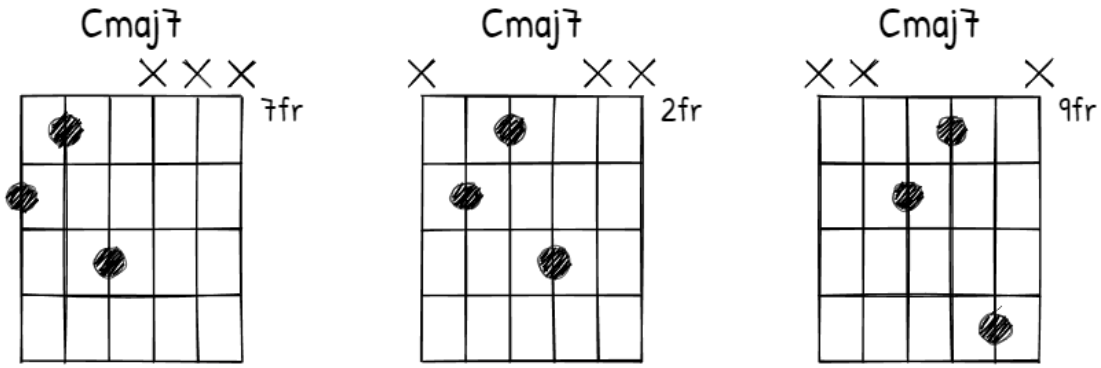
I find these types of chords work best in conjunction with other types of chords, so I will give some examples of their potential usages in the upcoming chord progression segment.

Extended Shell Chord Voicings

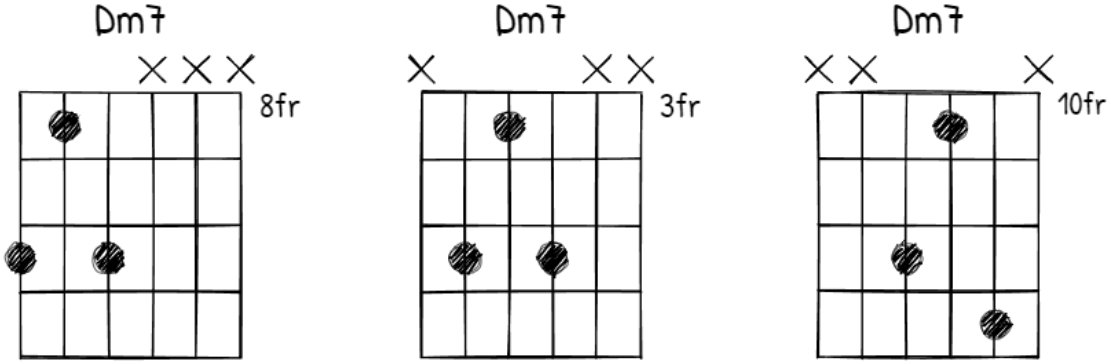
Extended shell guitar chord voicings are a unique type of chord voicing that simplify extended chords. They include the essential notes: the third, seventh, and root, which capture the core tonal quality of the chord (major, minor, etc). These chords have practical uses, especially in ensembles with complex harmonies. Shell voicings, or even just the third or seventh, can be a better fit in such situations. They also work well with distortion tones and are great for chord-based riffs, emphasizing the key notes of a chord.

Below are the shell voicing shapes for seventh chords that you will find used by many other mathy style bands that play in standard tuning.

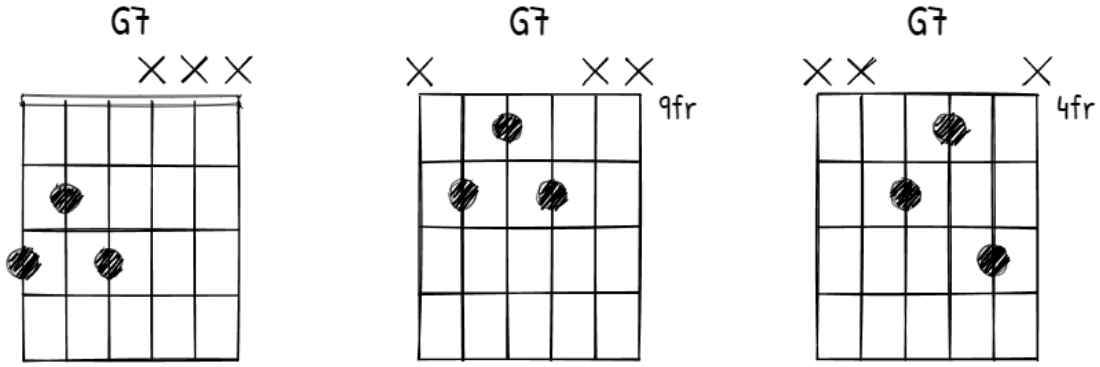
Major Seven Shell Voicings



Minor Seven Shell Voicings



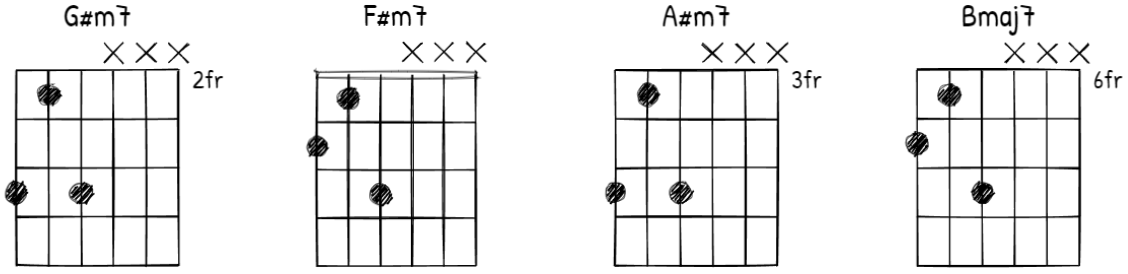
Dominant Seven Shell Voicings



Minor seven flat five chord shell voicings are the same as minor seven chord voicings. This is because these shell voicings are made up of the first, third, and seventh, and if you recall, a minor seventh flat five has a flattened fifth which is missing from these extended shell voicings. This is not a problem, nor should you worry about it, just use the shell voicing as normal in its place, it will still 'fit' by functioning as the seventh chord in a key.

Here are some exercises to start practicing shell chords. It is a good idea to incorporate them alongside regular full-sized chords. You can use them as the basis for crafting riffs or as shapes to spark your riff-writing creativity. Keep in mind that they have a strong and emotive quality, especially when combined with distortion or gain.

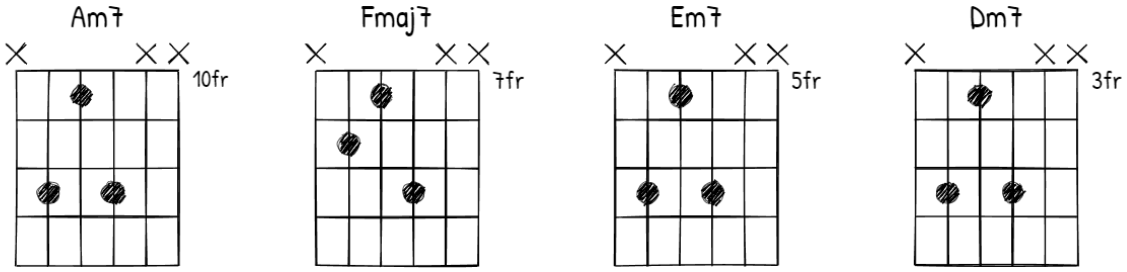
1



Try experimenting with this chord progression, for example, start in the order presented: G#m7 – F#m7 – A#m7 – Bma7 strumming in various patterns that you often use. Next, move to playing the progression as arpeggios (play the notes in succession), then combine the two. Lastly, mess with the order of chords, for example: Bma7 – A#m7 – F#m7 – G#m7 makes for a nice descending pattern or switch it around to be an ascending progression.

Moving on, try playing the chords in example two from left to right, do they sound familiar? They are the chords used in the intro to the song F.C.P.R.E.M.I.X by The Fall of Troy! An excellent example of shell chord voicings in use!

2



Chord Progressions & Mixing Chord Types

Not quite done with chords yet, sorry!

Let's move on to look at using these chords for writing chord progressions. First, we will look at chord progressions, and then I will move on to talk about embellishing chords. Neither of these two concepts are difficult to understand, but I will try to keep it as simple and clear as possible.

Chord Progressions

A chord progression is like the building blocks of a song, consisting of a sequence of chords. These progressions can serve as the backbone of a song or form its individual sections. Some progressions sound more pleasing than others, and they can vary from simple (just one or two chords) to highly intricate (involving substitutions, borrowing chords from other keys, using altered chords, and more).

As was presented earlier, knowing what key a piece of music is situated in assists us in knowing which chords are suitable to use for a potential chord progression. Think of keys as a guide, showing you what chords are available and, harmonically speaking, will fit together quite well.

If we refer to the key or C major, we can see seven chords we could use for a progression:

C Major I	D Minor ii	E Minor iii	F Major IV	G Major V	A Minor vi	B Dim vii

So how do we know which chords to use for a progression? While there are not any hard rules on which chords should go with others, there has been decades of experimentation in music as to which chord progressions work well together, and therefore may be the 'catchiest' for listeners. And, before we continue it is important to remember that when chords are extended to contain the seventh degree, the fifth chord (G major become a G dominant chord) becomes dominant, giving it a differing sound from its major counter parts. And the minor seven flat five chord differing from its minor counterparts:

CM7	Dm7	Em7	FM7	G7	Am7	Bm7b5
Major 7	Minor 7	Minor 7	Major 7	Dom 7	Minor 7	Min7b5
I	ii	iii	IV	V	vi	vii

Starting with major chords, if we do not change key or borrow chords from another key, we have two major extended chords at our disposal: a C major 7 (CM7), and an F major 7 (FM7).

In a major key, the first chord (In this case CM7) has the biggest pull because it resolves the key signature (brings it home to the main chord – how pretty much all popular songs end). The IV major is quite stable sounding on its own, but it tends to have a pull to change to another chord. It can sound great followed by the first or fifth chord.

For Minor chords, there are three chords to choose from: the second (ii - Dm7), third (iii - Em7), and sixth (vi - Am7). Of these three minor chords, the sixth sounds the strongest because it is also the relative minor of C major. This means the key of A minor shares the same notes as C major (refer to the circle of fifths and you can see this). Also, they almost share the exact same notes as each other. A six to one progression sounds quite satisfyingly resolved. Also going six, seven, to one creates this pleasant tension and release feeling.

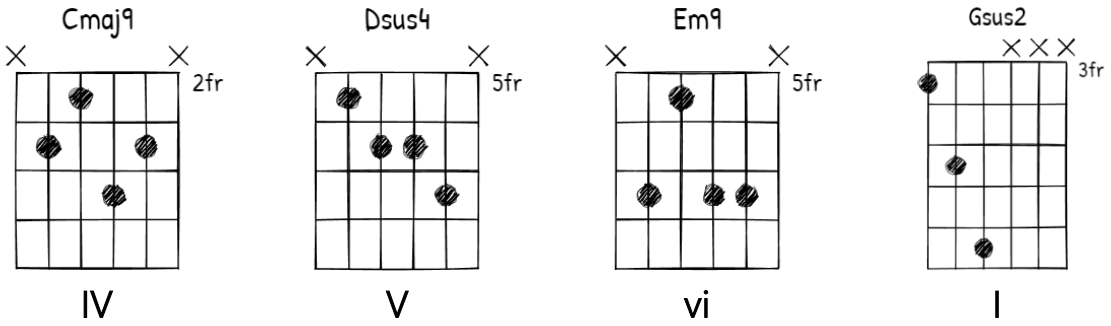
Below, you will find some useful chord progressions to start with. These progressions only use the chords we have covered so far. I have provided numeral symbols for each chord and given examples of different chord types. However, I encourage you to experiment by swapping chords, trying various strumming patterns, adjusting timing between changes, exploring fingerpicking patterns, and even creating riffs using the chord notes.

Keep in mind that you can play these progressions in different parts of the fretboard, explore different chord variations, and experiment with different key signatures. It might sound like a lot, but it is a great way to expand your musical creativity!

Functional Popular Diatonic Chord Progressions

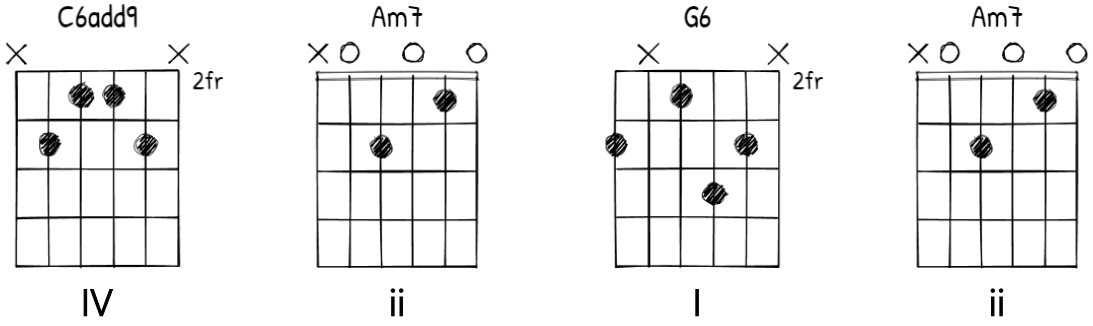
The following chord progressions are based of common popular mathy / emo songs. However, the chord shapes and intricacies of the harmony are often different to what is presented below, but they will be useful to get you started and to fulfil a particular function within a song. The progressions are all diatonic, meaning they only use chords from one key signature.

1.



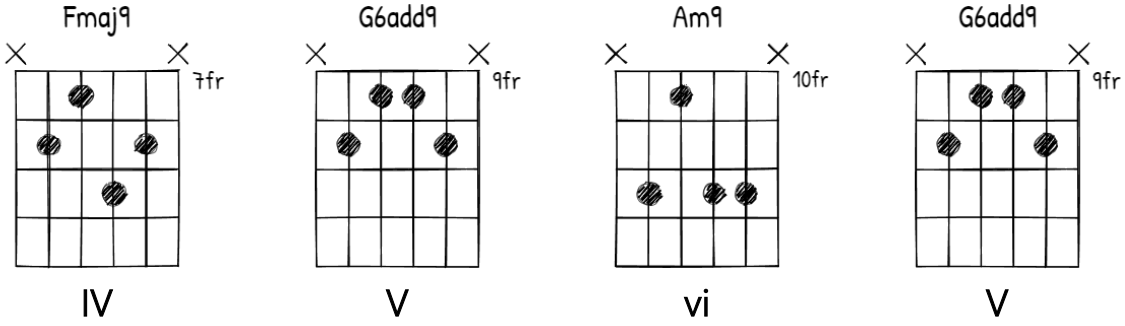
This chord progression has a satisfying ascending feeling with a satisfying resolution on the last chord because it resolves to the tonic chord. It is useful for an ending to a song, a middle section, or even a chorus.

2.



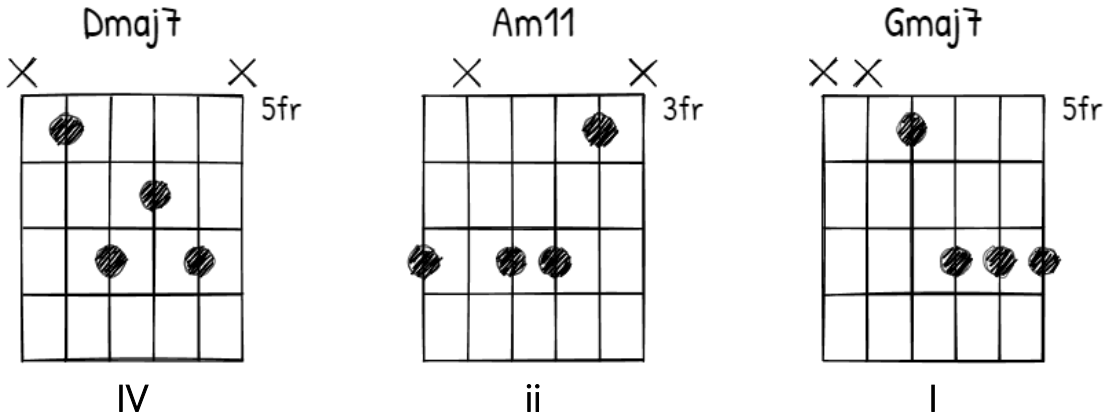
This chord progression is quite poppy in nature with a touch of melancholy, a feeling of longing, but hopeful longing. I find it works well for a chorusy sounding section.

3.



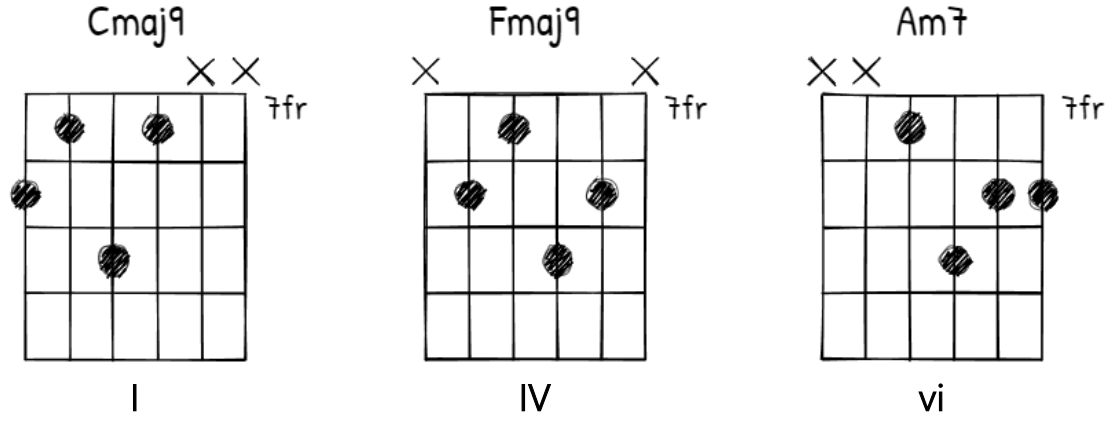
This chord progression works well as a middle section or for a section to build tension. This progression feels like it needs to 'go somewhere,' as in, it needs to progress to another section. Also, the 6add9 chord is a nice work around for the dominant seven sound.

4.



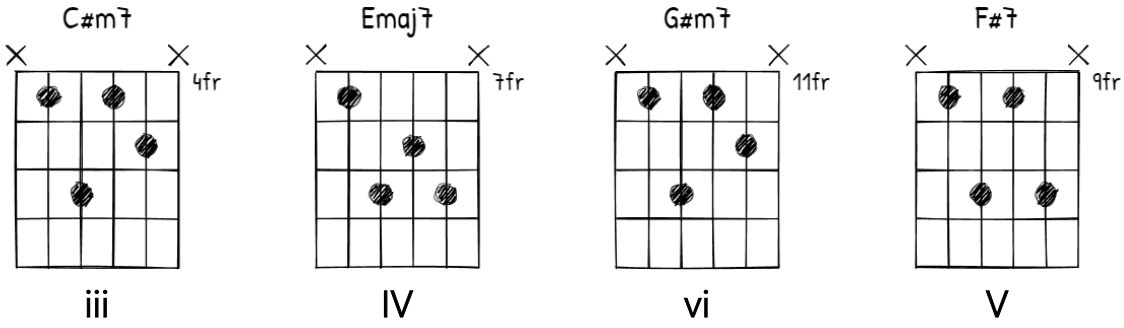
Chord progression four works nicely for a verse or a bridge section following a chorus, it has feel-good vibes to it.

5.



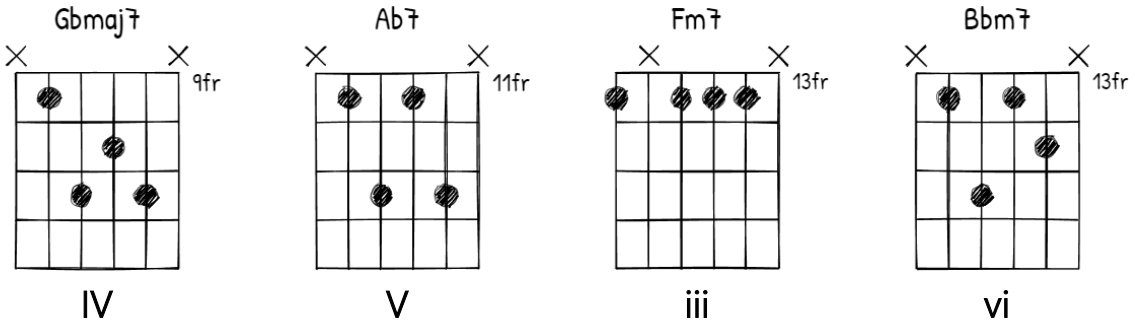
Chord progression five has this lovely uplifting feel to it, largely due to its ascending pattern.

6.



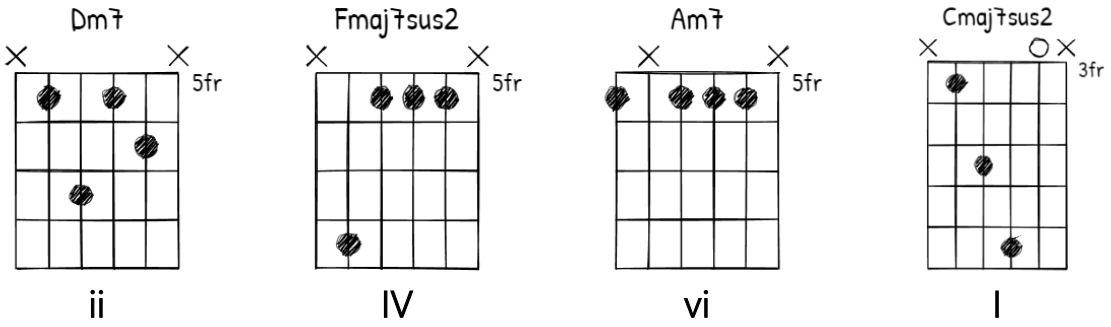
Chord progressions six and seven are perfect for that math rock meets neo soul vibe, and if you know you minor pentatonic scale then you can widdle about with that between the chord shapes or even include notes from the minor pentatonic in your chord shapes for interesting variations.

7.



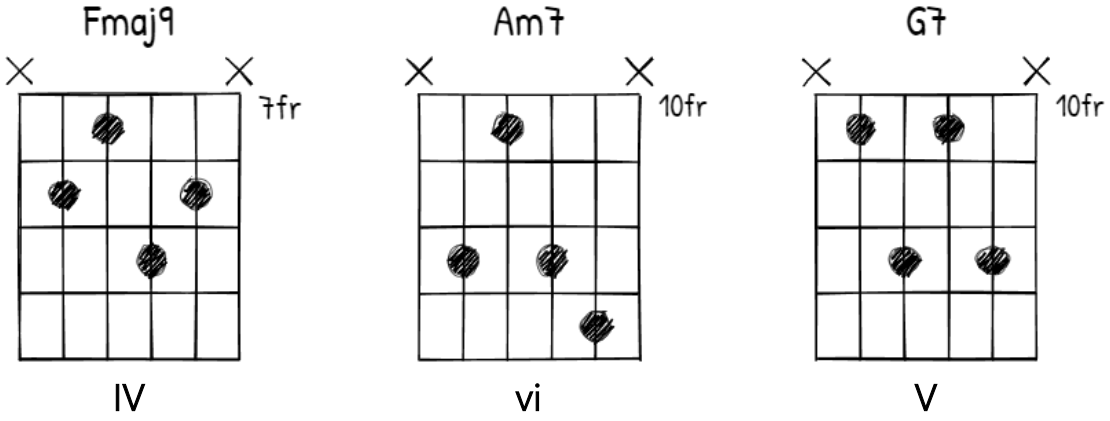
Chord progression eight is a perfect match for a suspenseful bridge section of a song, but of course you are welcome to try it in other places too!

8.



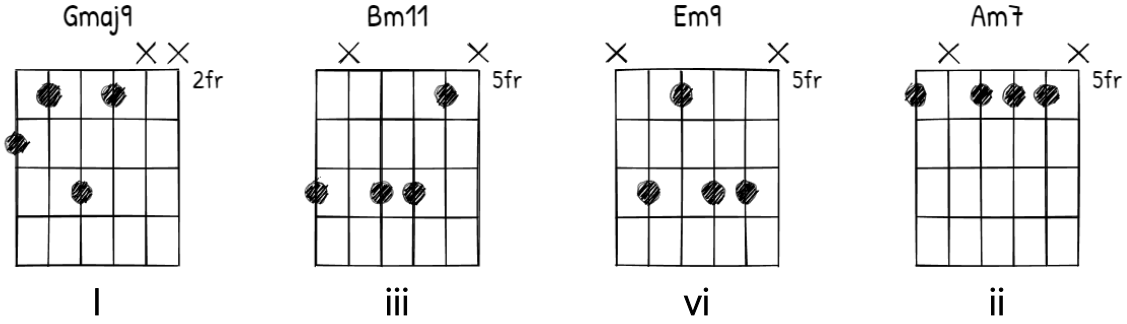
Chord progression nine is another take on the suspenseful bridge section but with a more math-pop feel to it.

9.



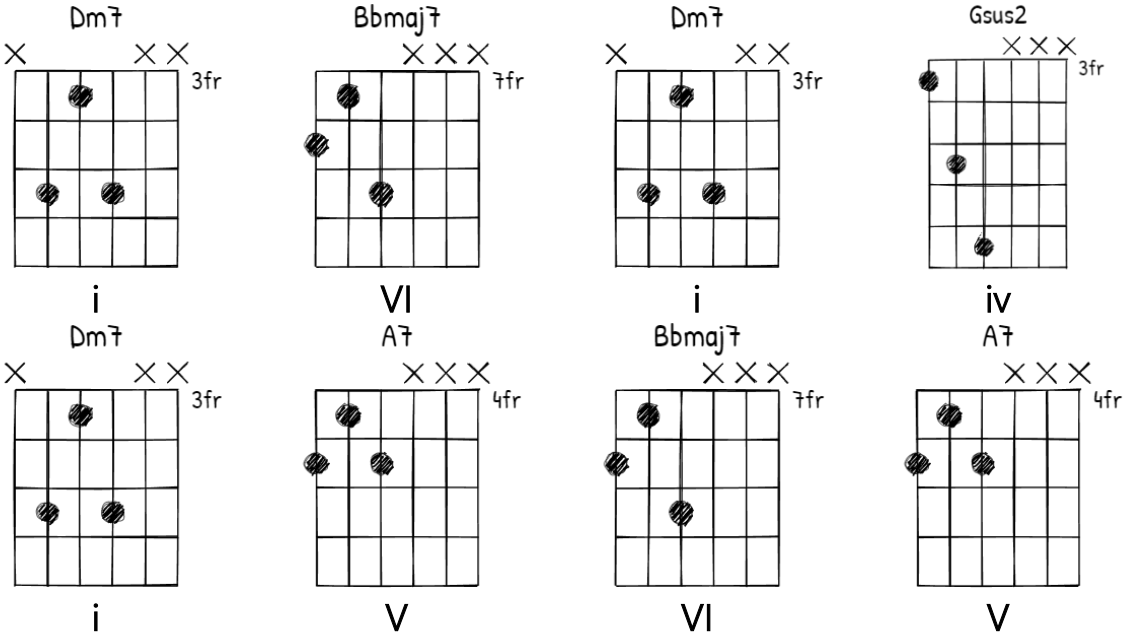
Lastly (for now), we have a chord progression for an epic outro to a song.

10.



11.

As a bonus (cannot help myself, sorry!), this chord progression is taken from a Leon Russell tune called Hummingbird. The progression sounds quite bluesy in its original form, however, when we change a few chords, it ends up being a rather appropriate progression for math rock. For this one we are in a minor key.

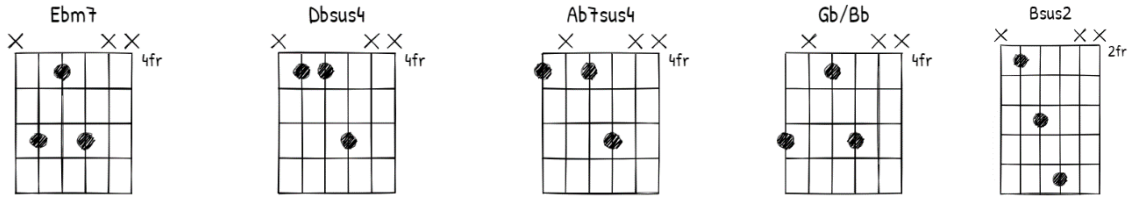


Chord Progressions in Context (Standard Tuning)

The following are some chord progressions from a bunch of math rock/midwest emo style songs. Notice how they make use of many of the chords we learned in the previous section. The examples progressions are by no means exhaustive, and they make some generalisations about each sub-style, but they give some good insight into where a lot of the chords in the guide were derived.

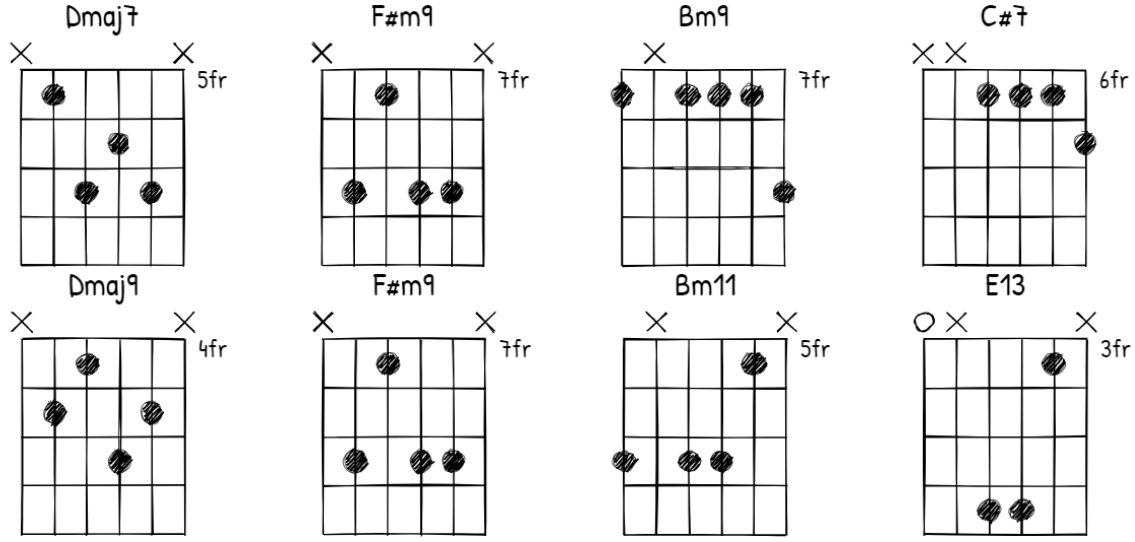
Midwest Emo

Races to April – Top Half of The Everglades



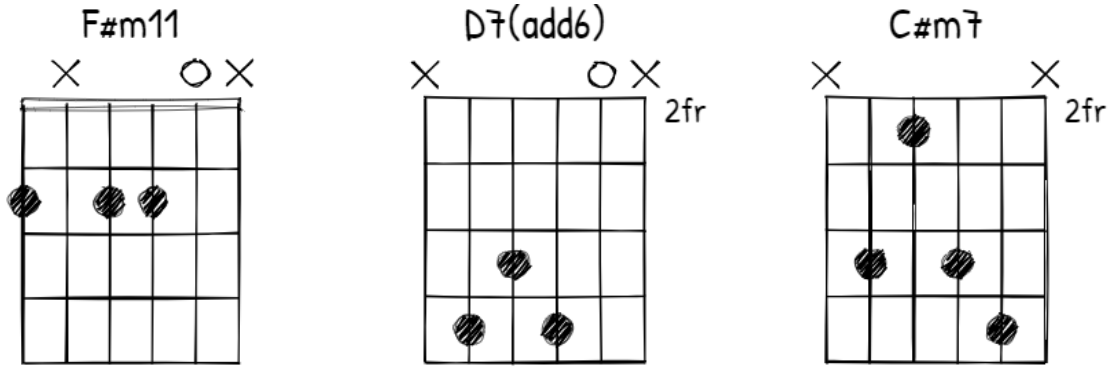
Progressive Math Rock

CHON – Sleepy Tea

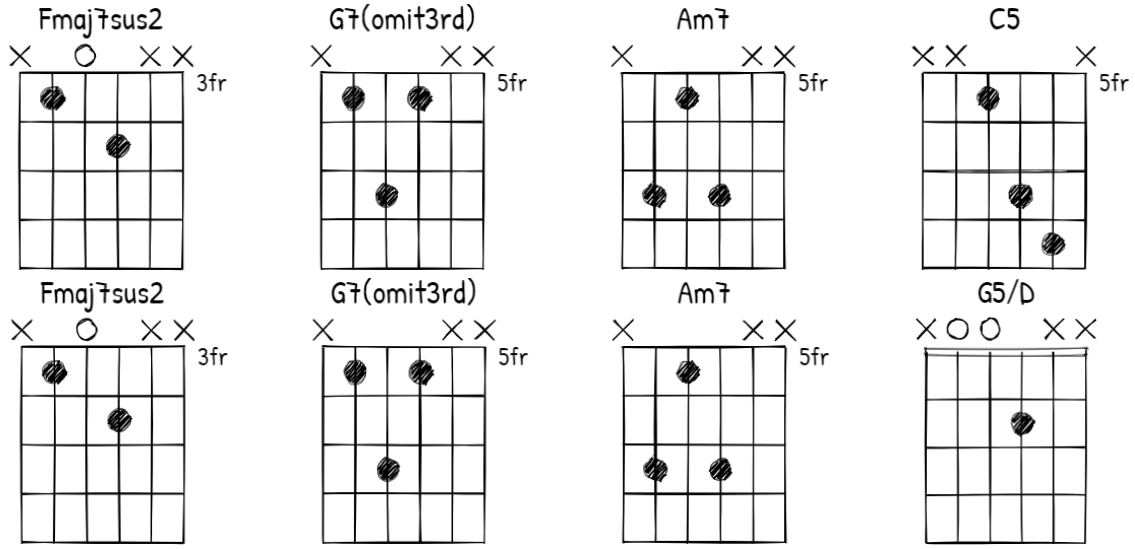


Japanese Math Rock

tricot – POOL

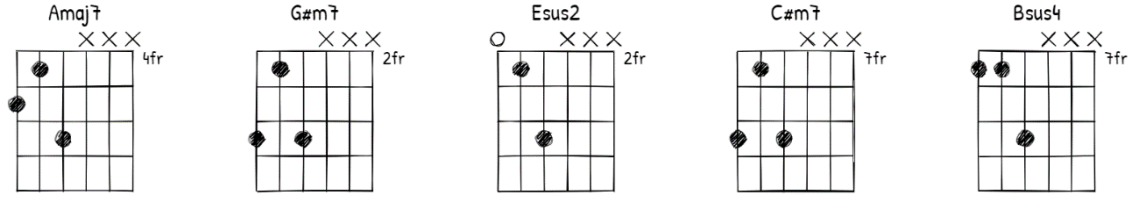


PENs+ SEARCH (capo 6th)



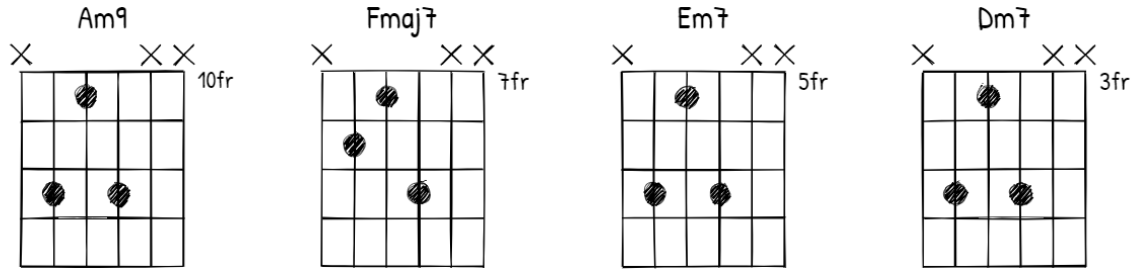
Emo Math Rock

Gulfer - Heat Wave



Post-Hardcore/Math Rock

The Fall Of troy FCPREMIX



Chord Progressions Containing Borrowed Chords

The following chord progressions contain borrowed chords and have a different feel and flare compared to the diatonic chord progressions previously looked at. I really recommend trying them out in your own song writing and playing, but first:

What are borrowed chords?

Put simply, it is a chord that is taken from another key other than the parent key (the key signature that is being used for a particular song or piece). Borrowed chords are frequently

borrowed from the parallel minor or major (the opposite key to which key you are using), hence the term, borrowed. For example, if we were in the key of C major, we could borrow chords from the key of C minor (the parallel minor)

Key of C major:

C	D	E	F	G	A	B
Major	Minor	Minor	Major	Major	Minor	Dim
I	ii	iii	IV	V	vi	vii

Key of C minor:

C	D	Eb	F	G	Ab	B
Minor	Dim	Major	Minor	Minor	Major	Major
i	ii	III	iv	v	VI	VII

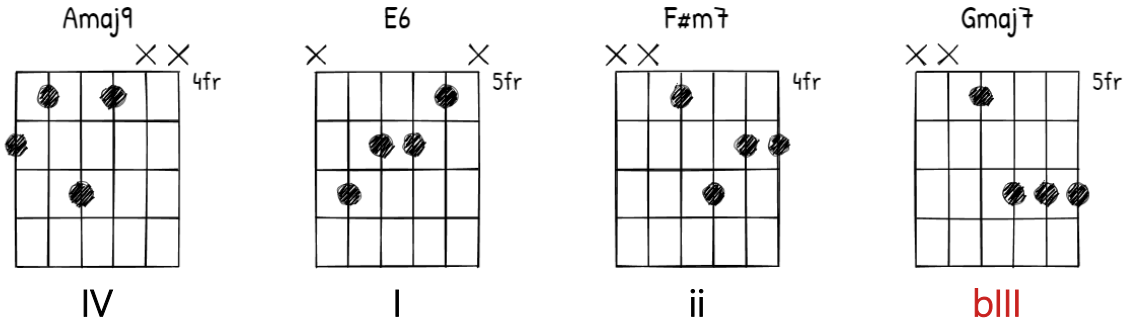
If we were to borrow a chord from C minor when in the key of C major, certain markings are used to indicate that the chord is a borrowed one:

C	D	Eb	F	G	Ab	Bb
Minor	Dim	Major	Minor	Minor	Major	Major
i	ii°	bIII	iv	v	bVI	bVII

Flat symbols are put before the major chords bIII, bVI, and bVII, because compared to their minor counterparts each note had been lowered a semitone. Next, the minor chords use a lower-case roman numeral: i, iv, v. Lastly, the second has the ii° diminished symbol.

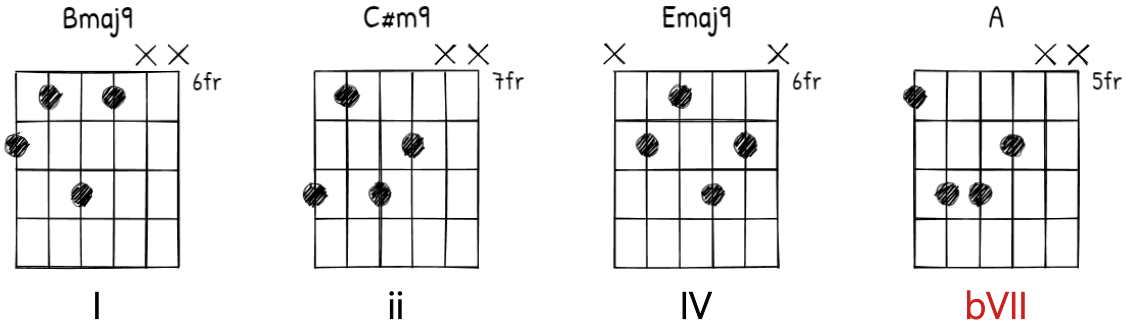
The following chord progressions utilize borrowed chords and I will explain the function of the borrowed chords used under each example.

1.



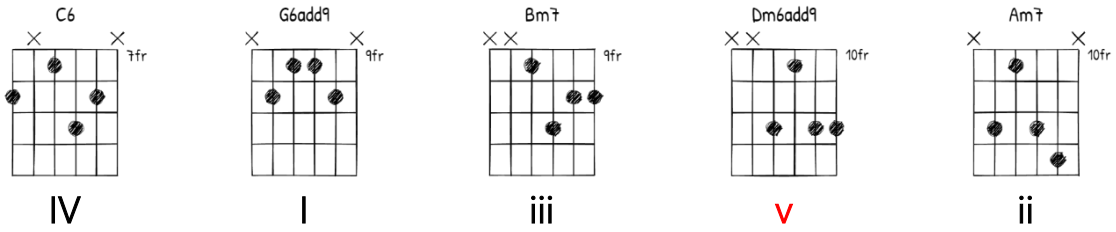
In this chord progression, the minor chord built from the third (G#m) has been replaced with a borrowed major third chord from the parallel minor key (E minor). It introduces an unexpected sound because your ears naturally want to hear the minor third chord instead. It is a cool trick and sounds great.

2.



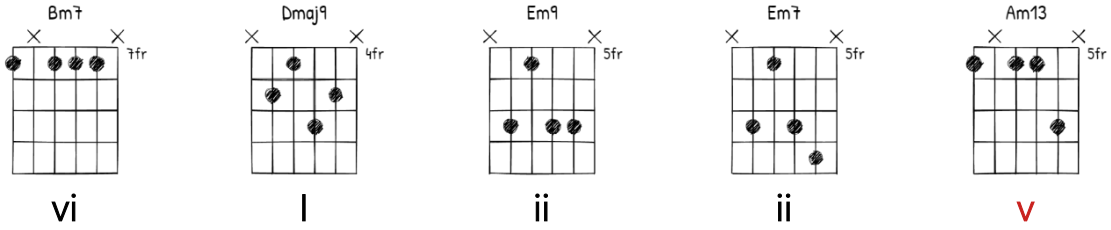
In chord progression two, the half-diminished chord (vii°) is replaced with a major seven borrowed from the parallel minor (B minor). This is a nice way of mitigating undesired tension provided by the diminished chord. However, you could include the half-diminished chord for some tasty voice leading; play the A followed by the A#m7b5 chord, followed by the Bmaj9 for a cool chromatic sound!

3.



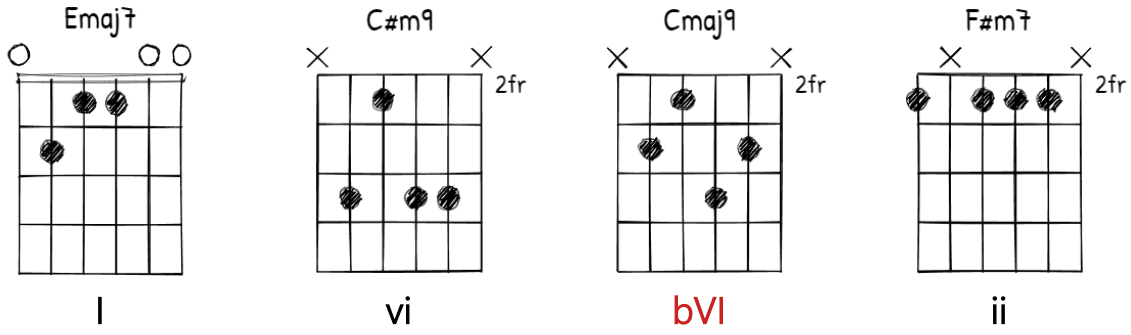
For this chord progression, the borrowed chord replaces the five major chord with a minor five chord, adding some real character to the progression. Oh, it is borrowed from the parallel minor again, if you had not worked that out already!

4.



The same thing as chord progression three is happening here, just a slightly different take on it.

5.



Lastly, I could not leave out an example from the masters of using borrowed chords: Delta Sleep. This is a progression based on the outro to a song called The Younger Years from their Younger Years EP. The inclusion of the bVI (Cmaj9) leads effortlessly to the F#m chord.

Tasks Answers

Keys Task

1. E major – A major – G# minor – F# minor – E major = E major: I – IV – iii – ii – I

2. D minor – G7 – C major = C major: ii – V - I

3. D major – E minor – F# diminished – G major = G major: V – vi – vii - I

4. C major – G major – A minor – E minor = either C maj: I – V – vi – iii or G maj: IV – V – ii – vi

5. G major – B minor – C major – C minor = G major: I – iii – IV – iv - *Okay, so I tricked you with this one, sorry! You may have been scratching you head over the C major and C minor. The C minor, minor iv chord, is borrowed (known as a borrowed chord, more on these later!) from the parallel minor G minor: G min, A dim, Bb maj, C min, D min, Eb maj, and F maj (i ii, III, iv, v VI, VII – minor key spelling).*

Segment 2C - Alternative Tunings for Math Rock

Alternative tunings are when you tune your guitar to something other than standard tuning (EADGCE). They are a key feature of math rock guitar and are often favoured by bands that blend emo and math rock. Using alternative tunings can be exciting, inspiring, and a lot of fun!

However, they come with challenges. One of the main challenges is that you lose your sense of familiarity on the fretboard. In other words, all the chord shapes, arpeggios, scales, and patterns you know in standard tuning will be in different places in alternative tunings. But, on the bright side, alternative tunings provide a unique foundation for your music. Unlike standard tuning, which is like a blank canvas, alternative tunings give your music a distinct colour and character from the start.

They are also great for ear training because you will rely on your ear more to find chords, melodies, and riffs in these tunings. So, despite the initial learning curve, they are worth the effort if you want to add depth and creativity to your music!

Examples of Alternative Tunings

This section will focus on two of the most common alternative tunings used; FACGCE and DAEAC#E or CGDGBD (same thing as DAEAC#E but tuned two steps lower).

Some examples of FACGCE tuning in action:

American Football – Never Meant

Colossal – Hot Probs

TTNG – Crocodile

Song Examples in DAEAC#E:

Sleepy Dog – Screamers

Algernon Cadwallader – Some Kind of Cadwallader

Tiny Moving Parts – Caution

What Makes Certain Alternative Tunings Sound Great?

You may be wondering 'what makes some alternative tunings sound so damn nice?'

The secret is in the note selection, and with a bit of simple music theory I can explain why.

If we look at the notes that make up standard tuning, E A D G B E and the distance between the notes we end up with a bunch of fourth intervals and one third interval (the distance between the notes). See the following for clarification:

E to A						
E	F	G	A	B	C	D
1	2	3	4	5	6	7
A to D						
A	B	C	D	E	F	G
1	2	3	4	5	6	7
D to G						
D	E	F	G	A	B	C
1	2	3	4	5	6	7
G to B						
G	A	B	C	D	E	F
1	2	3	4	5	6	7
B to E						
B	C	D	E	F	G	A
1	2	3	4	5	6	7

Which we can all agree upon when strummed does not really sound like much. Technically, we could say this is an Em11 chord:

E	G	B	D	A
1	b3	5	7	11

The notes and interval between them make up an E minor 11 chord. However, the way it is voiced does not really sound all that great.

If we take the tuning of F A C G C E, and analyse it in the same manner, we will see why it sounds more like a chord than a random assortment of intervals:

F to A						
F	G	A	B	C	D	E
1	2	3	4	5	6	7
A to C						
A	B	C	D	E	F	G
1	2	3	4	5	6	7
C to G						
C	D	E	F	G	A	B
1	2	3	4	5	6	7
G to C						
G	A	B	C	D	E	F
1	2	3	4	5	6	7
C to E						
C	D	E	F	G	A	B
1	2	3	4	5	6	7

As you can see, we have more third intervals in the voicing of the tuning. These make up an Fmaj9 (F major nine) chord and sound rather pleasant:

F	A	C	E	G
1	3	5	7	9

This means this tuning is an open voiced chord tuning; the tuning is based on a chord, in this case, an Fmaj9 chord. Meaning, this tuning sounds good because it is tuned to the sound of a major nine chord.

Let's repeat the same process for D A E A C# E tuning:

D to A						
D	E	F#	G	A	B	C#
1	2	3	4	5	6	7
A to E						
A	B	C#	D	E	F#	G
1	2	3	4	5	6	7

E to A						
E	F#	G	A	B	C#	D
1	2	3	4	5	6	7
A to C#						
A	B	C#	D	E	F#	G
1	2	3	4	5	6	7
C# to E						
C#	D	E	F#	G	A	B
1	2	3	4	5	6	7

This one is slightly more challenging to work out, but it is still based around a chord, that chord being an A/D (an A major chord with a D as the root note, aka a slash chord).

D	A	C#	E
4	1	3	5

This chord is slightly less complex (harmonically speaking) than what you get with FACGCE tuning, but that lower D note over and A chord sounds, just like FACGCE, inspiring. The difference between these two tunings is that FACGCE tuning contains a major 7th degree of the scale which gives this slight tension to the chord (a semi-tone between the seven and root), which gives the tuning its characteristic happy but with a pinch of sadness to its feel.

So, to summarize, these two tunings sound great because they are voiced around chords. This is something to keep in mind if you are attempting to create or use an alternative tuning. You could even voice a tuning around a minor chord for a more sombre feeling vs a happier brighter feeling you get from major variations.

String Gauges for Alternative Tunings

If you are only temporarily changing strings or adjusting certain strings by a semitone (like going from E to F or tuning to Eb), it is usually fine to stick with the same string gauge you use for standard tuning. This will not pose any issues for your guitar neck or playing.

However, for most alternative tunings, you will need more significant changes. In such cases, I suggest using your regular string gauge to experiment with the new tuning. And if you decide to use that tuning more frequently or permanently, then you can explore changing the individual string gauges accordingly.

If you plan to use a particular tuning for an extended period or if you are making larger tuning adjustments (more than a whole tone), here is what I recommend:

1. For strings you are tuning down (e.g., E down to D on the low E string), use a thicker gauge string. I suggest going one gauge higher than your current strings.

2. If you are tuning up (e.g., G up to A on the G string), use a thinner gauge string.

3. If the tuning changes are more significant than a whole tone, experiment with thicker or thinner gauges depending on whether you are tuning up or down.

This might require buying individual strings or mixing and matching sets, but it is definitely worth it for a better playing experience.

For example, using DAEAC#E tuning with a set of 9s:

Standard Tuning	Tuning Direction	Changes	Guage Change	Strings Needed
6 – E	Down	6 – D	42 up to 46	10s
5 – A	No change	5 – A	32	9s
4 – D	Up	4 – E	24 down to 22	8s
3 – G	Up	3 – A	16 down to 14	8s
2 – B	Up	2 – C#	11 down to 10	8s
1 – E	No change	1 – E	9	9s

To achieve the above you would need (depending on the manufacture of strings) a set of 8s, 9s, and 10s. You could also look at buying individual strings or hybrid sets of strings to keep the costs down. Ultimately it is about balancing the tension as evenly as possible across the neck.

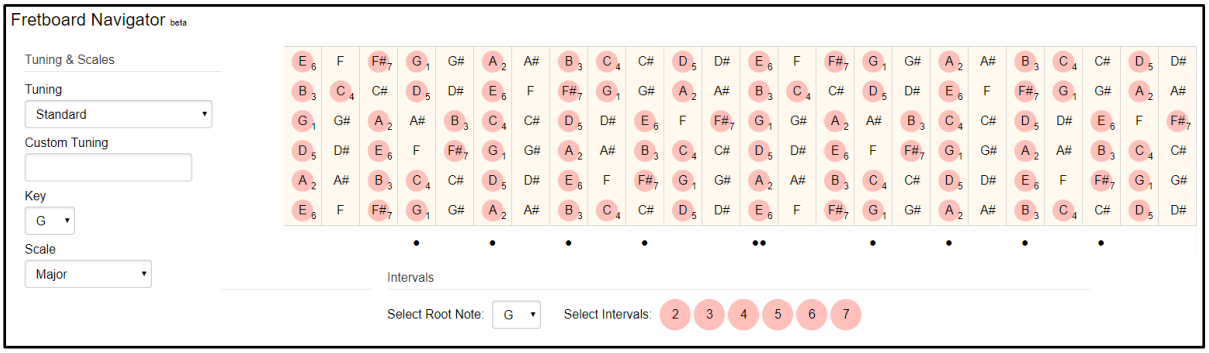
How to Write When Using Different Tunings?

There is no easy answer to give or easy way of approaching writing in different tunings. I have a solid knowledge of where chords and notes are in standard tuning but when I change to a different tuning, I lose that knowledge. You could look at this as off putting and not bother with it, but I recommend seeing this as a good challenge.

Exploring alternative tunings can also serve as an excellent ear training exercise. You will be challenged to find the notes you hear in your mind on the guitar, and this process can lead to discovering unique and appealing chords.

If you are someone who wants to know where all the notes are in many different tunings, then you will find this tool super helpful: https://fretboard-navigator.com/#

On this website you can choose a tuning, a key and even scales! Meaning you can see where they are in a tuning and any key on the fretboard. Awesome stuff indeed! Get it booked marked if you have not already.

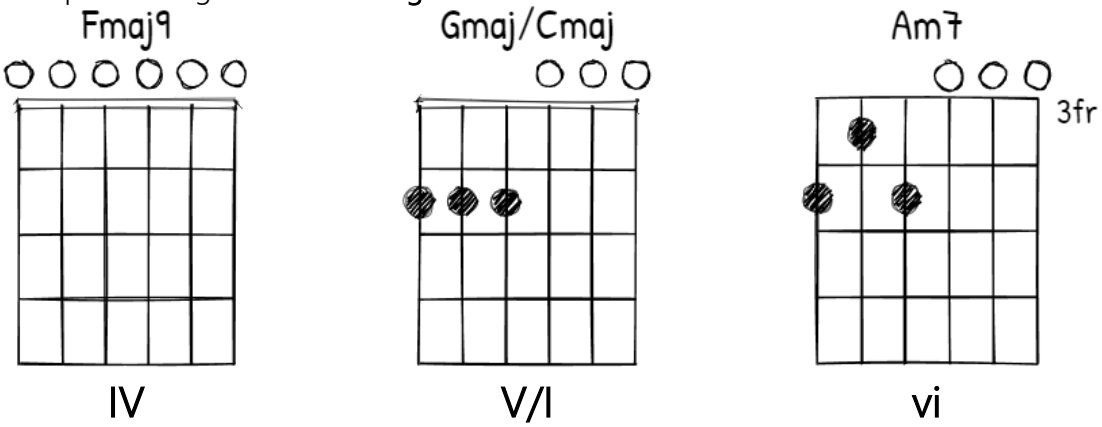


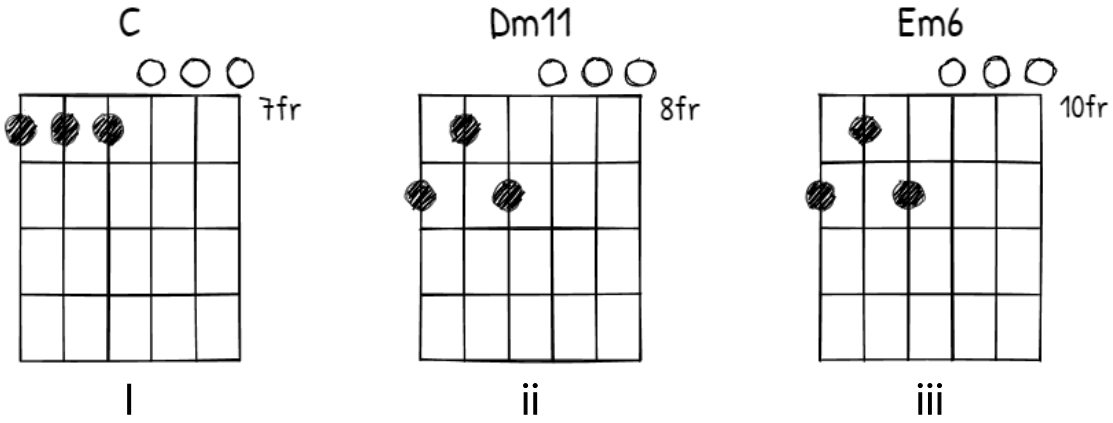
Remember that open tunings offer a specific sound and feeling because they are tuned to a chord. You can use this to your advantage by incorporating the open tuning (the chord when strummed open) into your musical ideas and by including open strings within your chord choices. If you want to alter that feeling, you can simply use a capo to change the key or opt for a different tuning altogether. There is no need to make one alternative tuning sound like another; it is easier to switch to a tuning that suits your desired sound.

Chords for FACGCE Tuning

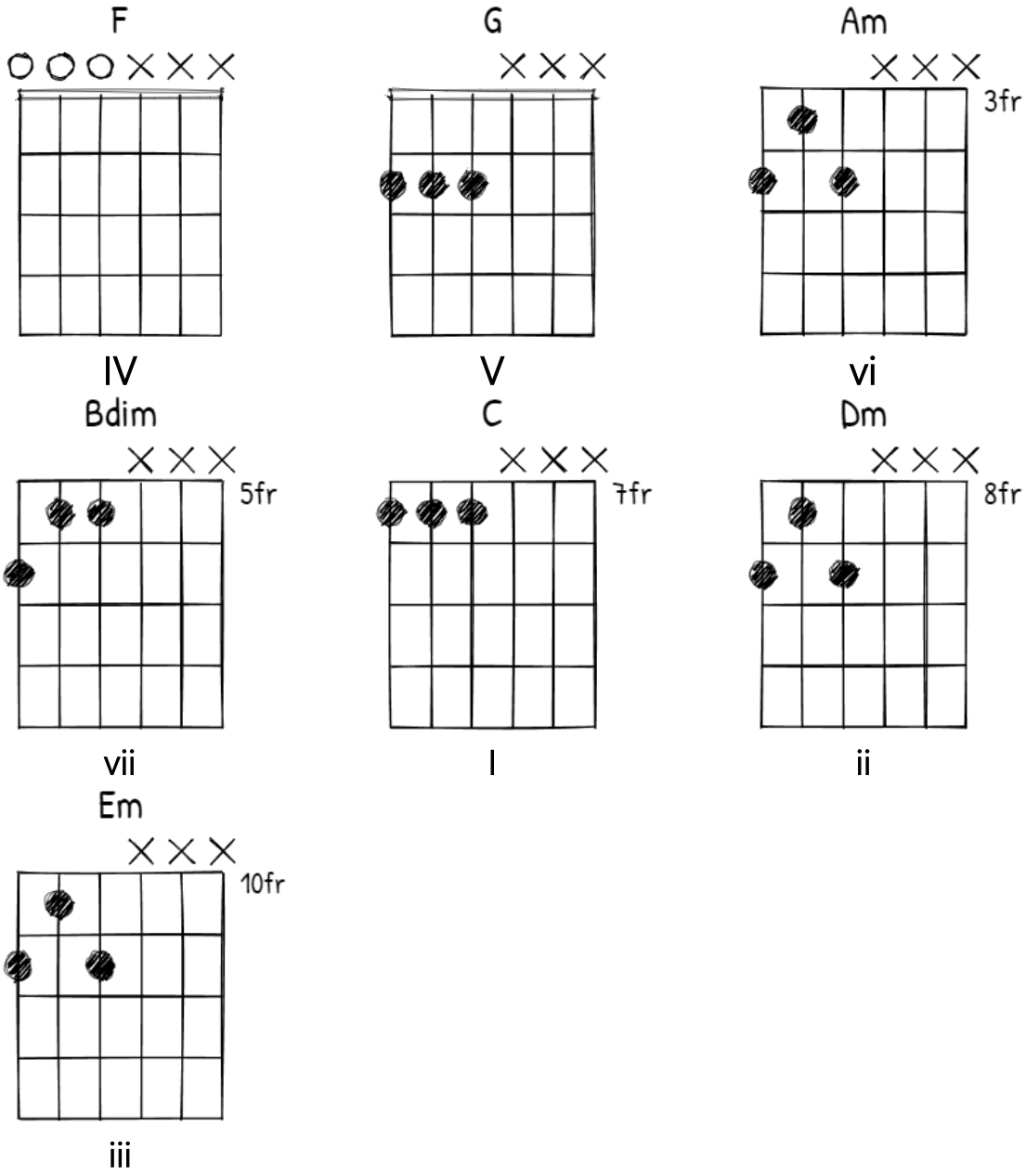
The approach to the following chords is twofold. First, I wanted to provide you with chords that capitalize on the sound of the tuning itself, meaning, making use of the available open strings to create interesting open-string chord sounds but within the confines of a key signature. Therefore, you will notice there is a chord for each note of the major scale. These chords sound wonderful in clean or with some light gain applied. To change the key, just apply a capo in the desired place. Also, I wanted to provide shapes that work well with higher gain therefore, you do not need to learn two sets of chords; you can simply play the fretted notes and leave out the open strings from the chords (in the case of the Fmaj9 chord play the top three strings). With higher gain you will find these chords sound quite mushy and undesirable because of all the harmonic information clashing together.

Chords shapes starting from **sixth string:**

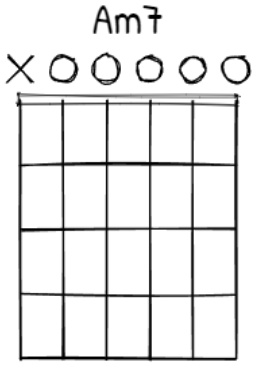




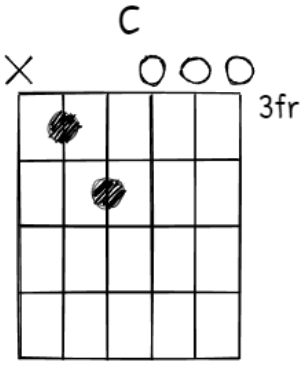
Chords shapes starting from **sixth string (triads)**:



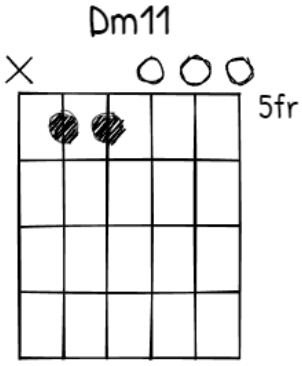
Chord shapes starting from **fifth string:**



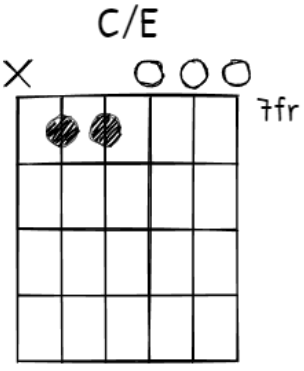
vi



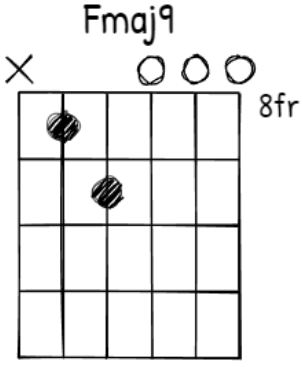
I



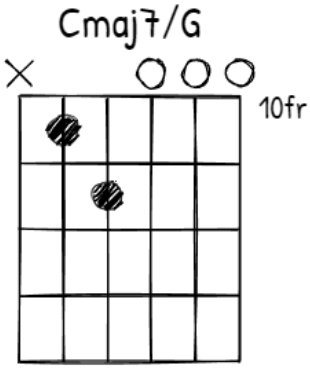
ii



I (1st inversion)

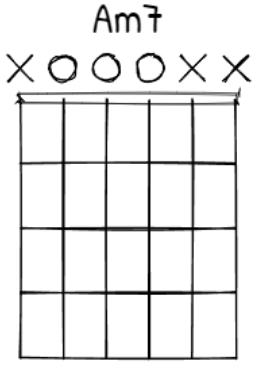


IV

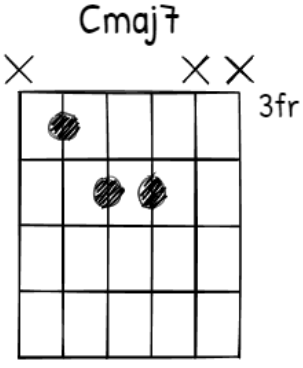


I (2nd inversion)

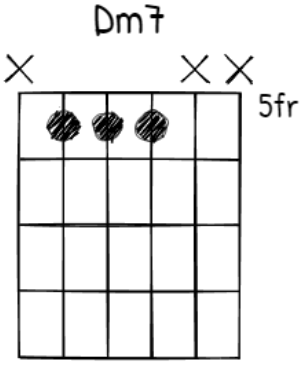
Chord shapes starting from **fifth string (triads):**



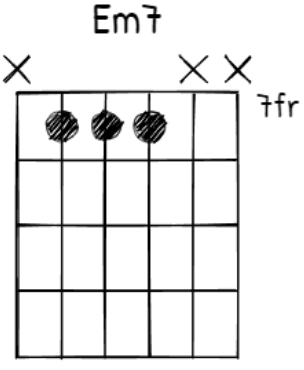
vi



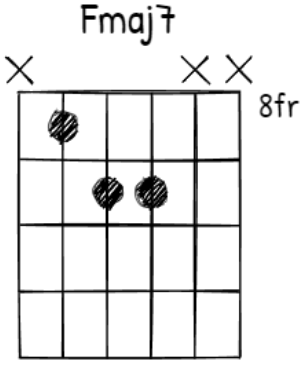
I



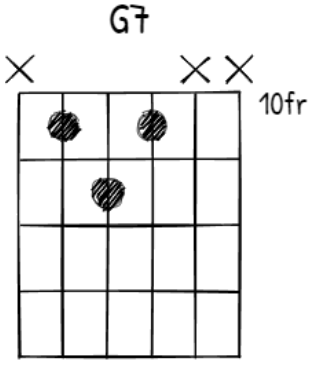
ii



iii



IV

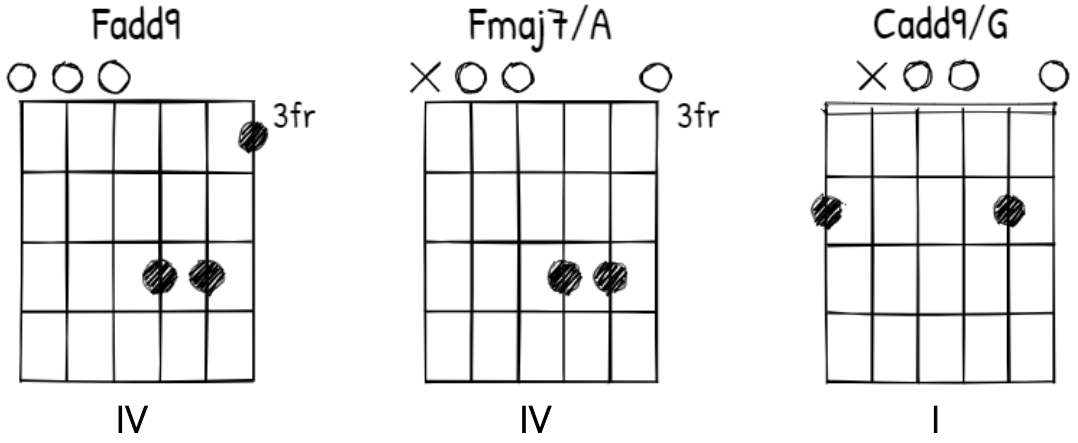


V

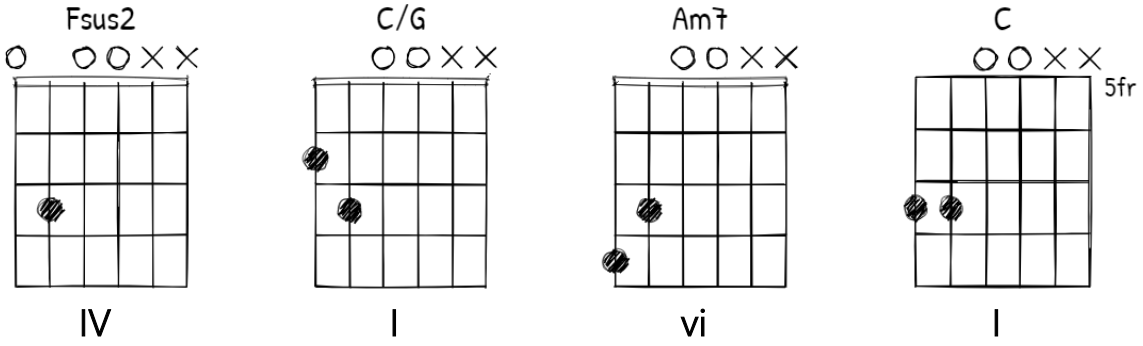
FACGCE Chord Progressions in Context

Below, you will find chords and chord progressions from popular songs that use the FACGCE tuning. It is important to note that not all these chords are meant for strumming; some were originally played with picking patterns (arpeggiated). However, they can still serve as fantastic sources of inspiration for your own musical ideas!

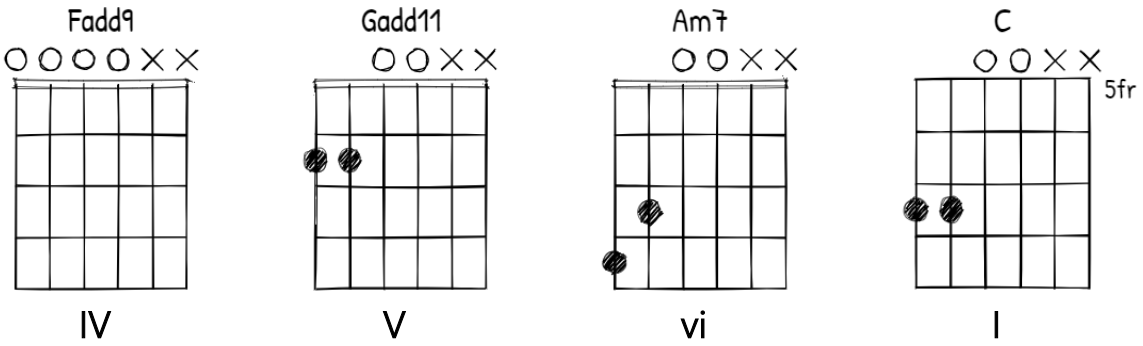
Covet – Parachute



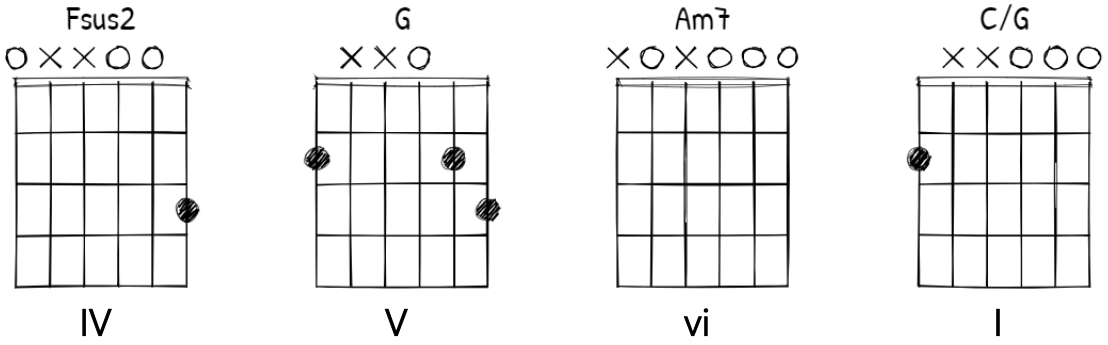
Tiny Moving Parts – You Lost Me



Standards – Smile



American Football – Never Meant

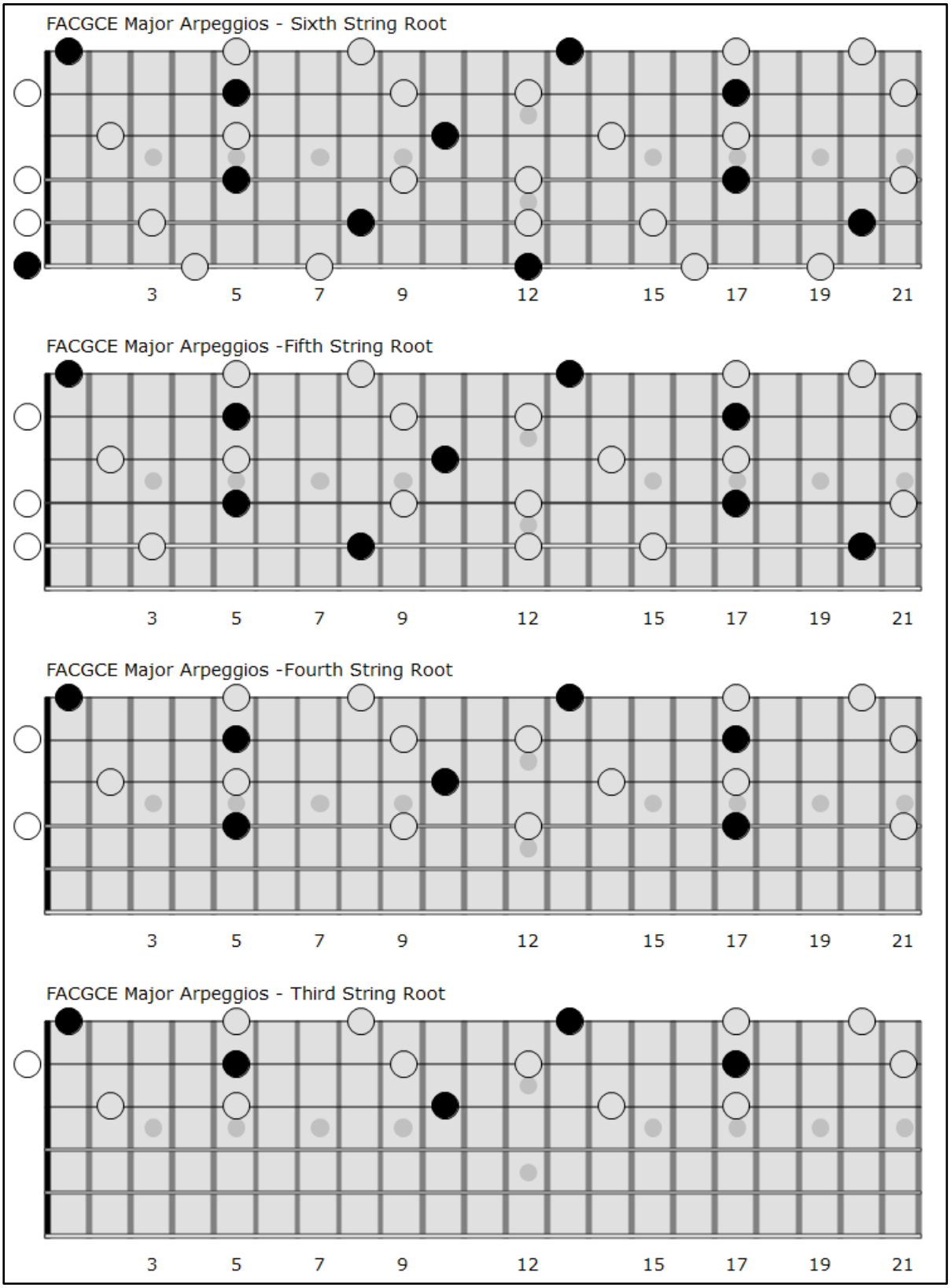


FACGCE Arpeggios

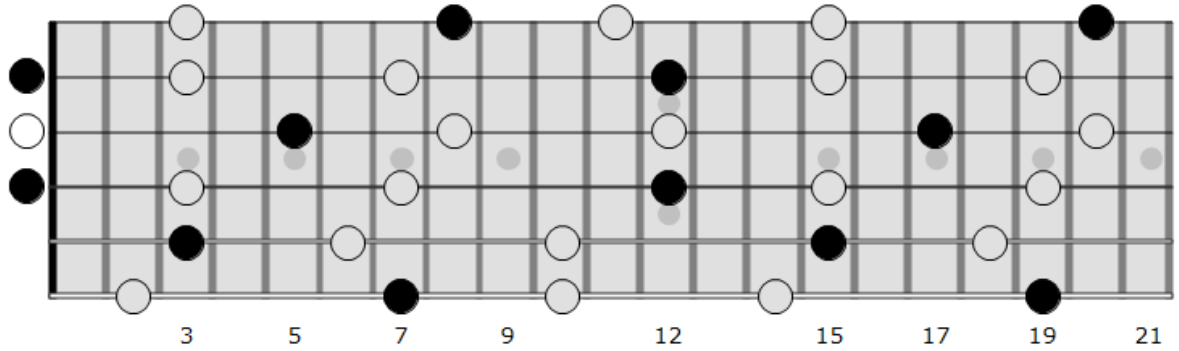
Below, you will find major triad (1, 3, 5) and minor triad (1, b3, 5) arpeggio shapes for FACGCE tuning. The major arpeggio shapes are based on F major, while the minor shapes are based on C minor. These shapes are movable, so you can adjust them based on the root of the major or minor chord you are using. Solid black notes on the charts indicate the root notes of each chord. For each chord, you will find three arpeggios spanning an octave to cover all possible inversions (root position, 1st inversion, and 2nd inversion).

As you study the charts, you will notice that the arpeggio shapes repeat when you reach the octave (usually around the twelfth fret). I have also marked where each arpeggio starts on each string. Although the root is positioned differently on each string, you will always have three arpeggios: the root triad and the two possible inversions.

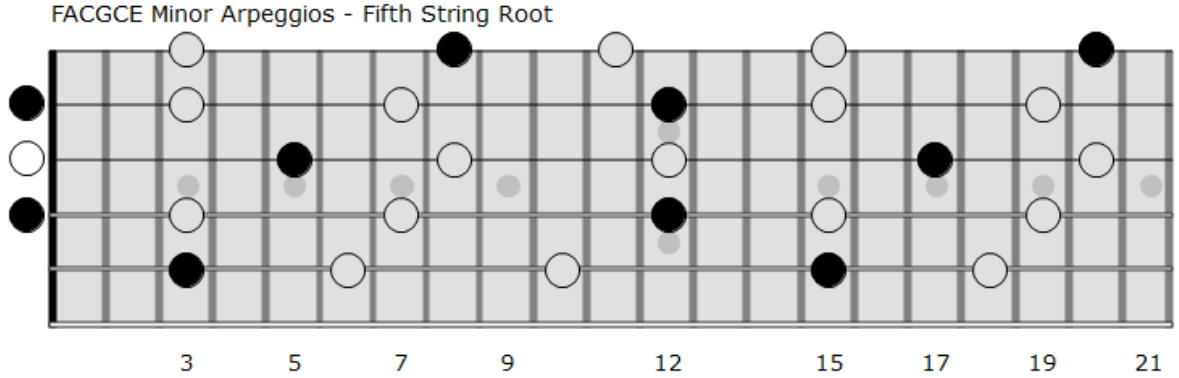
These arpeggios will come in handy when writing melody, riffs, song ideas, and so on, so it is worth investing the time to learn them. As you learn them, practice moving between different major and minor chord arpeggios to really ingrain them into your memory. I recommend identifying where the root arpeggio is on each string for the both the major and minor type as this will assist in learning them much quicker. If it is proving too challenging to learn the expanded shapes provided below, then try learning each arpeggio across three strings at a time. For example, from the sixth string, learn all three possible arpeggio shapes for the F, A, and C strings. Then repeat the process ascending through the other strings: ACG, CGC, and lastly, GCE.


FACGCE Major Arpeggios - Sixth String Root
3 5 7 9 12 15 17 19 21
FACGCE Major Arpeggios -Fifth String Root
3 5 7 9 12 15 17 19 21
FACGCE Major Arpeggios -Fourth String Root
3 5 7 9 12 15 17 19 21
FACGCE Major Arpeggios - Third String Root
3 5 7 9 12 15 17 19 21

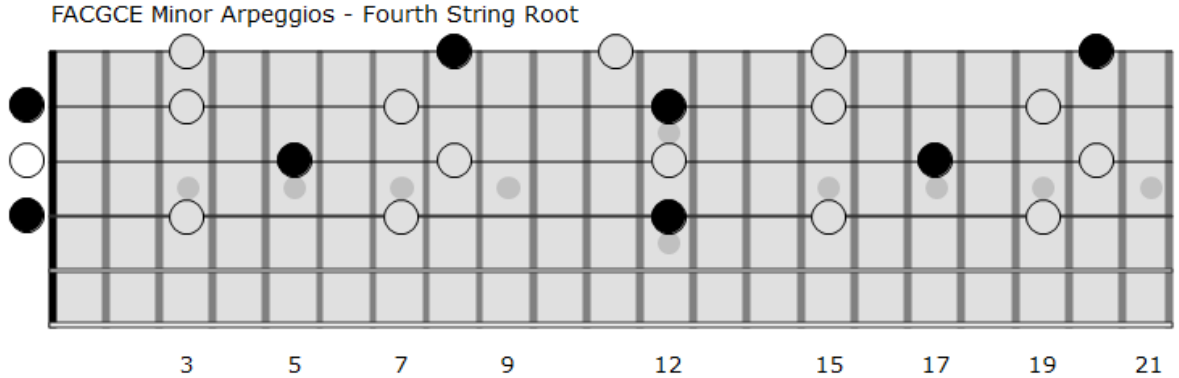
FACGCE Minor Arpeggios - Sixth String Root



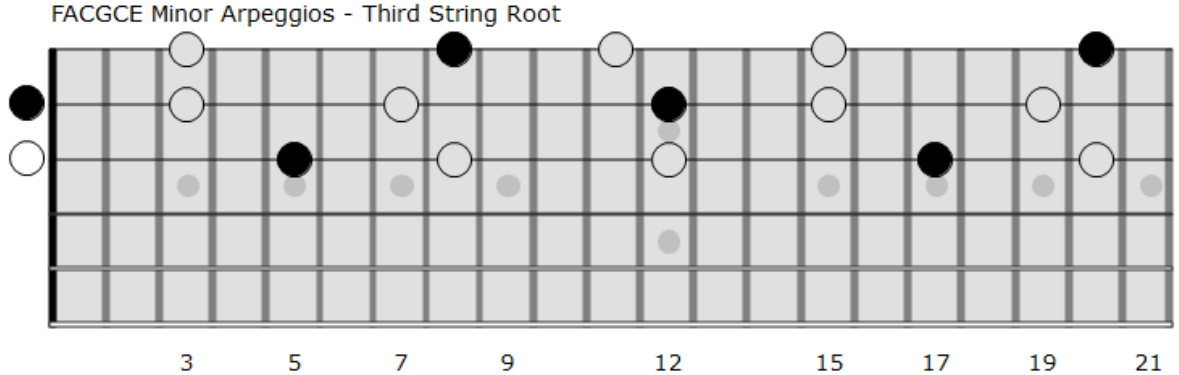
FACGCE Minor Arpeggios - Fifth String Root



FACGCE Minor Arpeggios - Fourth String Root

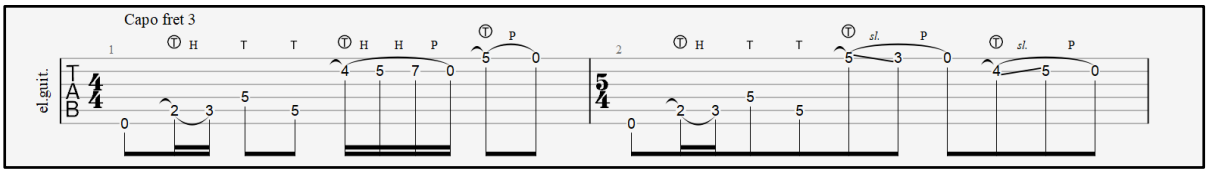


FACGCE Minor Arpeggios - Third String Root

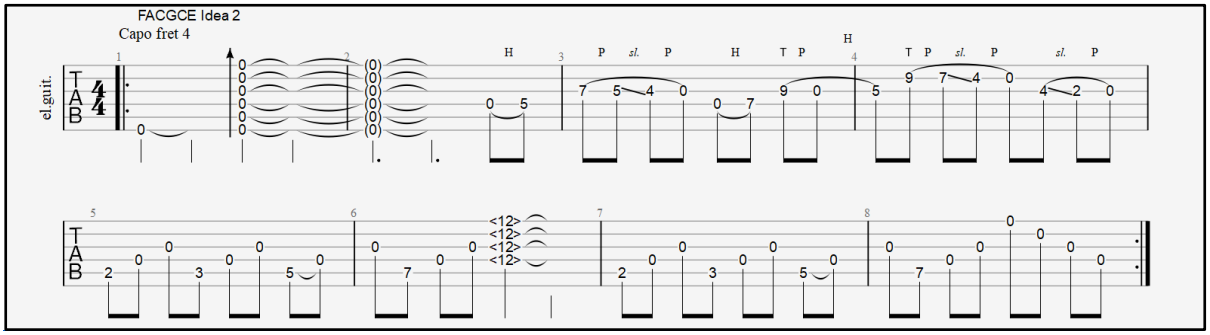


FACGCE Ideas in Context

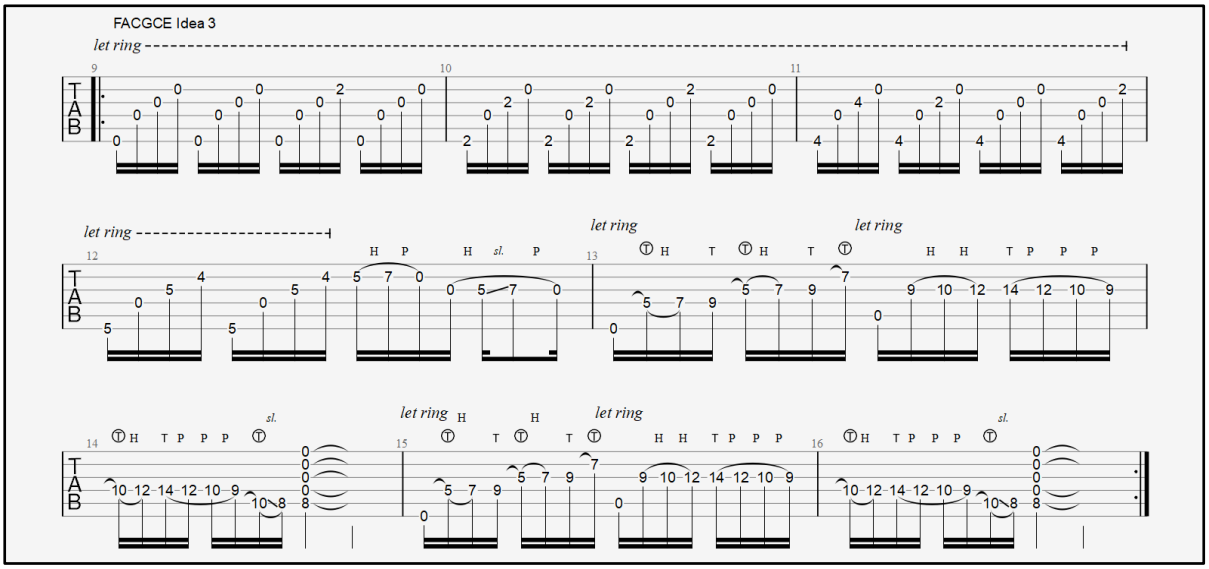
The following are five ideas I have written in FACGCE. I have included lots of noodling to get your brain working and to start noticing patterns and shapes available to you for writing licks in between your chords. Take your time with each one, and I recommend taking it phrase-by-phrase and build up the idea till you can play it at a slow pace. I have included video file links for each idea so you can get a better understanding of each of them. Examples two through five are all played with a capo applied to the fourth fret.



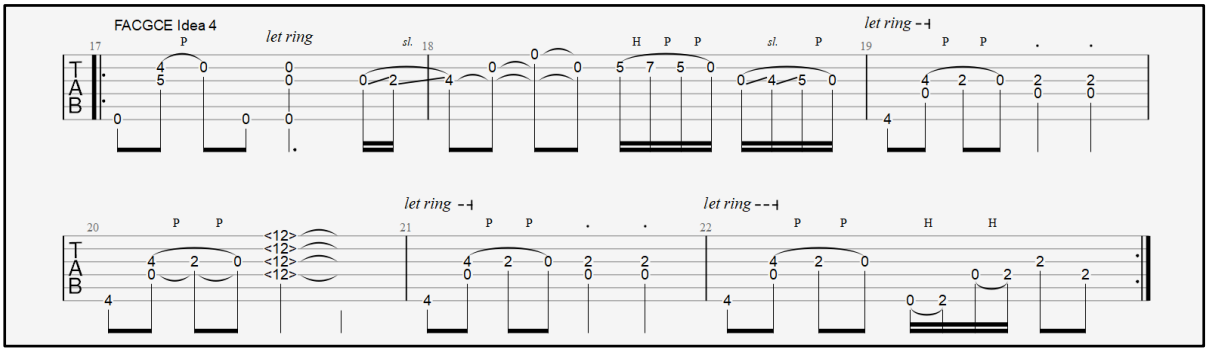
FACGCE idea 1



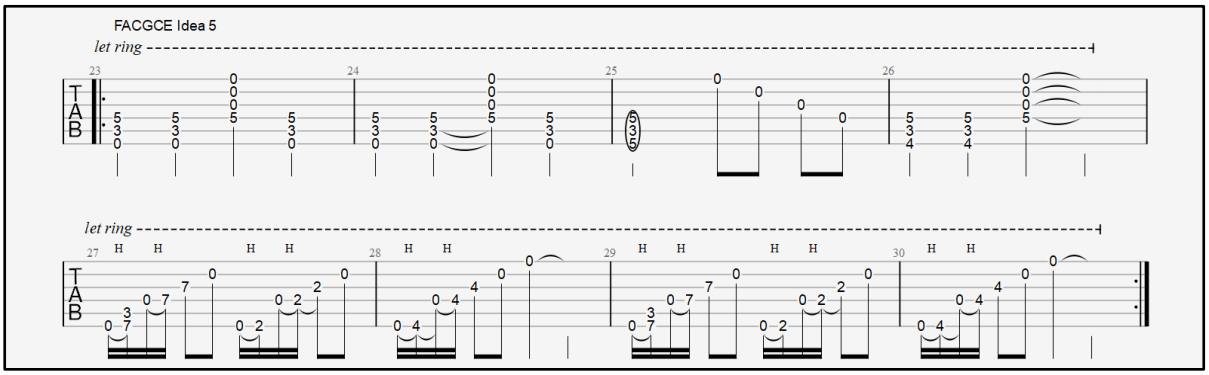
FACGCE idea 2



FACGCE idea 3



FACGCE idea 4



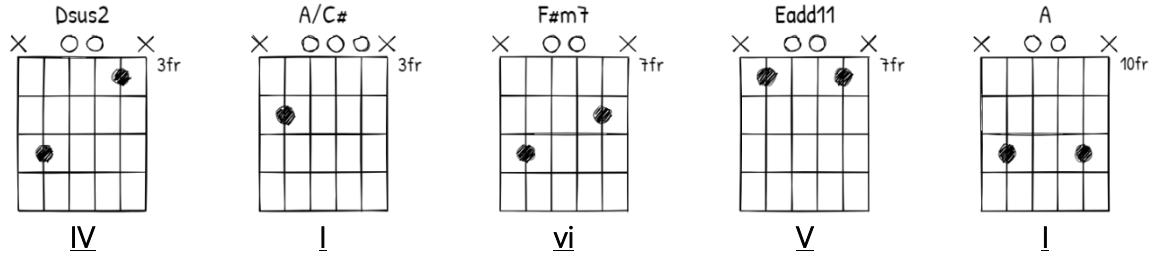
FACGCE idea 5

Chords for DAEAC#E Tuning

Just like FACGCE tuning, I want to share chords that take full advantage of the unique sound DAEAC#E tuning offers. However, this tuning differs significantly from standard tuning, meaning it is even less aligned with standard tuning, so it is much more difficult to find chords that align with every note in a key signature. So, I take a slightly different approach by providing approachable chords and chord progressions that make the most of the open strings. I also offer several examples of chord progressions to illustrate how they can work harmoniously together. Try playing through these chord shapes and progressions to discover what you like. Then, I encourage you to create your own progressions and explore possible licks and riffs that can complement them (the arpeggio section below will help you generate ideas). These chords sound great when played cleanly or with a touch of light gain. If you want to change the key, simply use a capo. Additionally, if you wish to use these chords with more distortion, you can play the fretted notes and exclude the open strings from the chords. However, keep in mind that this may not work for every chord shape, and that some chord shapes may even work using the full shape.

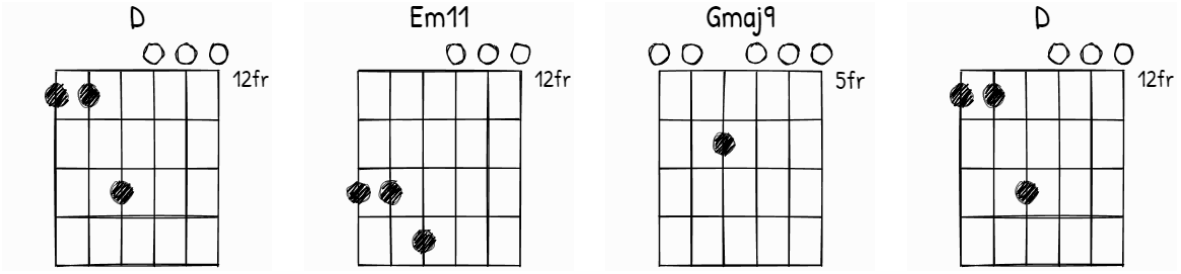
DAEAC#E Chords and Progressions

'For the Sake of Brevity Style Chord Progression Idea'



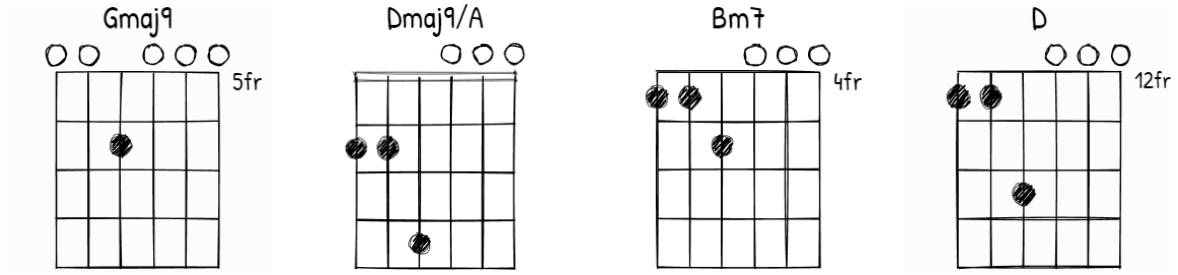
IV I vi V I

'Night of The Pencils Chord Progression Idea' (Capo 5th Fret)



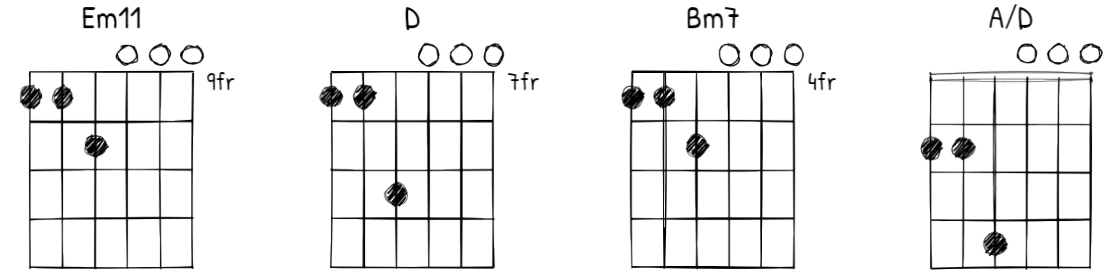
I ii IV I

'Upbeat Midwest Emo Style Chord Progression' (Capo 5th Fret)

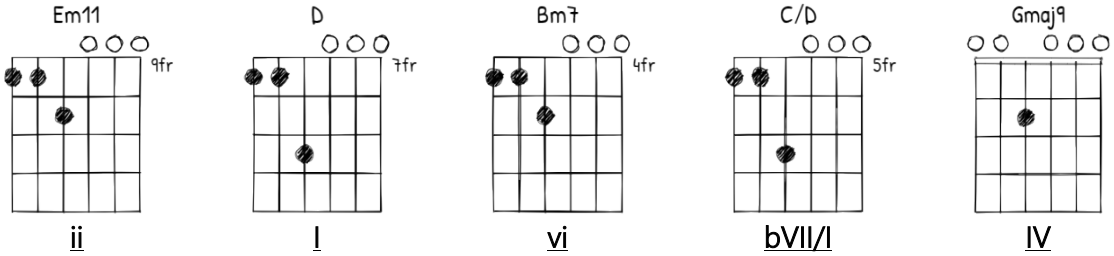


IV I/V vi I

'Descending Style Chord Progression with A Twist' (Capo 5th Fret)



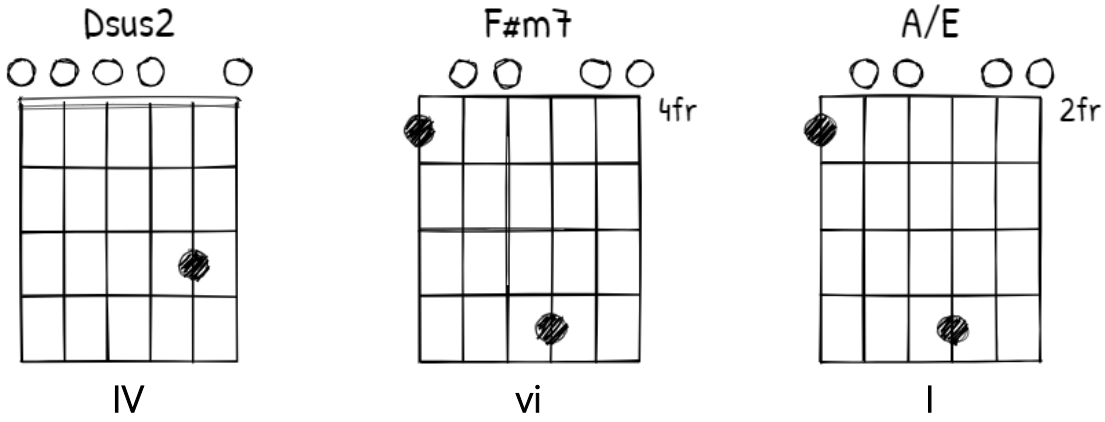
ii I vi V/I



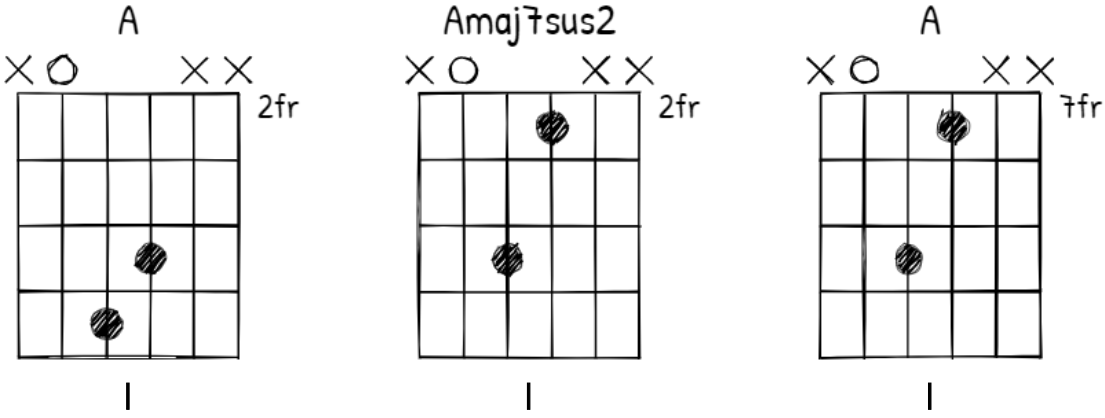
DAEAC#E Chord Progressions in Context

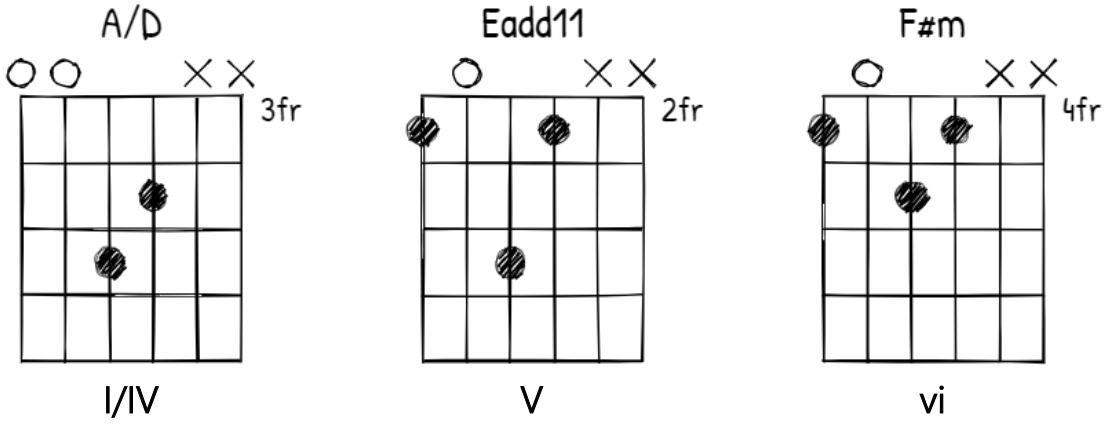
The chords and chord progressions below are derived from some of the most popular songs that utilize this tuning. Consider the chords above as a reliable foundation and use the following chords to spark more ideas for chords and chord progressions.

Tiny Moving Parts – Always Focused

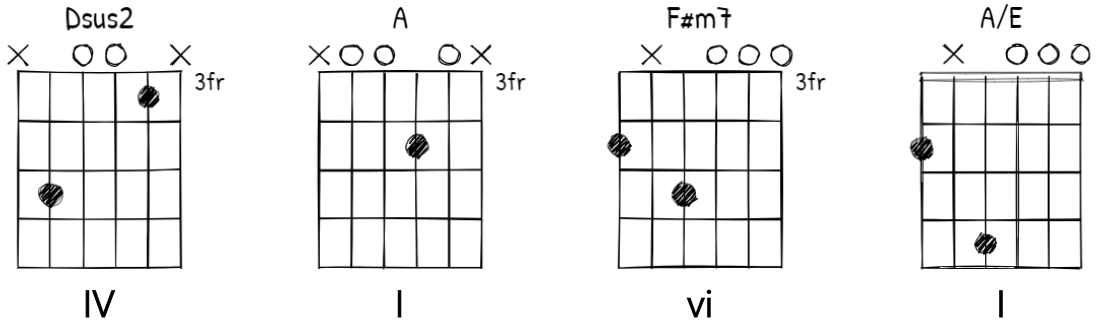


Tiny Moving Parts – Caution

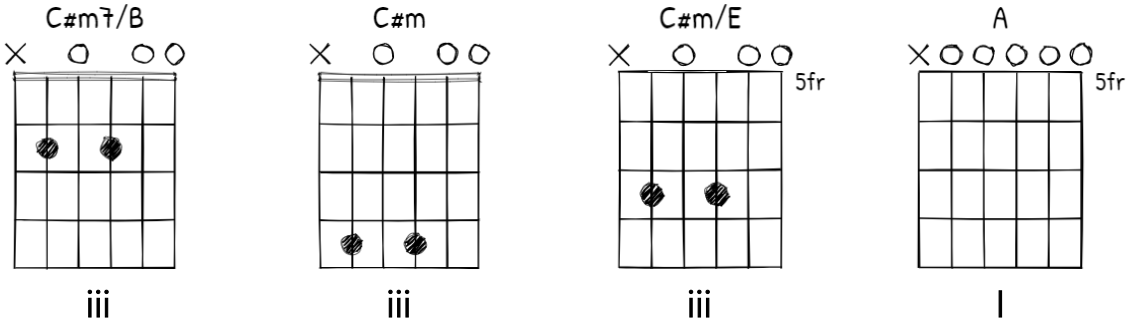




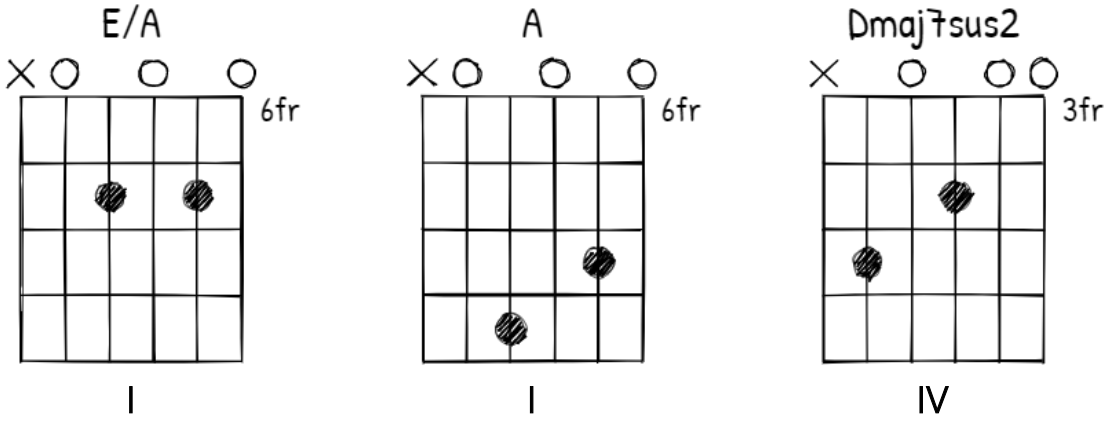
Tiny Moving Parts – Dakota

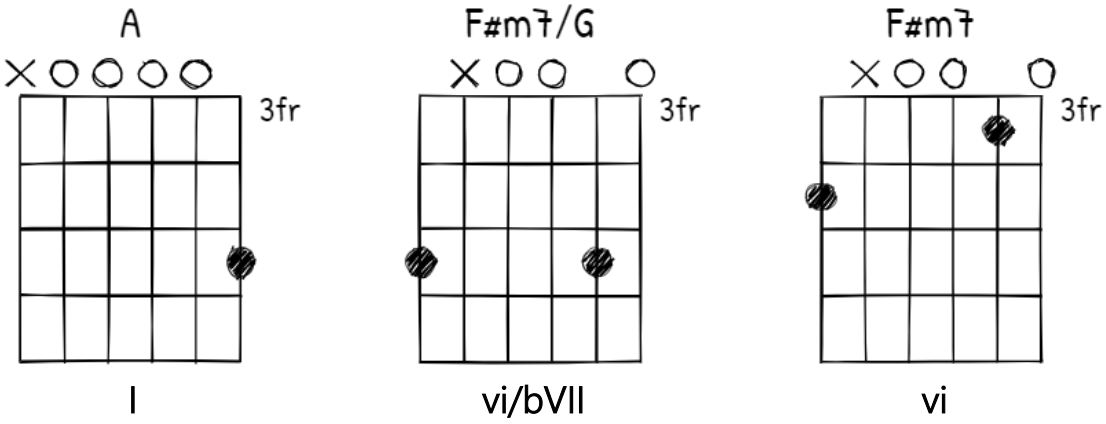


Tiny Moving Parts – Sundress

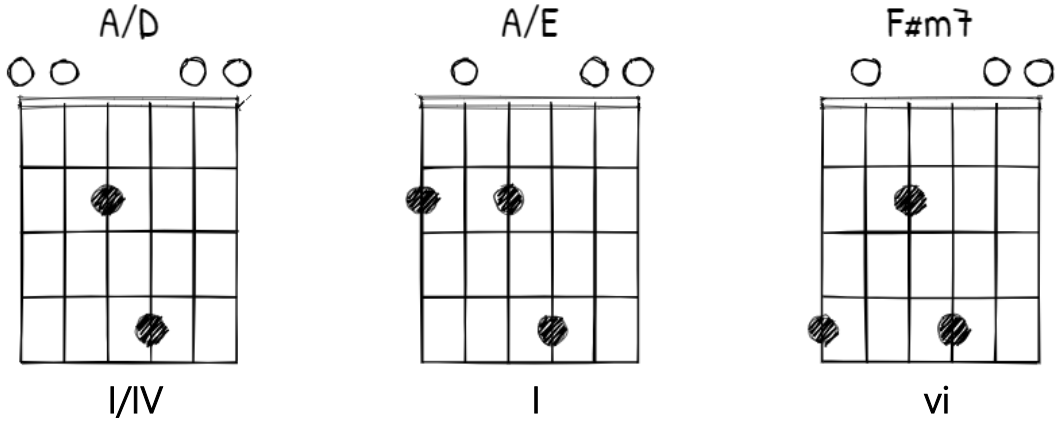


Algernon Cadwallader – Spit Fountain (implied progression in verse & pre-chorus)

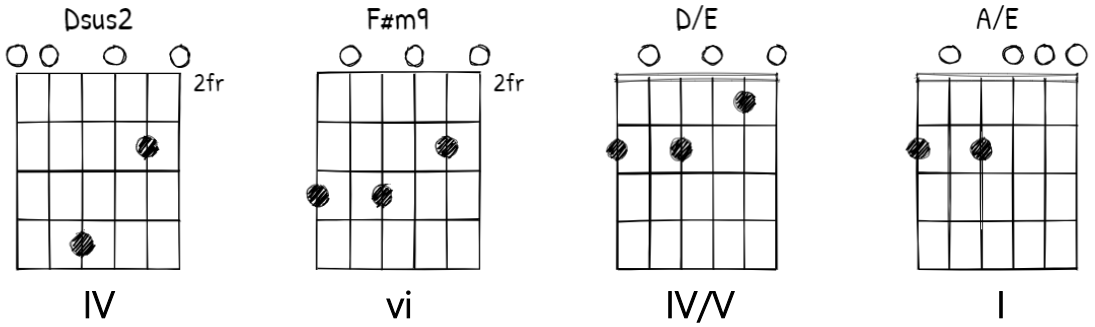




Sleepy Dog – Screamers

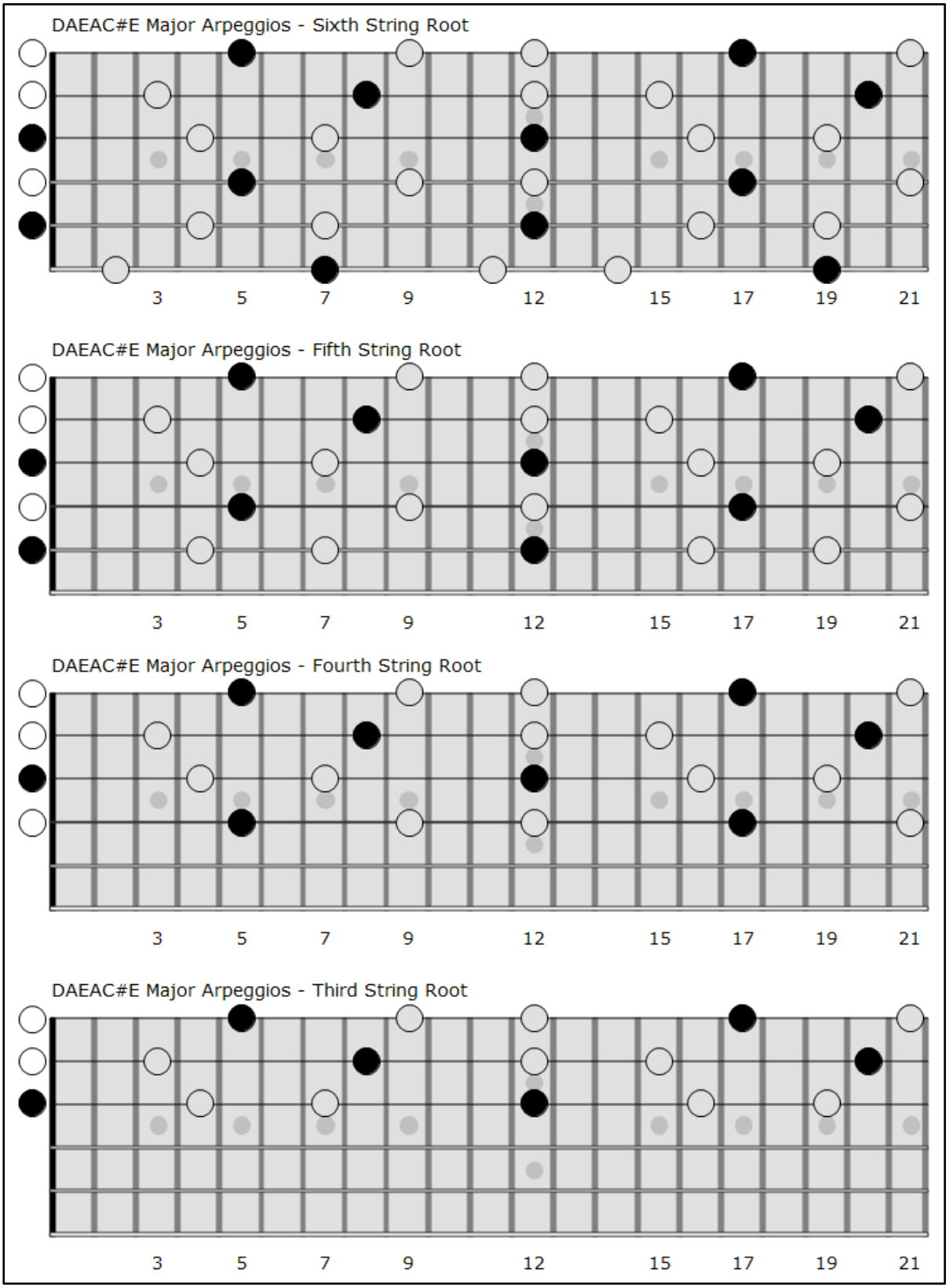


Mountains - Stride

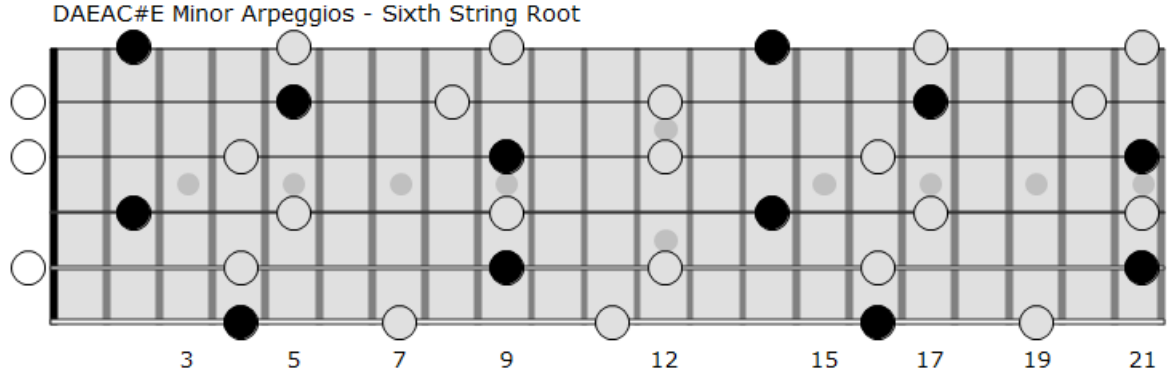


DAEAC#E Arpeggios

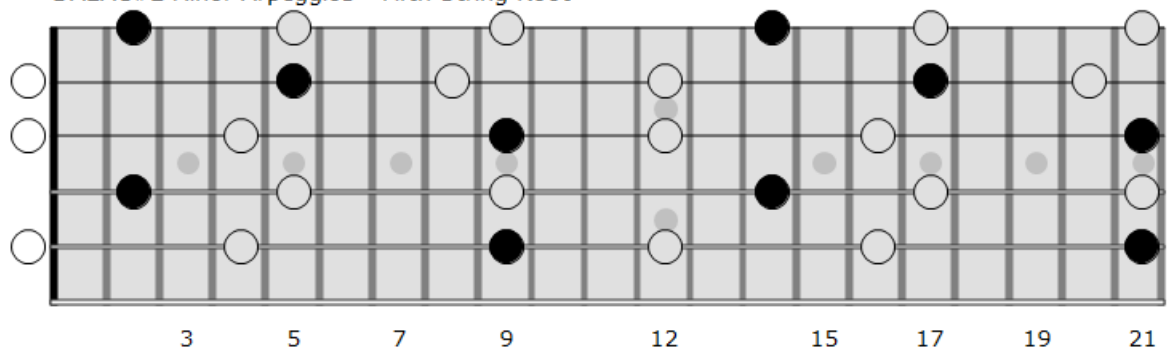
Here are major triad (1, 3, 5) and minor triad (1, b3, 5) arpeggios for DAEAC#E tuning. The provided major arpeggio shapes are in A major, and the minor shapes are in F# minor. These shapes can be shifted to match the root of the major or minor chord you are using. You will find solid black notes on the charts indicating the root notes of each chord. Each chord has three arpeggios spanning an octave to cover all possible inversions (root position, 1st inversion, and 2nd inversion). When you study the charts, you will notice that the arpeggio shapes repeat at the octave, typically around the twelfth fret. I have also indicated where each arpeggio starts on each string. Keep in mind that the root position varies per string, but there are always three arpeggios: the root triad and the two possible inversions.


DAEAC#E Major Arpeggios - Sixth String Root
3 5 7 9 12 15 17 19 21
DAEAC#E Major Arpeggios - Fifth String Root
3 5 7 9 12 15 17 19 21
DAEAC#E Major Arpeggios - Fourth String Root
3 5 7 9 12 15 17 19 21
DAEAC#E Major Arpeggios - Third String Root
3 5 7 9 12 15 17 19 21

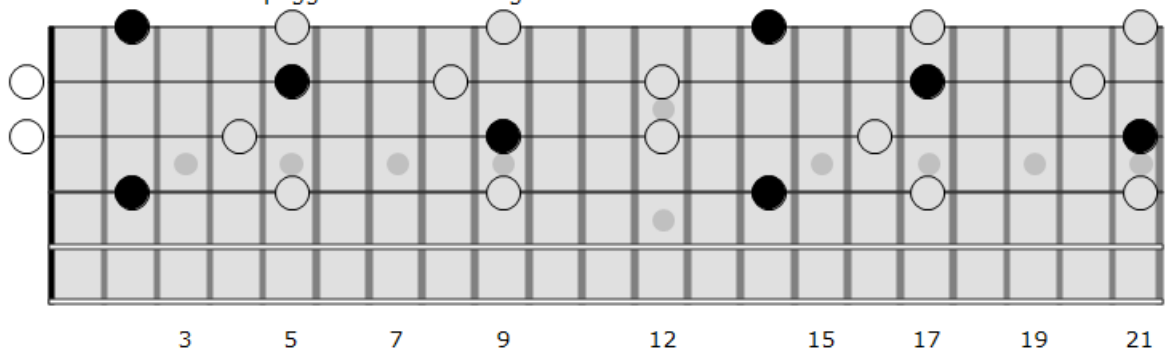
DAEAC#E Minor Arpeggios - Sixth String Root



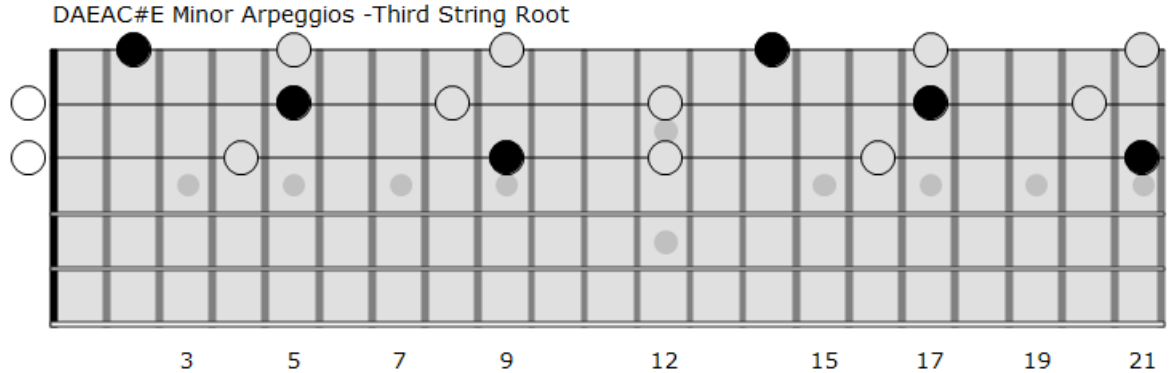
DAEAC#E Minor Arpeggios - Fifth String Root



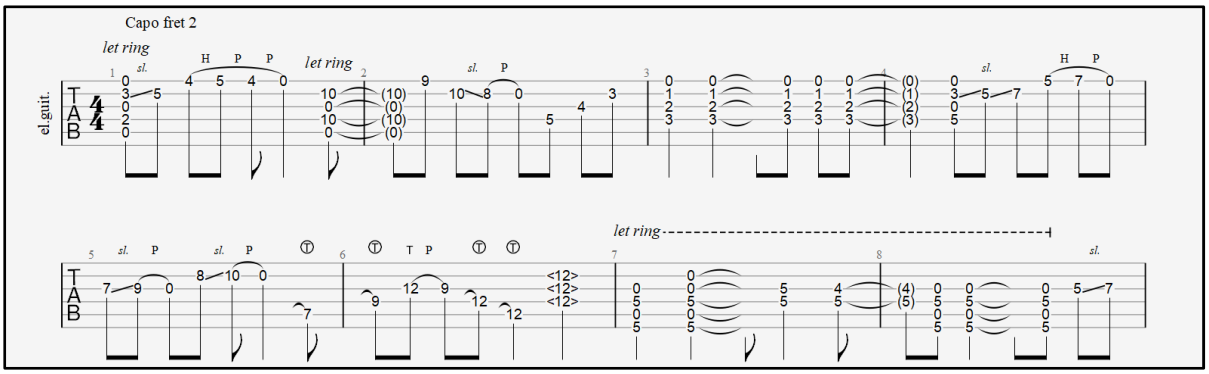
DAEAC#E Minor Arpeggios -Fourth String Root



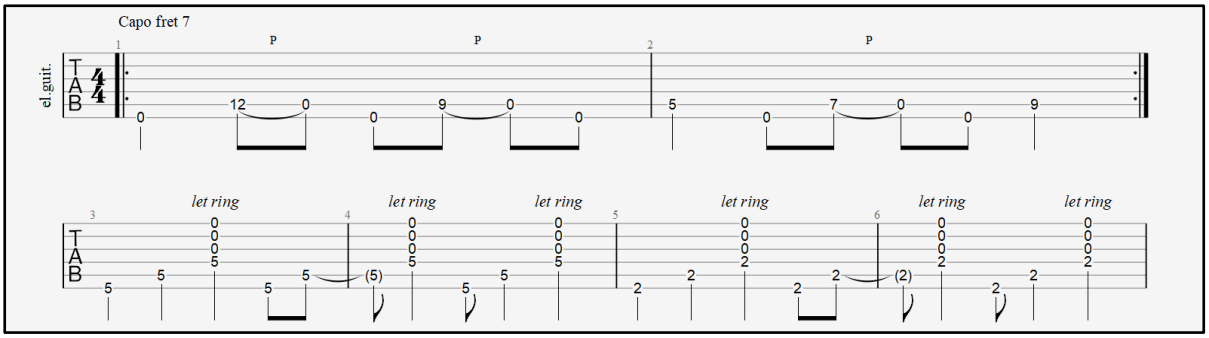
DAEAC#E Minor Arpeggios -Third String Root



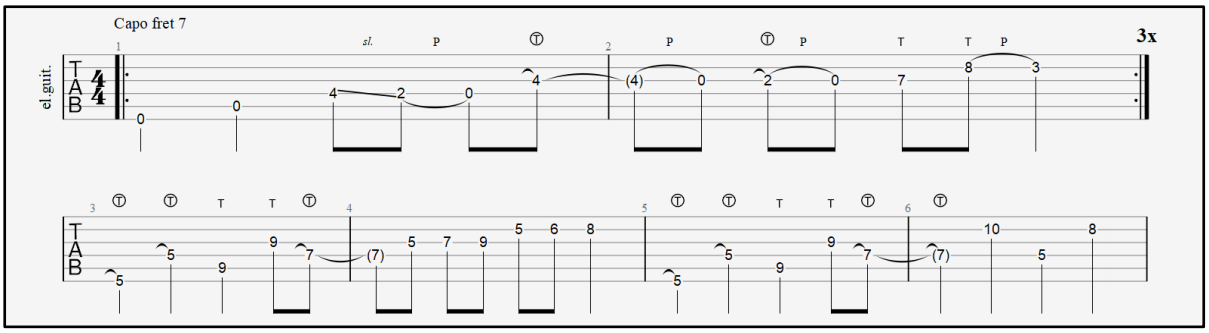
DAEAC#E Ideas in Context



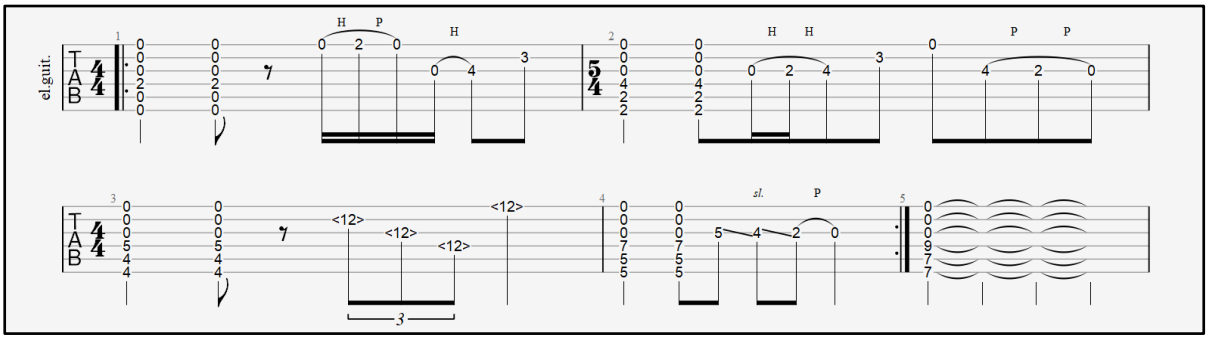
DAEAC#E idea 1



DAEAC#E idea 2



DAEAC#E idea 3



DAEAC#E idea 4

Bonus Content

Congratulations! You have successfully grasped the fundamental concepts necessary to begin crafting math rock-inspired guitar ideas. Your dedication and effort are commendable, and I want to express my gratitude for making it this far. The bonus content will delve into some of the challenges you are likely to encounter next, offering valuable insights and guidance to further enhance your skills. Keep an eye out for the next piece of content, which will aid you in your quest to learn, write, and play math rock guitar music. Thank you.

Tools for Writing

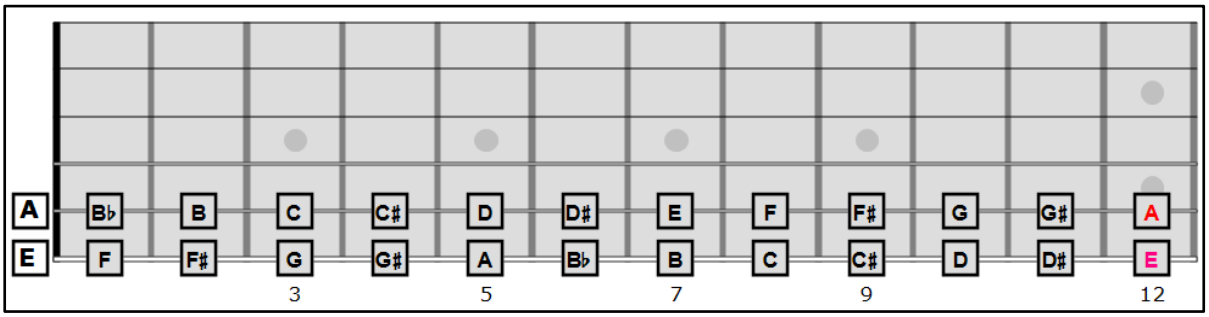
Up to this point, we have explored guitar techniques commonly associated with math rock, delved into the basics of music theory, acquired practical chords and chord progressions, and embarked on our exploration of alternative tunings. In the upcoming section, you will discover an array of valuable tools, methods, and creative approaches for composing music. These insights and techniques are insights into my own guitar journey, and my aim is to share them with you in hopes of providing you with a clear sense of purpose and direction in your playing. You will also gain a versatile set of tools to aid you in crafting songs, riffs, and musical ideas.

Segment 3A – Mastering the Fretboard

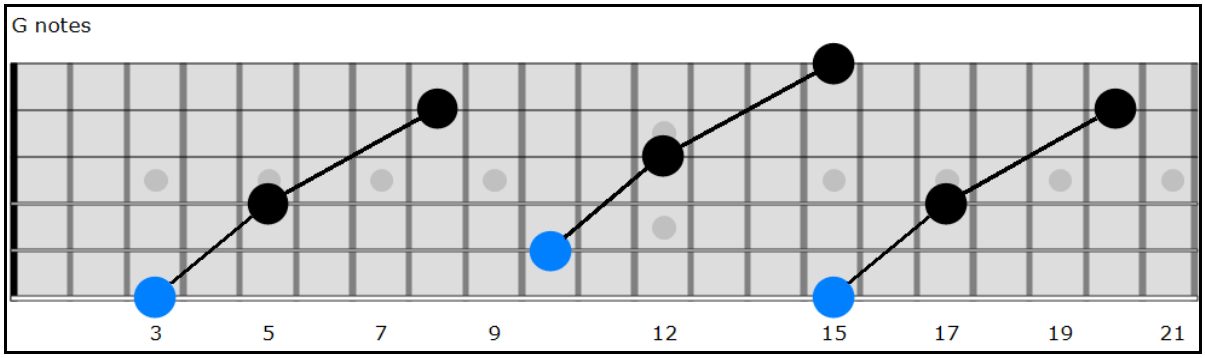
There are many methods out there to help you become familiar with note names and their locations on the neck: arpeggios, scales, movable chord shapes, chord shapes starting from different strings, the list goes on. I encourage you to try out different methods to find what works best for you. The following are methods I found rather useful for finding my way around the guitar neck more easily, more logically, and helped me on my way to mastering the fretboard.

Finding Notes Quickly – The Octave Method

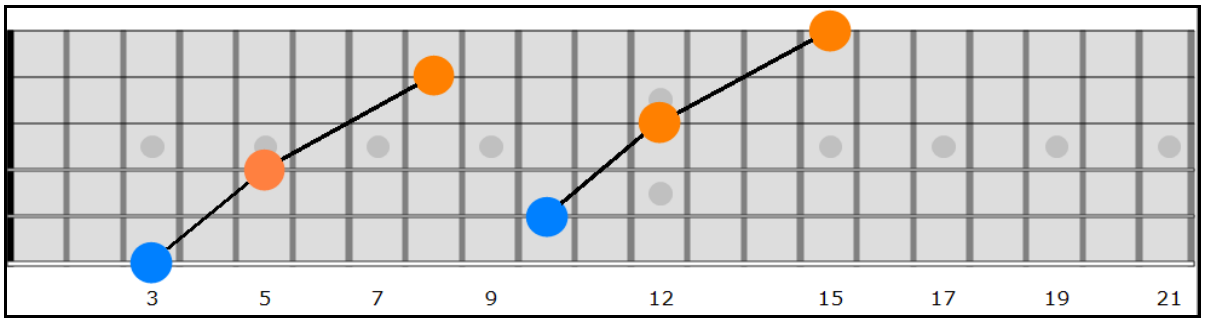
This method is a useful way to quickly locate notes all over the fretboard. It proves valuable for identifying root notes in chord shapes, arpeggios, scales, and more. I call it the "octave method," although it might have another name. Initially, focus on learning the positions of notes on the Low E and A strings (in standard tuning). While this may seem daunting if you are new to it, with practice, it becomes less challenging than you might expect. I have listed the notes starting from the open E and A strings up to their respective octaves on the twelfth fret (beyond the twelfth fret, you'll find the same notes but an octave higher).



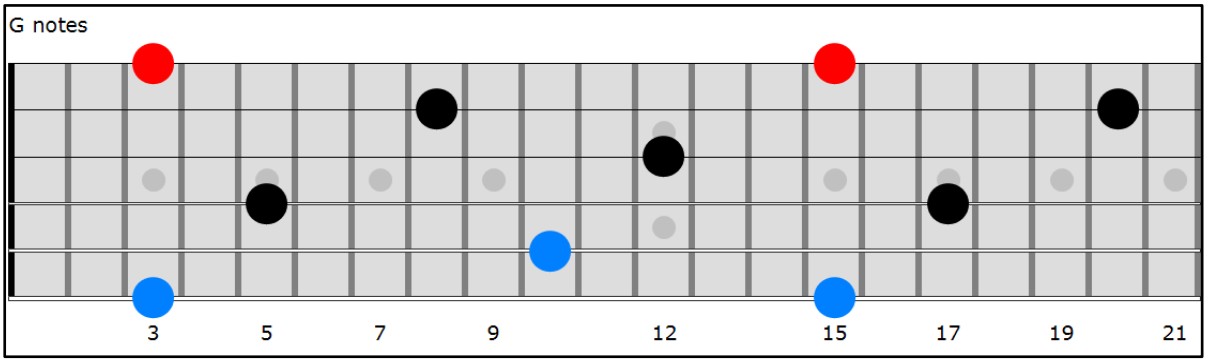
To illustrate how to apply this concept, I have provided a diagram. I have marked the note G on the low E and A strings with blue circles. Additionally, I have indicated the octaves of these notes (the same note but higher in pitch) with black circles. This can help you visualize how notes and their octaves are located on the guitar neck.



Notice how they have a particular spacing between each octave. From the G note (third fret) on the low E string, we jump a string (the A string) and find the octave G three frets higher (fifth fret on the D string). The same applies for the note G played on the A string. You may be familiar with these octave shapes if you have ever learned a pop punk or punk song. However, the spacing does change slightly when we start from the G note on the D string, and the same for the G note starting from the G string, the spacing is now four frets instead of three. The orange notes in the example below highlight this. I would not worry about getting this to stick in your mind because when played correctly you will hear them as octaves. If you play three fret spaces instead of four for octaves starting on the D and G strings you will get this major seven interval that clashes in sound, so you will instantly realize you need to slide your higher note up one fret to get the octave sound.



Another handy trick is to notice that the low E and high E strings follow the same note order. Meaning you can mirror any note you play on either string to find the same note. I have indicated this with the red notes on the chart below.



This system, I believe, is suited to players who have yet to have a built-up knowledge of note locations on the fretboard, so by following this method, you will slowly but surely start building your knowledge of note names and where they are located across the fretboard.

Visualizing the Neck in Shapes - The Arpeggio Method

By learning this method, you will be able to conjure up something both quickly and easily anywhere on the neck. Even something to play over fast changing or complex chord changes, and if time permitted you can really write something you will be proud of. Not only will it help you achieve the former, but it will also provide another useful approach to writing riffs, licks, melody lines, and so on. How and why this method will help you will become clearer as you work through this section.

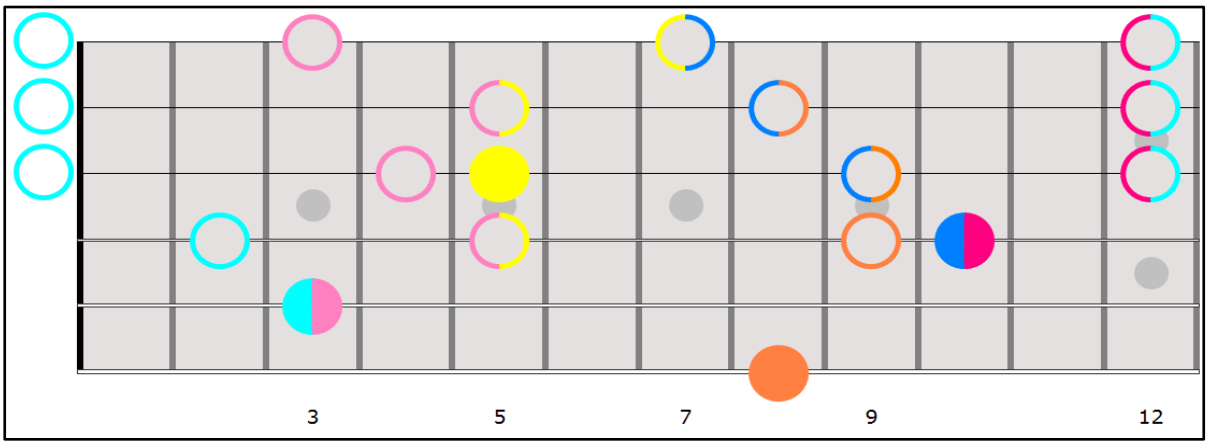
In this section, four types of extended arpeggios will be presented: major seven, minor seven, dominant, and minor seven flat five. Compared to arpeggio shapes you may have already learned; I will be taking a slightly different approach by teaching you arpeggios shapes that combine to form a pattern across the fretboard that you can easily learn. Also, I have included examples of how you could apply them.

Extended seven arpeggios give us an extra note to work with when compared with basic chord arpeggios (refer to the theory section for more detailed information) and will 'fit' over any iteration of a particular unaltered chord type. For example, a C major 7 arpeggio will work over a C major chord, C sus 2 or sus 4 chord, a C major 6 chord, C major 7 chord, C major 9 chord, and so on. If the chord or implied harmony is not altered, you should be safe to use it. A Cmaj7b5 chord is an example of an altered chord, the flattened 5th changes the quality of the chord, meaning it would NOT be harmonically sound if you were to play a Cmaj7 arpeggio along with it. Chords that contain notes such as b9, #9, b5 #5 are referred to as being altered chords, there is a lot to unpack there, but just think of it as making a chord sound more 'out there' in terms of sound.

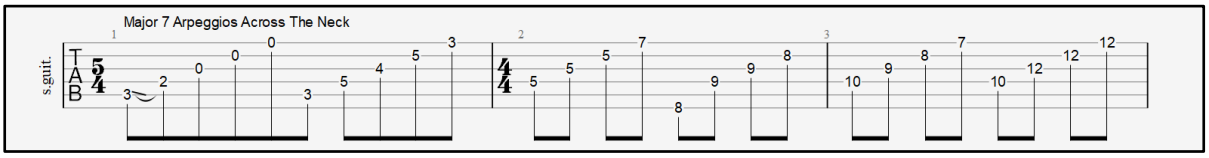
Major Seven Arpeggios Across the Fretboard

Below you will find major seven arpeggios spanning one octave, meaning that when you get to the twelfth fret on your guitar, you should repeat the same patterns again. I have colour coded each shape to show how they are connected to each other and so you can see each individual arpeggio (chord) shape. Take your time working through each shape and find the most

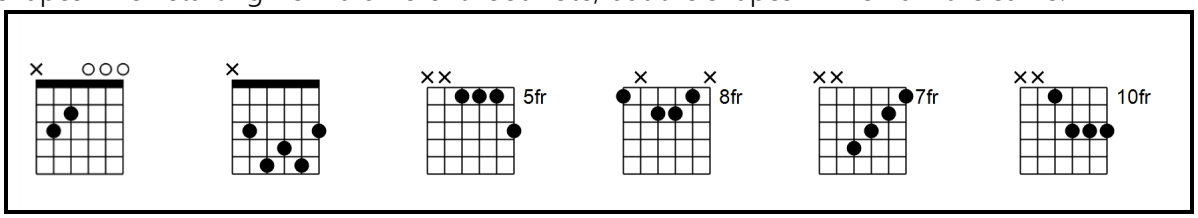
comfortable fingering, next, experiment with stringing them together into a chain, like in the transcribed version below, to help ingrain them into your memory and muscle memory.



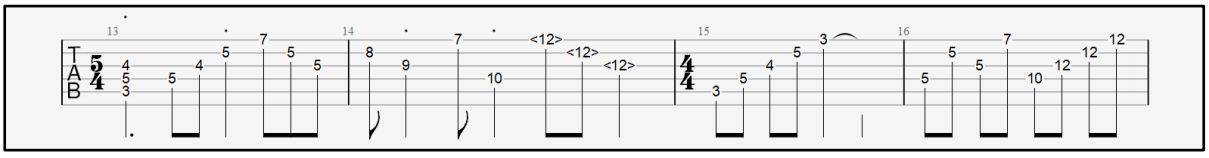
Here is how they look transcribed for better clarity.



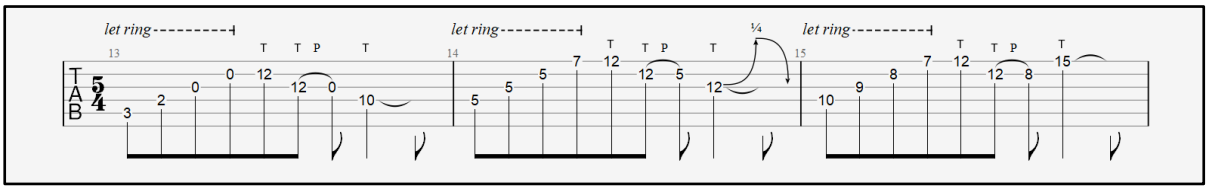
Each individual chord shape used for the arpeggios are listed below. You can use any of these chord shapes interchangeably for major seven chords. Meaning, you will have to adjust the shapes when starting from a different root note, but the shapes will remain the same.



The following are a couple of examples of ideas you could potentially write. You could extend the examples here, but you will soon run into problems of things sounding too similar, thus these arpeggios are best suited to being combined with other arpeggios, scales and so on. At the end of this section, I will show examples of how you can combine a chain of arpeggios together for some musical phrases.



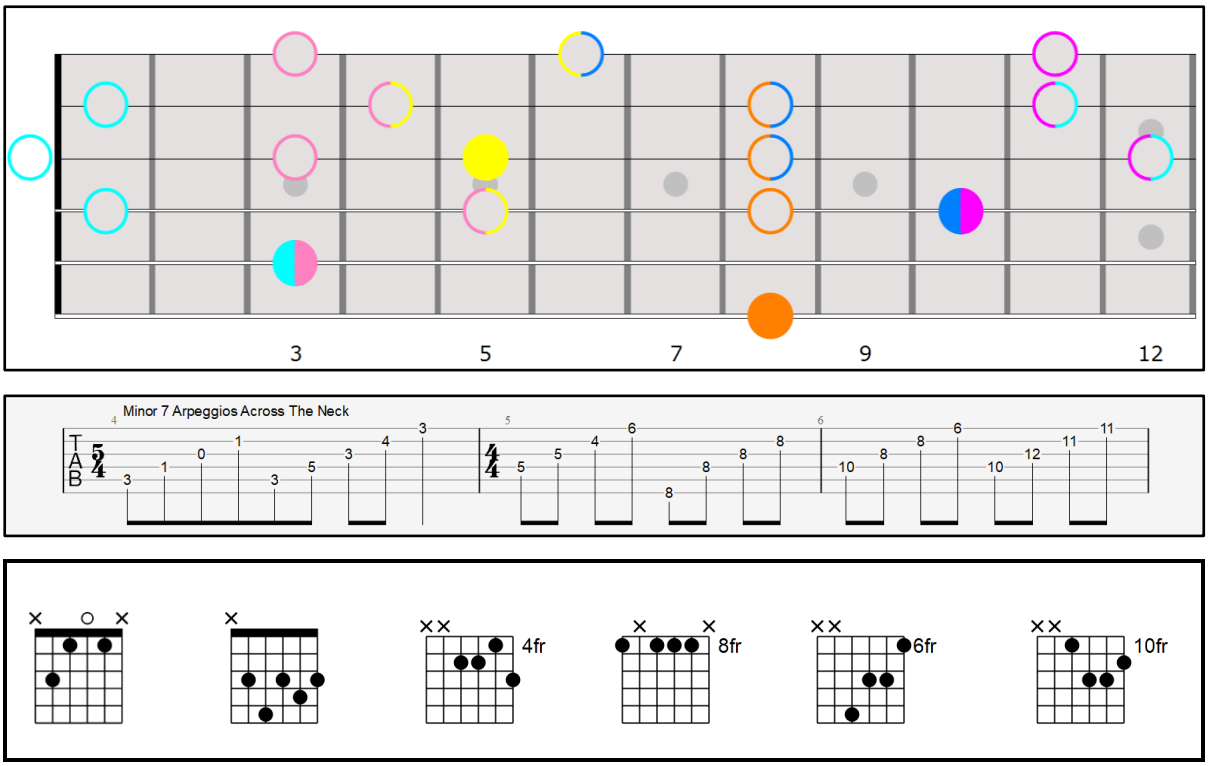
4A - Major 7 Arpeggios Example 1



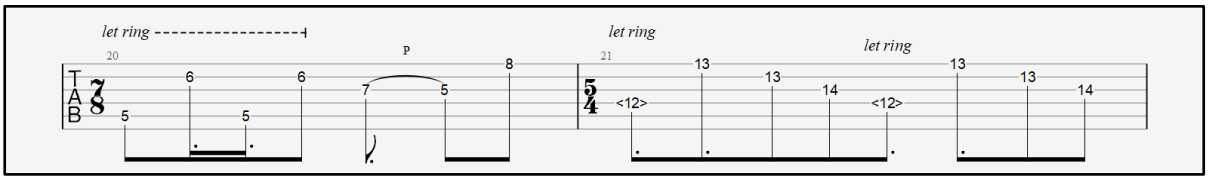
4B - Major 7 Arpeggios Example 2

Minor Seven Arpeggios Across the Fretboard

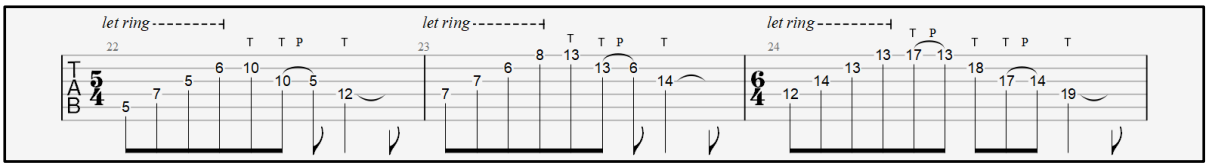
Moving on to minor seven arpeggios, again they span one octave, meaning when you get to the twelfth fret, repeat the same patterns again. I have colour coded each shape to show how they are connected and so you can see each individual arpeggio (chord) shape. Take your time to work through each shape and find the most comfortable fingering for you, next, experiment with stringing them together into a chain like in the transcribed version below to help ingrain them into your memory and muscle memory.



Again, here are a few examples of short ideas you could potentially write.



4C - Minor 7 Arpeggios Example 1



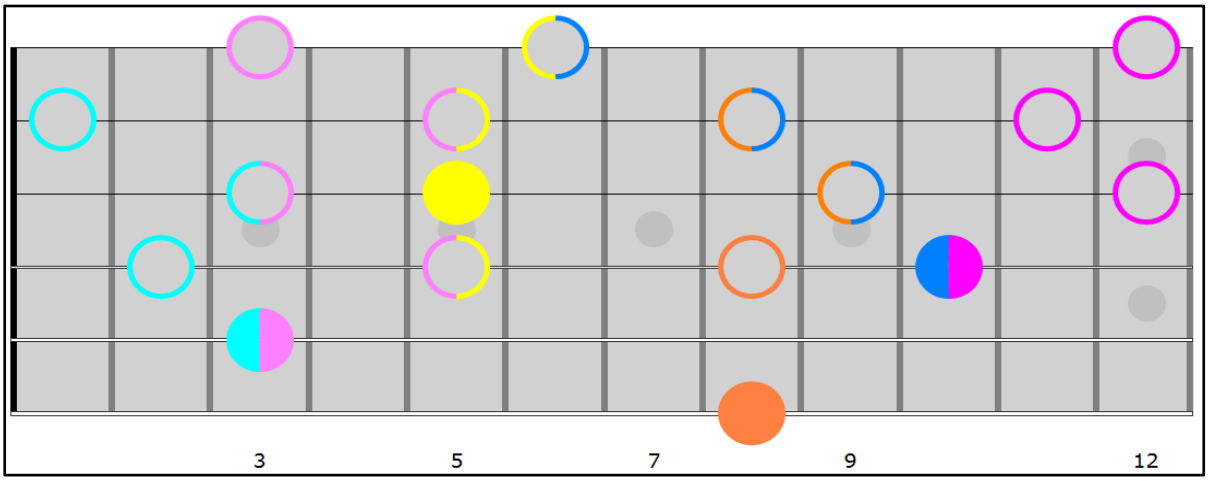
4D - Minor 7 Arpeggios Example 2

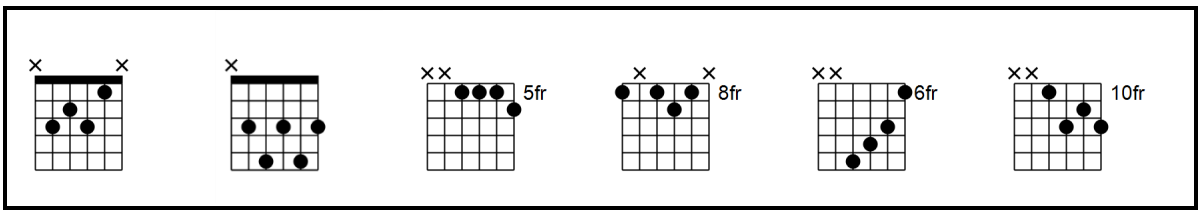
In example, 4D, there are three individual finger tapping licks (even though they are on the same stave, I see these as three individual licks). They are designed to show you what you can potential achieve when writing with these arpeggios. They are fairly limited to what you can do

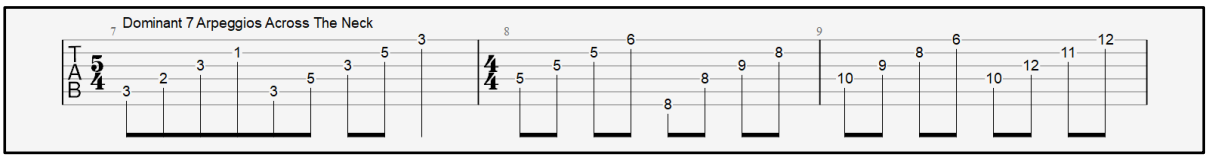
because they are the same four notes, therefore you could either use these short passages as a repeating phrase, or use other arpeggios, scales and so on in conjunction. It really depends on the context of the music and what you are aiming to achieve.

Dominant Arpeggios Across the Fretboard

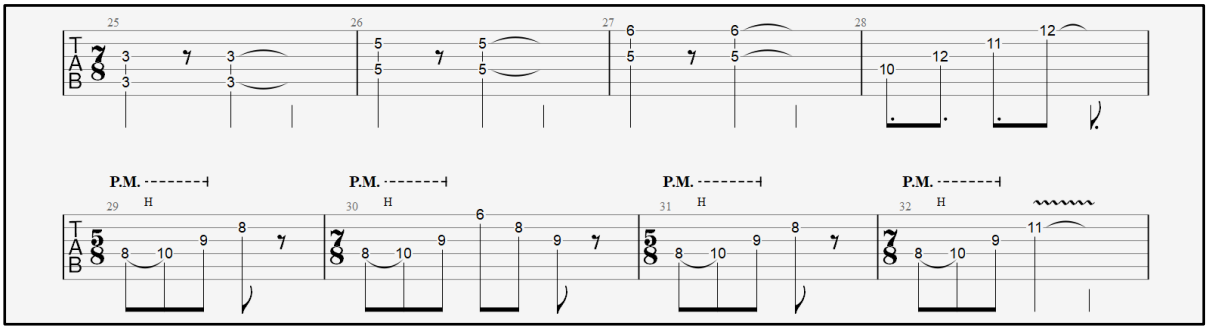
Next, we have dominant arpeggios spanning one octave. Again, I have colour coded each shape to show how they are connected so you can see each individual arpeggio (chord) shape. Take your time to work through each shape and find the most comfortable fingering for you. transcribed version below the former will aid in ingrain them into your memory and muscle memory.







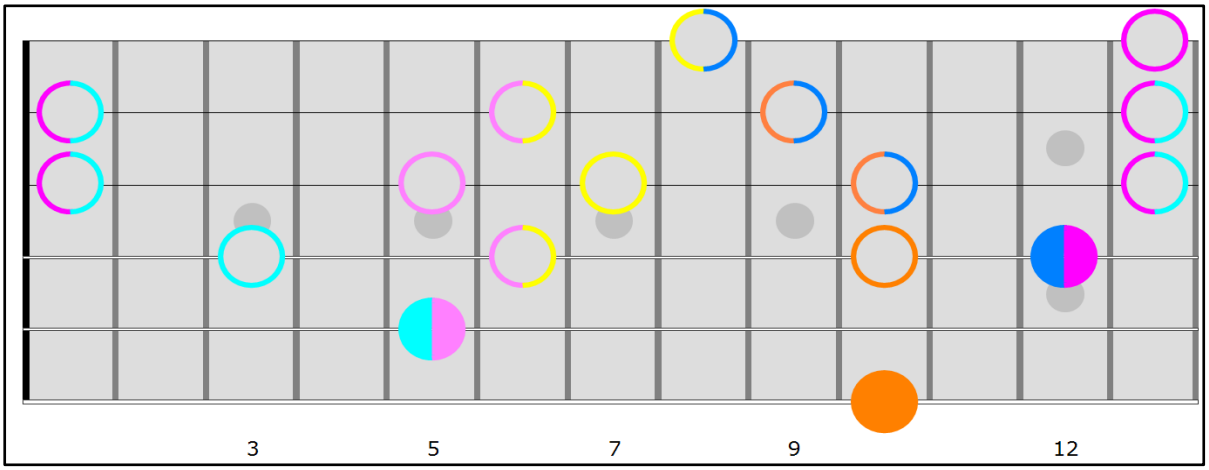
Again, here are a few examples of short ideas you could potentially write. They sound a little odd in isolation, but they do work as individual licks or riff ideas. They sound better with the relevant dominant chord behind them (C7 for these examples), and the perhaps the following examples feel limited because they would work better in conjunction with a changing chord progression, meaning the arpeggios would change too. However, this is math rock so do whatever suits best!

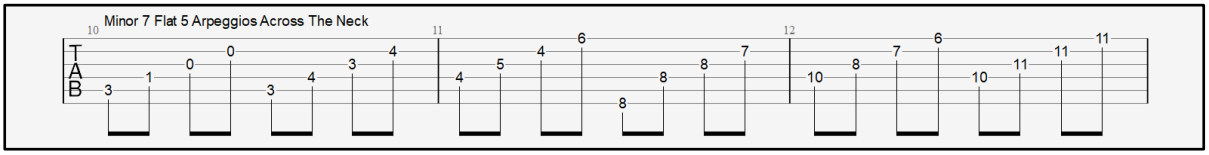


4E - Dominant Arpeggios Examples

Minor Seven Flat Five (Half Diminished) Arpeggios Across the Fretboard

Lastly, let us conclude with minor seven flat five arpeggios spanning one octave. Again, I have colour coded each shape to show how they are connected so you can see each individual arpeggio (chord) shape. However, I have changed the root note from C to D to better fit them in the examples. Take your time to work through each shape and find the most comfortable fingering for you. From there, experiment with stringing them together into chunks. The transcribed version below will aid in ingraining them into your memory and muscle memory.



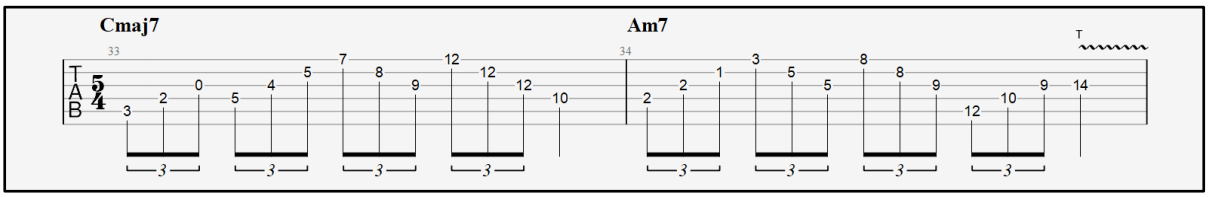


Arpeggios - Examples Riffs & Licks

Before moving forward, it is important that you are already familiar with the extended arpeggios we have just explored. In the upcoming section, the goal is to provide you with a few examples of how to combine different types of arpeggios to create riffs and licks. These can be played alongside a chord progression or used independently. These examples are intentionally designed to be relatively simple as this will make it easier for you to learn and grasp the concepts. As you work through them, try to identify which arpeggio shapes are being used.

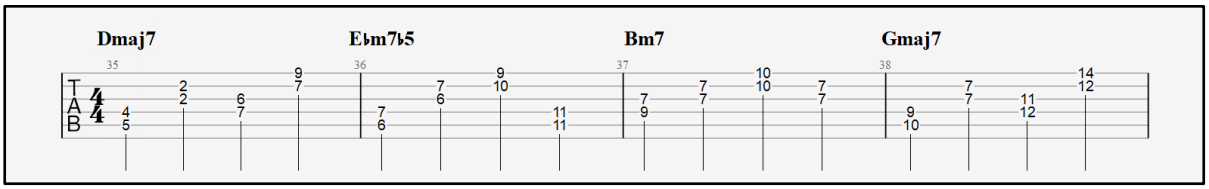
The following is an example of how you could write some ascending style licks. To make things

easier notice the triplet groupings (grouping of three notes). Seeing it as groupings of three notes it will make it much easier on the fingers and your mind. Try them backwards too, for a descending feel.



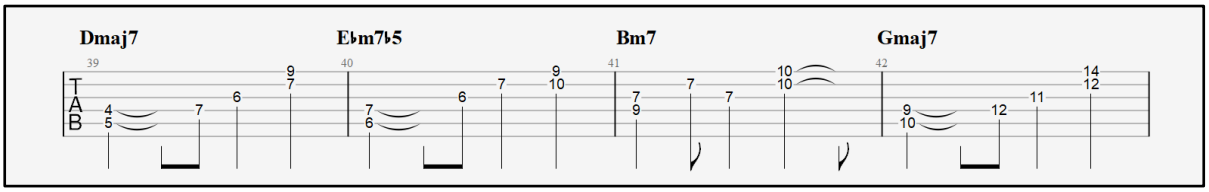
4F - Ascending Triplet Example

Example 4G is a rather crude one, but I designed it to get you bouncing around the arpeggio shapes in two note groupings. Can you work out which arpeggio shapes the note groupings are derived from? Noticing them will help cement ideas in your memory better.



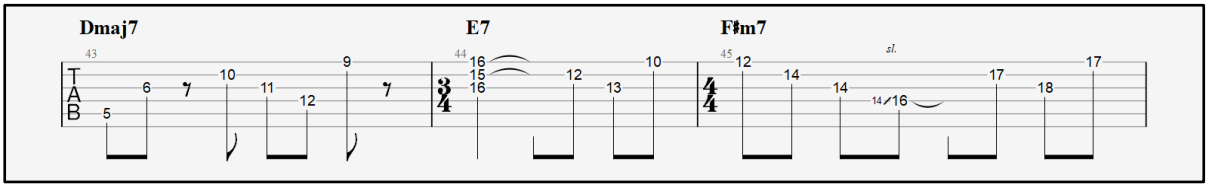
4G - Two Note Groupings Example 1

Example 4H takes the same idea but hashes it out into a passage. Sounds much more musical, right?



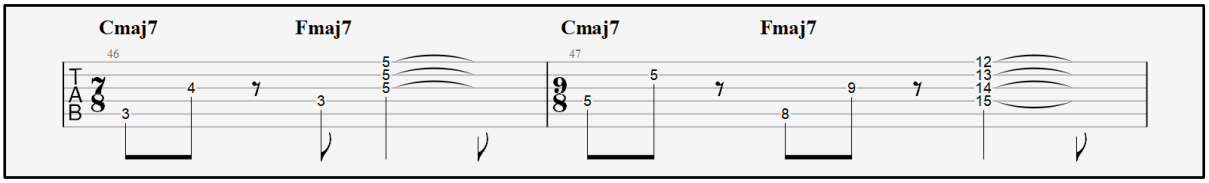
4H - Two Note Groupings Example 2

Example 4I is another more musical idea. It is based around a IV, V, vi progression. You will hear the band Delta Sleep use this kind of chord progression quite regularly and to great effect in their music.



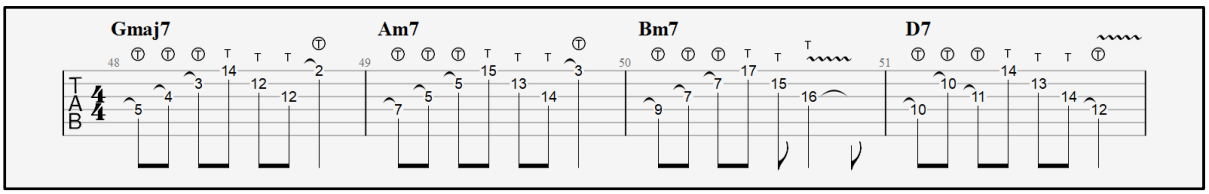
4I - IV, V, vi Progression Example

Example 4J is another mathy riff sounding riff based round a I-IV progression. This time, I wanted to have quicker changes and some odd timing to make things more musical and an example of what you are likely to see in math rock styles of guitar. If you count the first bar as a grouping of three and four beats, and the second bar as three groupings of three it will make things easier.

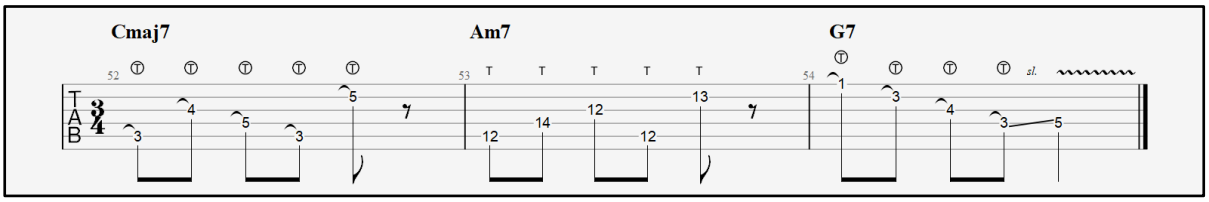


4J - I, IV Progression Example

Lastly, I could not help but include a couple of tapping examples. This is math rock after all!



4K - Tapping Example 1



4L - Tapping Example 2

As time passes, and the more time you dedicate to these methods, you will better internalize where notes are on the fretboard and formulate your ideas to greater effect. This concluded this segment. Next, we will look at functional strumming patterns.

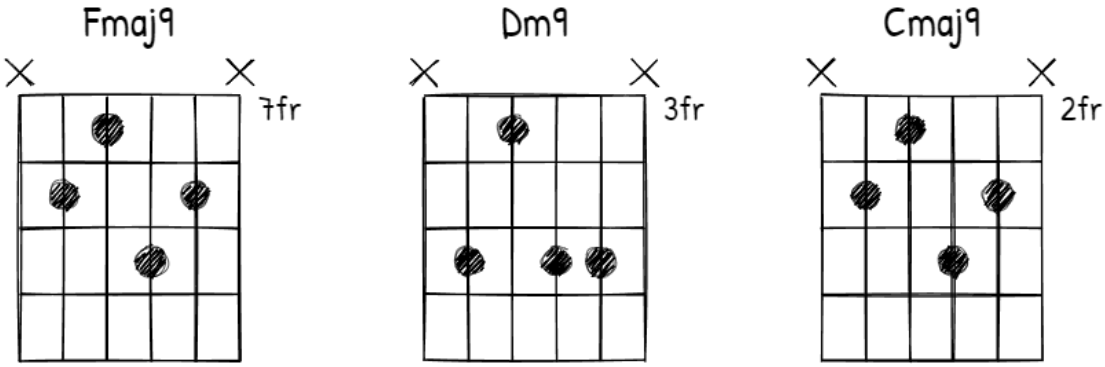
Segment 3B – Functional Strumming Patterns

In this section, we will explore some functional strumming patterns for math rock. We will start by looking at simple time: 4/4, compound time: 3/4, and then include a more complex time signature of 5/8.

You can use the following chord progression to practice these patterns, but do not hesitate to experiment with your preferred chords! I have included example videos for each pattern, so if you are having trouble with the rhythm, watch the video to grasp the feel of each strumming pattern. Additionally, I have added counts above each note in the bar to assist you. There is also a rhythm line above the transcription to make the details of each pattern clearer. Notes enclosed in brackets are for count clarification and should not be played. Having a basic understanding of rhythm counting will benefit you here, but you can still learn them by watching/listening to the example videos.

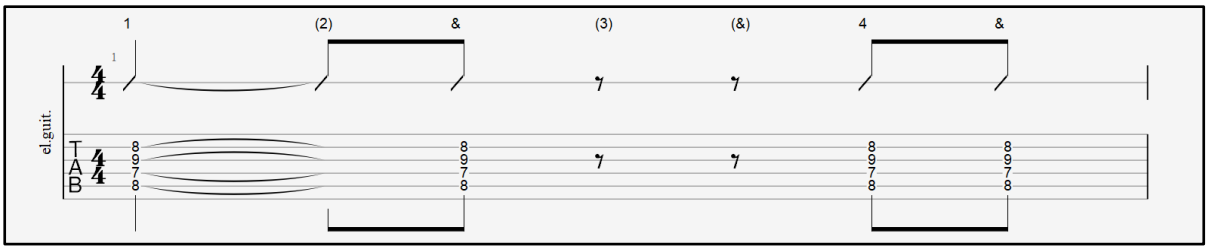
Here is a handy chart to make note division counts easier and clearer for you:





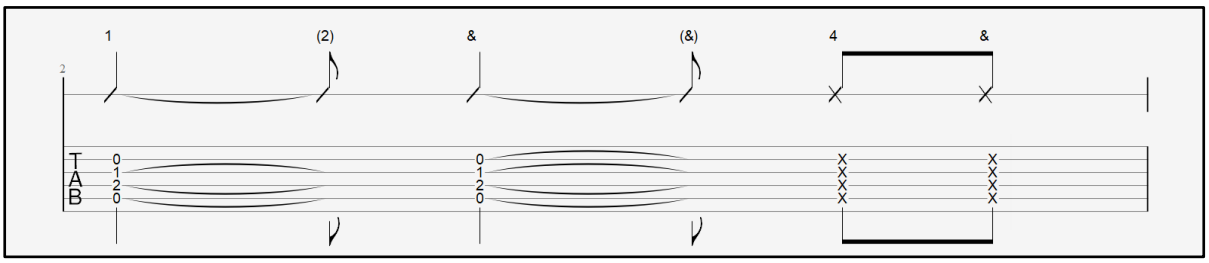
Functional Strumming Patterns for Math Rock in 4/4 Time

You might have heard strumming pattern 5A or something similar at the beginning of or in the breakdown section of math rock/Midwest emo songs. It is quite a common pattern but there are usually slight variations on this theme. For example, see the strumming pattern to the intro of the song Heat Wave by the band Gulfer, below in example 5.B.



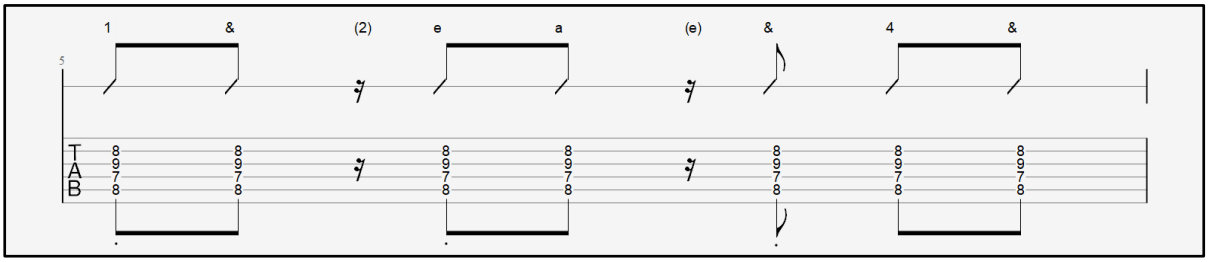
5A – 4/4 Functional Strumming Pattern 1a

In example 5B, the second strum is longer and the last two notes (4 &) should be played as dead notes (muting the strings as you strum them).



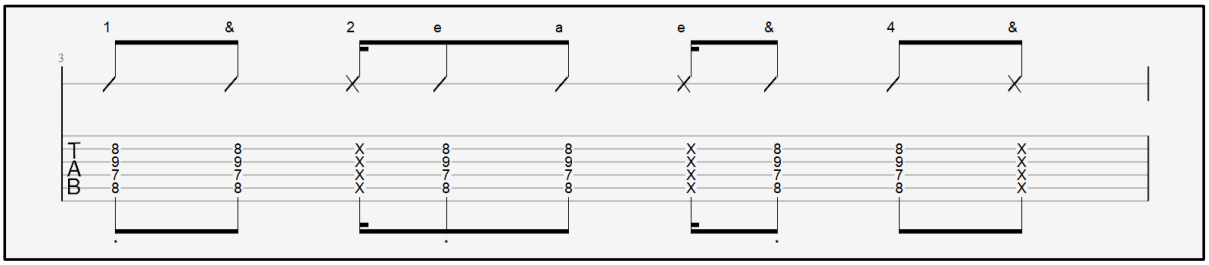
5B – 4/4 Functional Strumming Pattern 1b

Strumming pattern 5C may also be a pattern you have heard many times, albeit with slight variations to the one presented below. The key to this pattern is the short strum on the first note (notice the staccato mark) and a regular eighth note strum on each of the following strums.



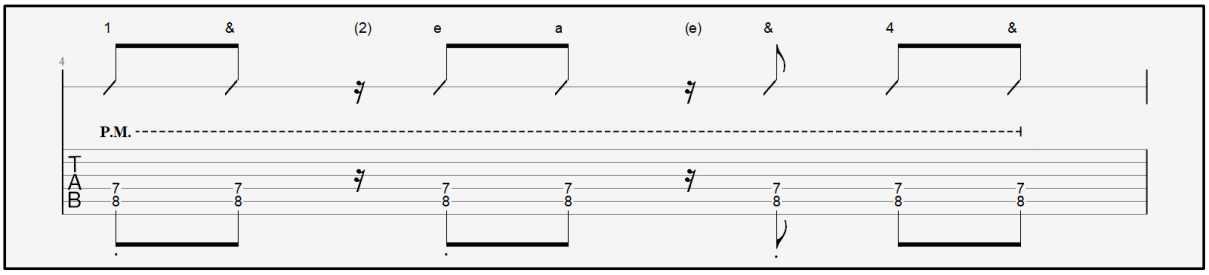
5C – 4/4 Functional Strumming Pattern 2a

Adding some dead notes before each phrase adds an interesting feeling to the piece.



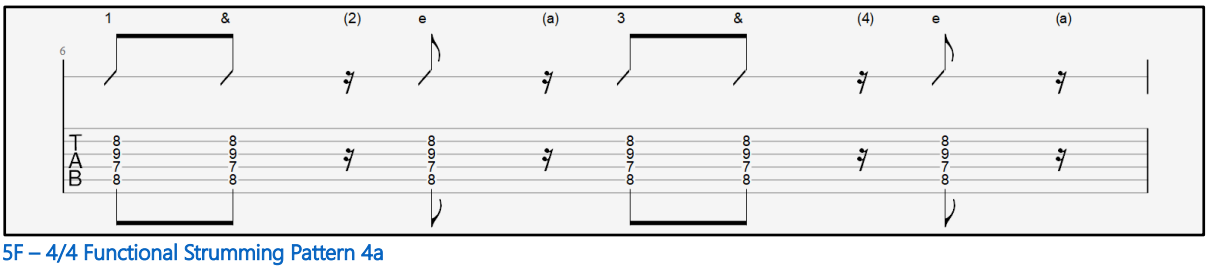
5D – 4/4 Functional Strumming Pattern 2b

This rhythm is also commonly used with palm muting. Usually when a part of a song is building to something. Try out Example 5E below, and I am sure some songs that uses this pattern will come to mind.



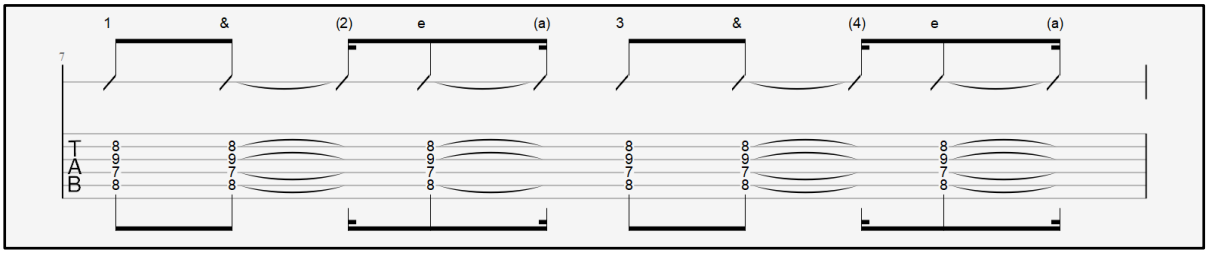
5E – 4/4 Functional Strumming Pattern 3

Strumming pattern 5F is perfect for a math-pop idea. It is based on many of the strumming patterns you will hear used by British math-pop bands, especially if you play it at a quick tempo. It has a poppy grove, but the syncopation makes it a less conventional strumming pattern than you would hear in most pop music.



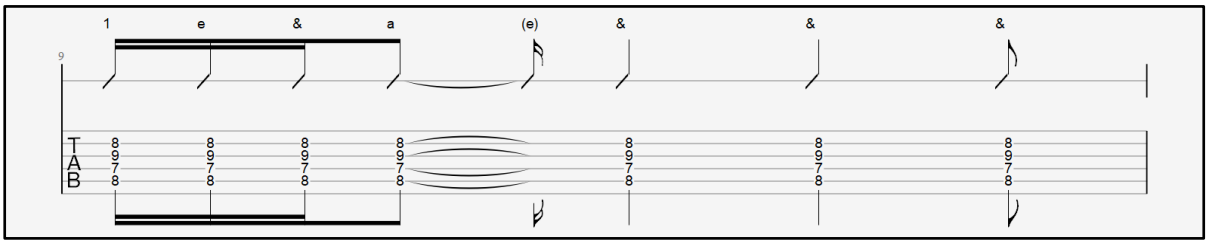
5F – 4/4 Functional Strumming Pattern 4a

You could also play it without the rests for a slightly different feel.



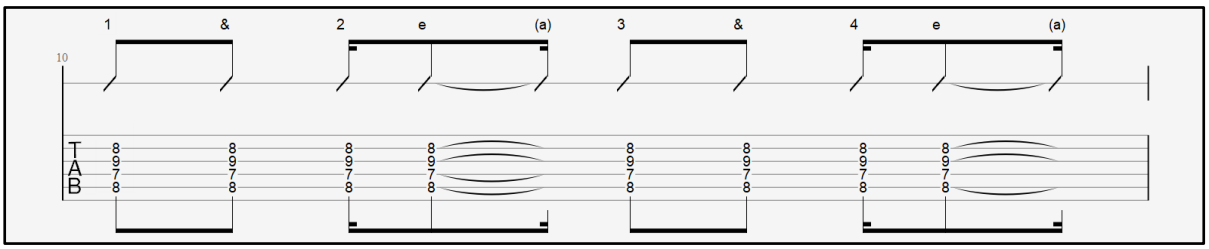
5G – 4/4 Functional Strumming Pattern 4b

Leaving the frequent choppy patterns aside for a moment. Strumming pattern 5H fills in all the gaps but still retains the characteristic syncopated feeling that you will frequently hear. This pattern works well for an upbeat intro to a song. That is not to say you cannot use it anywhere else though!



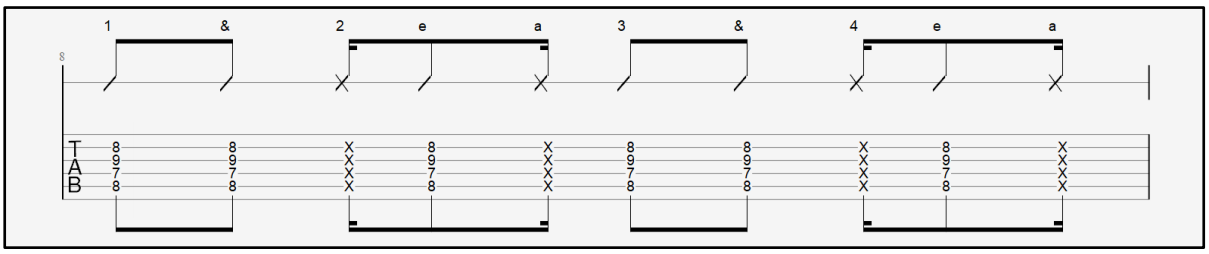
5H – 4/4 Functional Strumming Pattern 5

Strumming pattern 5I could work for a multitude of different parts of a song. It has quite a bit of energy so experiment with where you could use it. For example, if you palm-mute some intervals instead of the full chord, it could work nicely as a verse or bridging section in a song.



5I – 4/4 Functional Strumming Pattern 6a

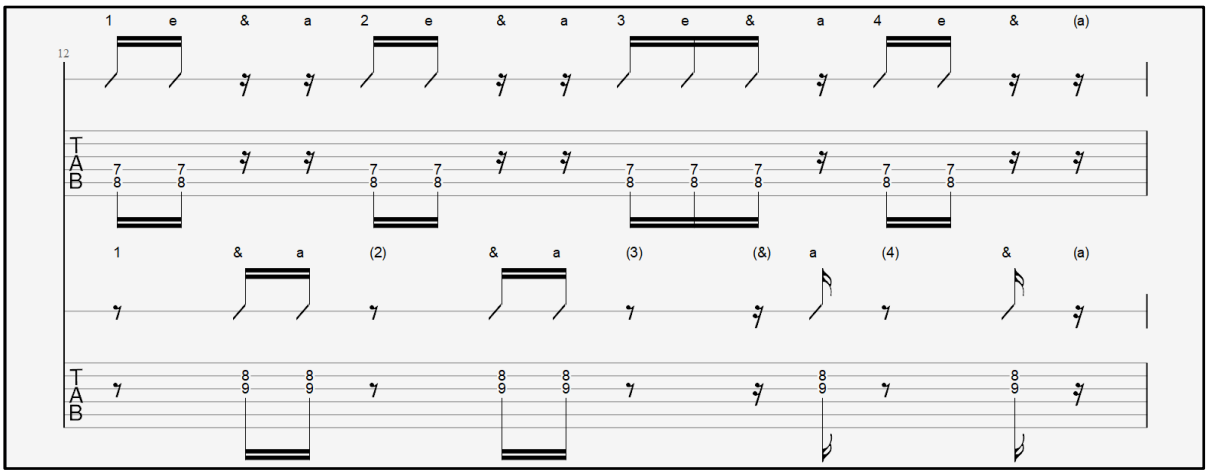
Example 5J replaces some of the strummed notes with strummed dead notes. This gives a more upbeat rhythmic feel. Like the start of the song Birdhouse by Tiny Moving Parts.



5J – 4/4 Functional Strumming Pattern 6b

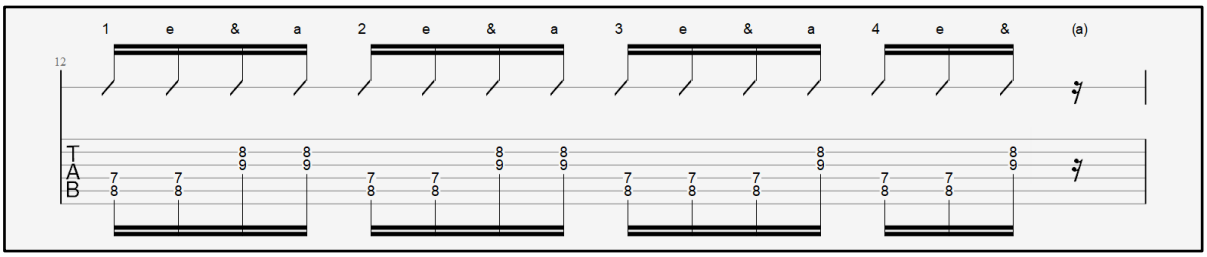
Let's move on to look at some functional strumming patterns for two guitars. Example 5K is based around a pattern used by the band Meet Me In St. Louis and a song called Right This Way, You Maverick Renegade. However, you will hear this alternating style of strumming patterns all over the same record. The alternating pattern is the same for the first two beats

then some variation is used from beat three onwards. Panning the guitar parts left and right with these kind of ideas sounds excellent.

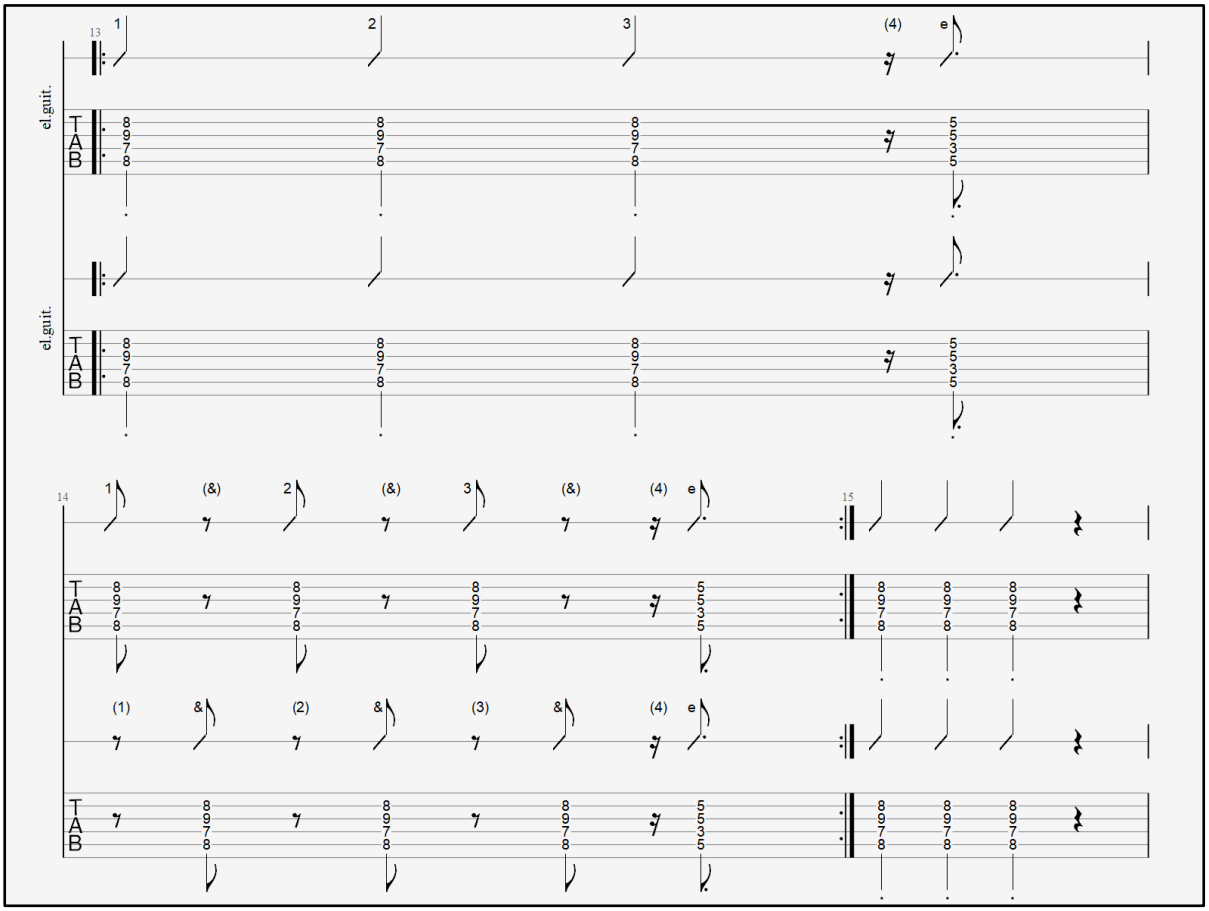


5K – 4/4 Functional Strumming Pattern 7

You could play the above example on one guitar (as indicated below), but you would not get the effect the separation of two individual guitars would provide. Nevertheless, it is still a useful strumming pattern for one guitar!

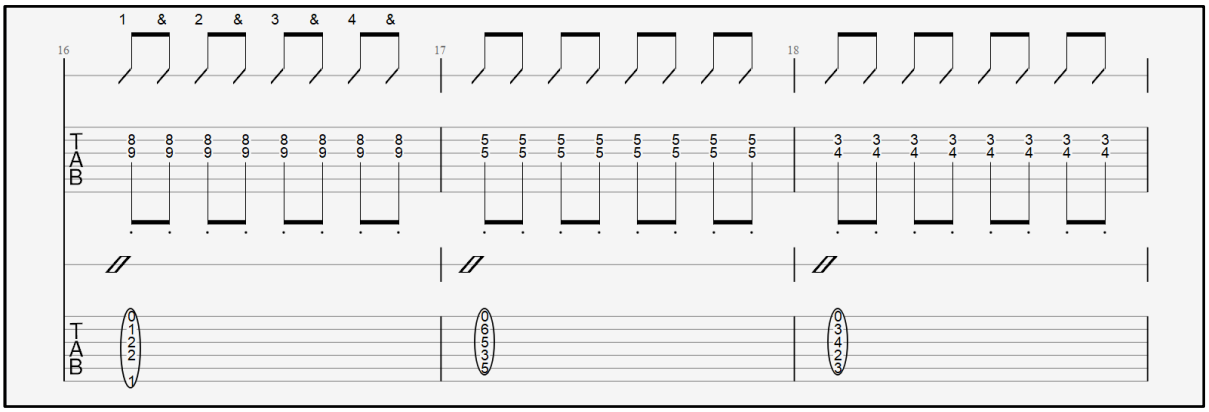


You will often hear variation of strumming pattern 5L used to establish a groove in a song. Usually, one of the beats in the bar will be syncopated, in this case, the last beat of the bar. Also, the groove is usually reinforced by a unity strumming pattern of two guitars, as demonstrated in the first bar, then an alternating strumming pattern between the two guitars will be used to add some interesting variation. I have given an example of this in the second bar.



5L – 4/4 Functional Strumming Pattern 8

Lastly, strumming pattern 5M is excellent for creating some space in a song. However, it retains an upbeat feeling through the energy provided by the staccato eight notes played by one of the guitars. I encourage you to experiment with the placement of the guitar playing the full chords for some interesting variation. You could even play the second guitar part as arpeggios for an interesting rhythmic contrast. Also, as you are only strumming the chord once per a bar here, it is a good opportunity to add reverb, delay, modulation etc to add to fill out the space, if desired.

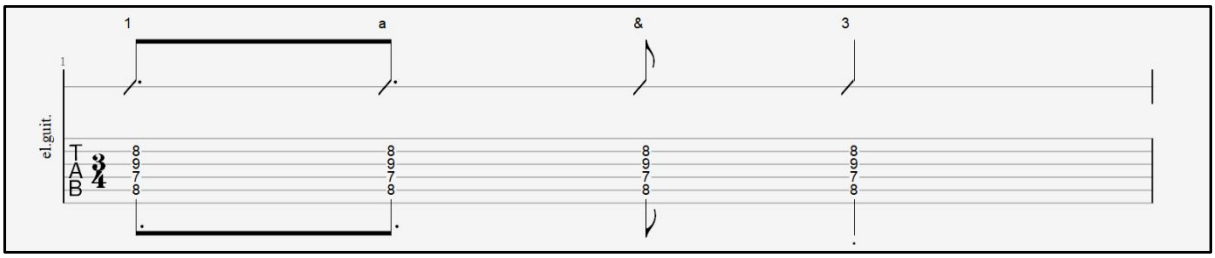


5M – 4/4 Functional Strumming Pattern 9

Functional Strumming Patterns for Math Rock in 3/4 Time

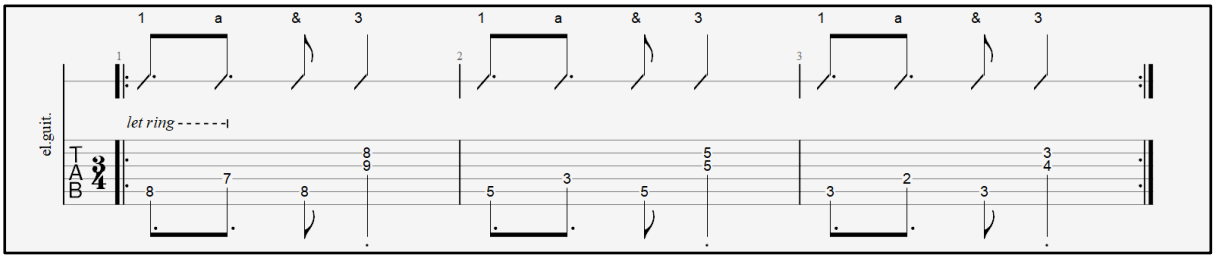
Moving on to 3/4 time. This is one of my favourite time signatures (along with 7/8). They are a great way to inject some energy into a song because the bar changes usually 'feel' quicker due to them begin shorter (Check out the record 'See You in Another City' by the band Blakfish for some excellent examples of quick 3/4 changes in action). The following are some common 3/4 strumming pattern variations you will hear used by math rock bands. Again, just as the 4/4 examples, I have included the count on the transcription and included accompanying demonstration videos to help you learn the patterns.

To being with (6A), we have a strumming pattern that accents beat three. This idea is taken from the song Pretend by the band Via Luna (and excellent math rock band you should check out). This one is a lot of fun, especially if you really accent the final strum (the three).

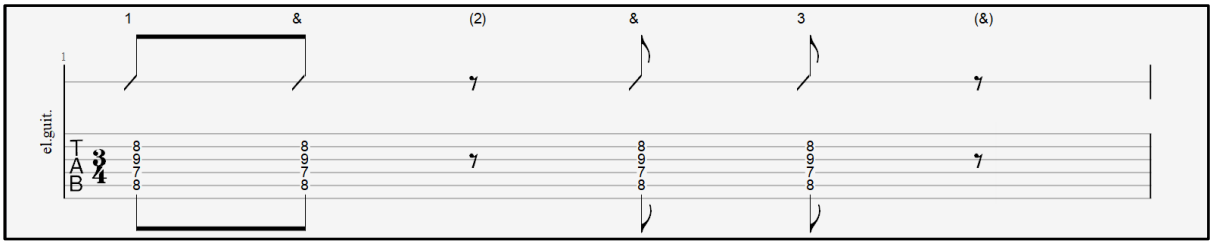


6A – 3/4 Functional Stumming Pattern 1

Strumming pattern 6B takes the same idea as above (6A), but arpeggiates the chord in the same pattern, and uses the same accent on beat three. I find this to be a nice variation that you could use in tandem to help the idea not be too repetitive.

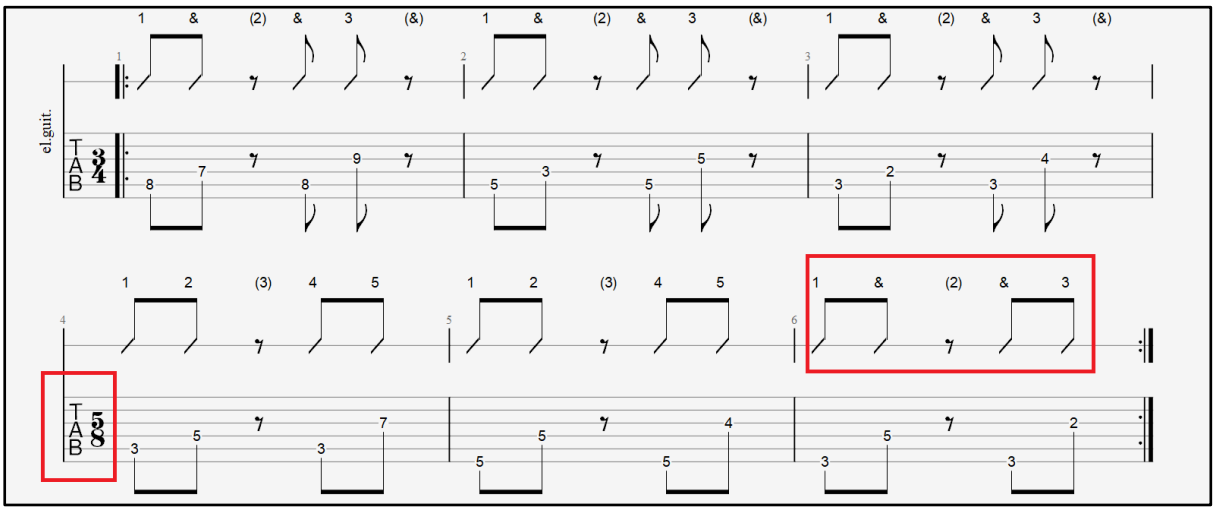


Next, strumming pattern 6C is inspired by the song Home Alone by the band Totorro. It is a fantastic 'driving-the-song' patten that pretty much works well in any song. I highly recommend adding this one to your go to strumming patterns.



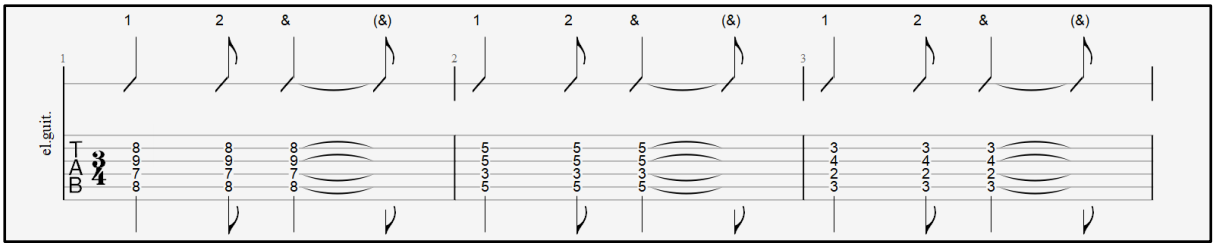
6C – 3/4 Functional Stumming Pattern 3

Again, like strumming pattern 6B, 6D takes the same idea as above, but arpeggiates the chord in the same pattern. Also, I used the same idea but in 5/8 time for an interesting variation. You could count the bar of 5/8 (1, 2, 3,4, 5) or you can count 1 & 2 & 3 (equals a count of 5) to continue the feel of 3/4 but with an abrupt ending to each bar. I indicated this in the example below to make it clearer. And just as before, you could use in tandem to help your idea not be too repetitive.



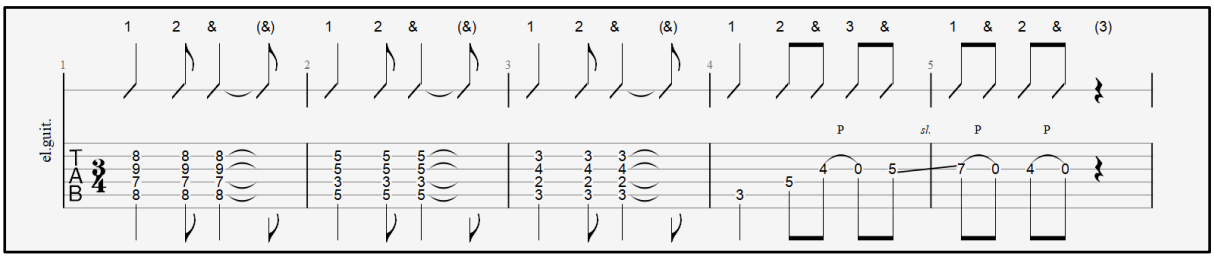
6D – 3/4 Functional Stumming Pattern 4

Strumming pattern 6E is inspired by the song Preparing for Guests by the band Blakfish. It is an excellent pattern for injecting some energy into a song.



6E – 3/4 Functional Stumming Pattern 5

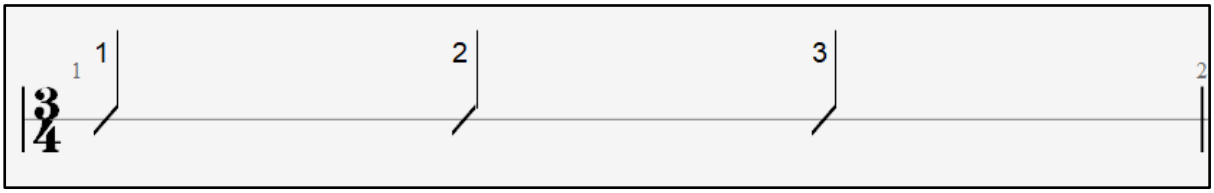
Lastly, pattern 6F takes the same idea and shows how you could complete the phrase by ending with an arpeggio.



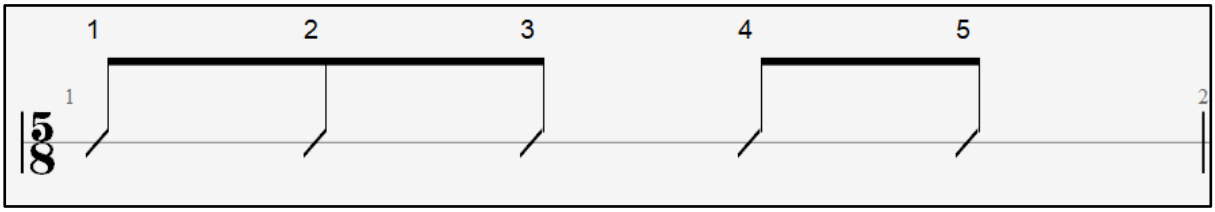
6F – 3/4 Functional Stumming Pattern 6

Functional Strumming Patterns for Math Rock in 5/8 Time

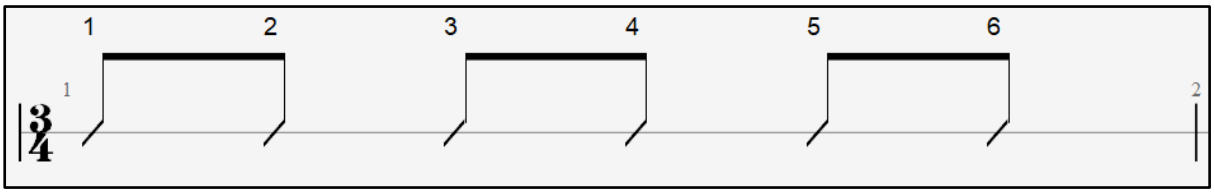
Now, let's delve into an unusual time signature: 5/8 time. I've included this because it's a logical progression from 3/4 time. To clarify, in a bar of 3/4 time, we count three quarter notes like this:



In a bar of 5/8, there is a five eighth note count:



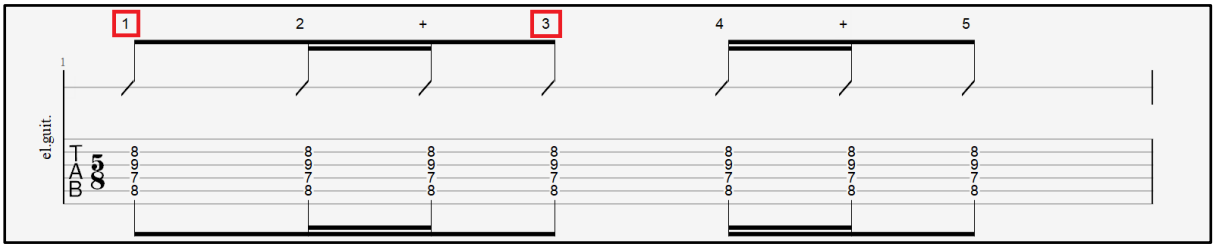
If you were to subdivide 3/4 time into eighth notes count, we would get a count of six eighth notes (could be seem as 6/8 time). Notice now that there is one more beat compared to 5/8.



Therefore, you can think of 5/8 as a chopped short bar of 3/4; try counting like this 1 & 2 & 3, 1 & 2 & 3, and so on. Without the & of 3, the bar will feel somewhat cut short; this is the feeling of 5/8 time. Hopefully now, you can begin to see why I recommended learning 5/8 following the timing of 3/4, and why the inclusion of the bars of 5/8 time in example 6D is a suitable variation.

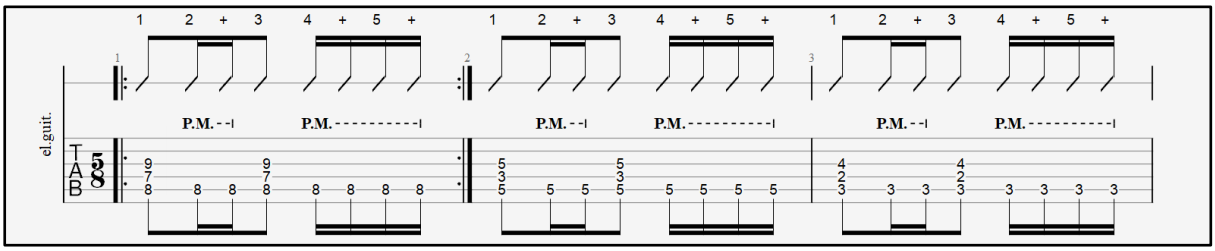
With that out the way with, let's start learning some strumming patterns. Because we have switched from quarter note counts to eight note counts, sixteenth notes will now be noted as + symbols on the examples. You can still count these as &s, but I wanted to make the difference clear to you. Also, if you have trouble working out any of the rhythm, then check the example videos.

In 5/8 (and 5/4 time), it is common to accent the 1st and either the 3rd or 4th beats. In strumming pattern 7A, we begin by accenting the 1st and 3rd beats.



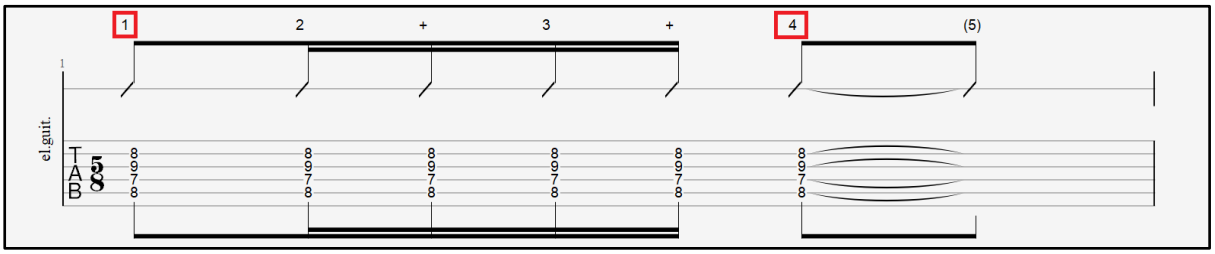
7A – 5/8 Functional Stumming Pattern 1

Following the same theme, we can accent the 1 and 3 while palm-muting notes in-between.



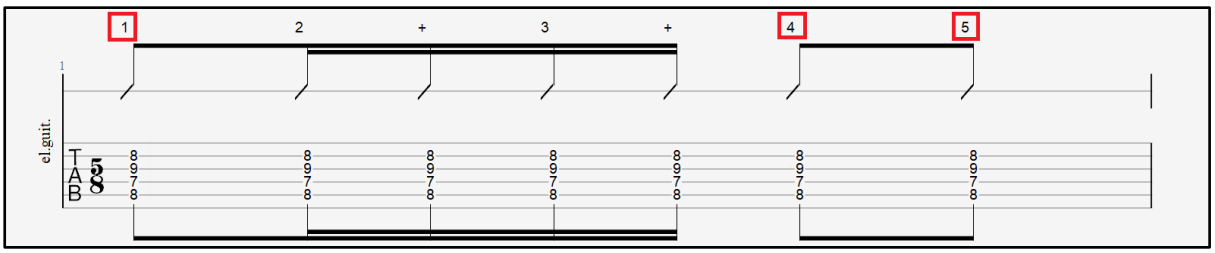
7B – 5/8 Functional Stumming Pattern 2

Moving on, strumming pattern 7C accents the 1st and 4th beats.



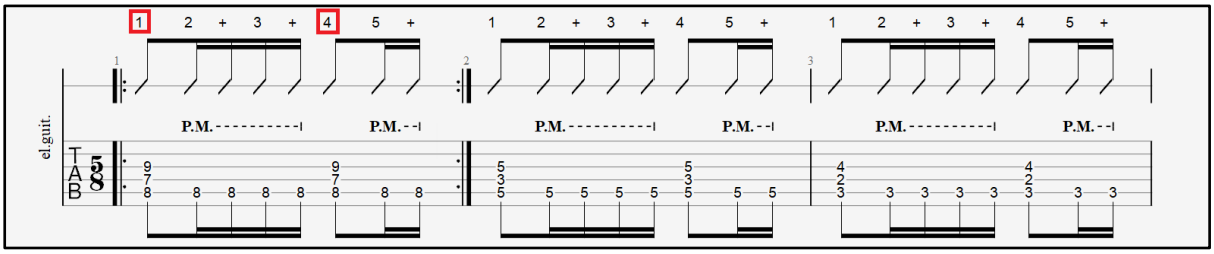
7C – 5/8 Functional Stumming Pattern 3

Strumming pattern 7D continues the theme by adding another accent on beat 5. Combining this pattern with pattern 7C is a great place to start for including some variation.



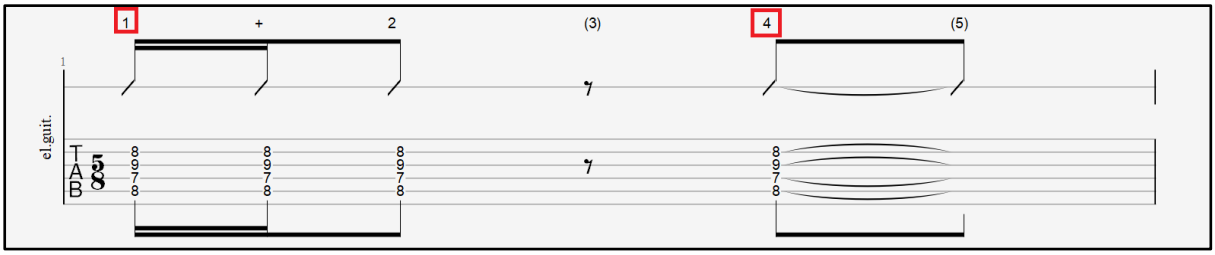
7D – 5/8 Functional Stumming Pattern 4

Just like example 7B, we can also apply a palm-mute pattern and accent the 1 and 4.



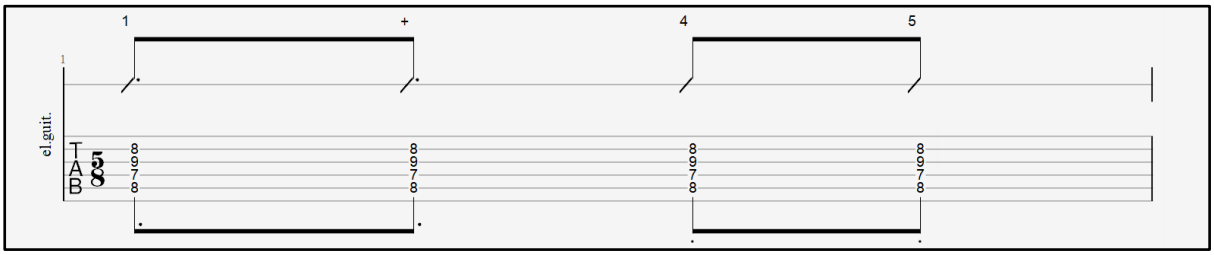
7E – 5/8 Functional Stumming Pattern 5

Continuing the theme of 1 and 4 accents. Strumming pattern 7F is an excellent way to introduce some space to the feeling of 5/8 time.



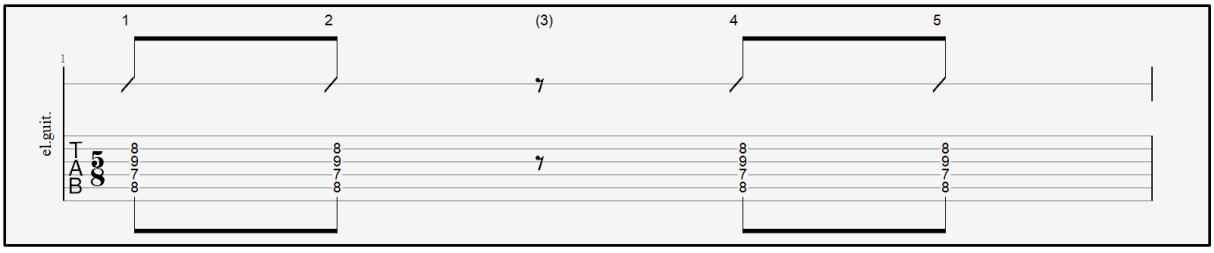
7F – 5/8 Functional Stumming Pattern 6

Strumming pattern 7G is a pattern famed for its use in the James Bond movie theme song. Movie trivia aside, it is a super useful strumming pattern for you to know.



7G – 5/8 Functional Stumming Pattern 7

Lastly, we have a pattern that accents the 1, 2, 4, and 5. I am quite found of using this pattern because as each bar repeats, it feels stranger to play.



7H – 5/8 Functional Stumming Pattern 8